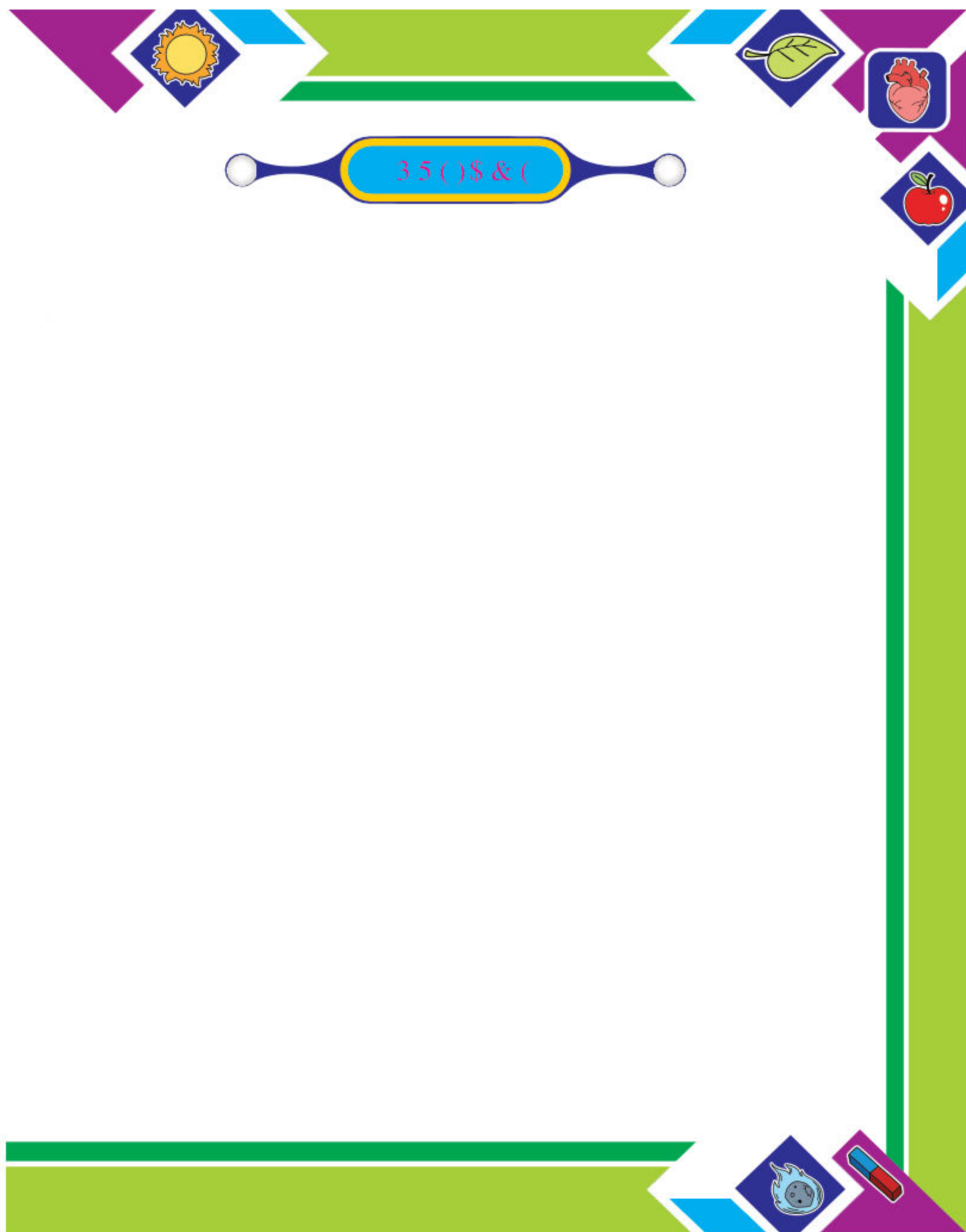
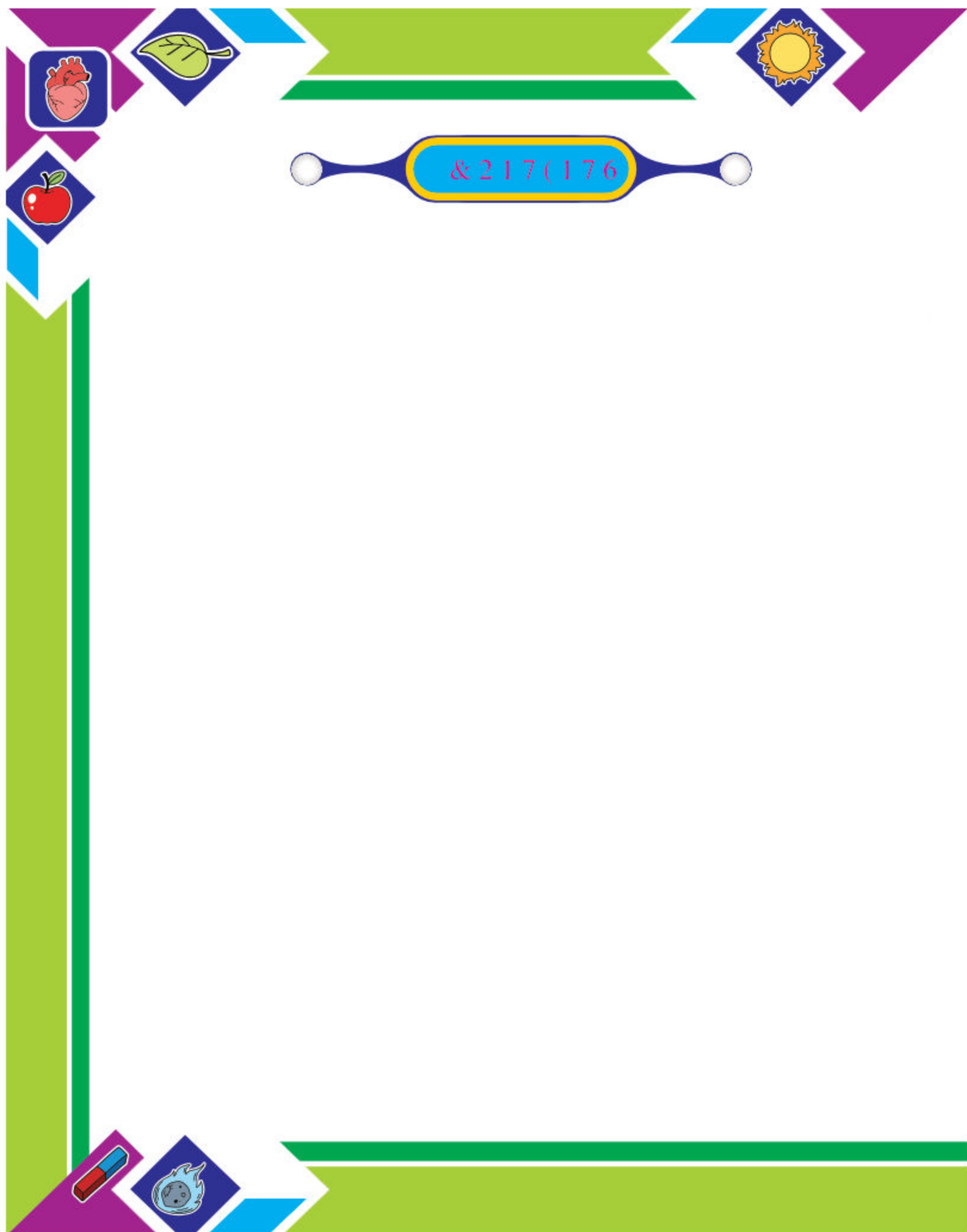




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◆ 'U 0XKDPPDG -XQDLG .KLOML
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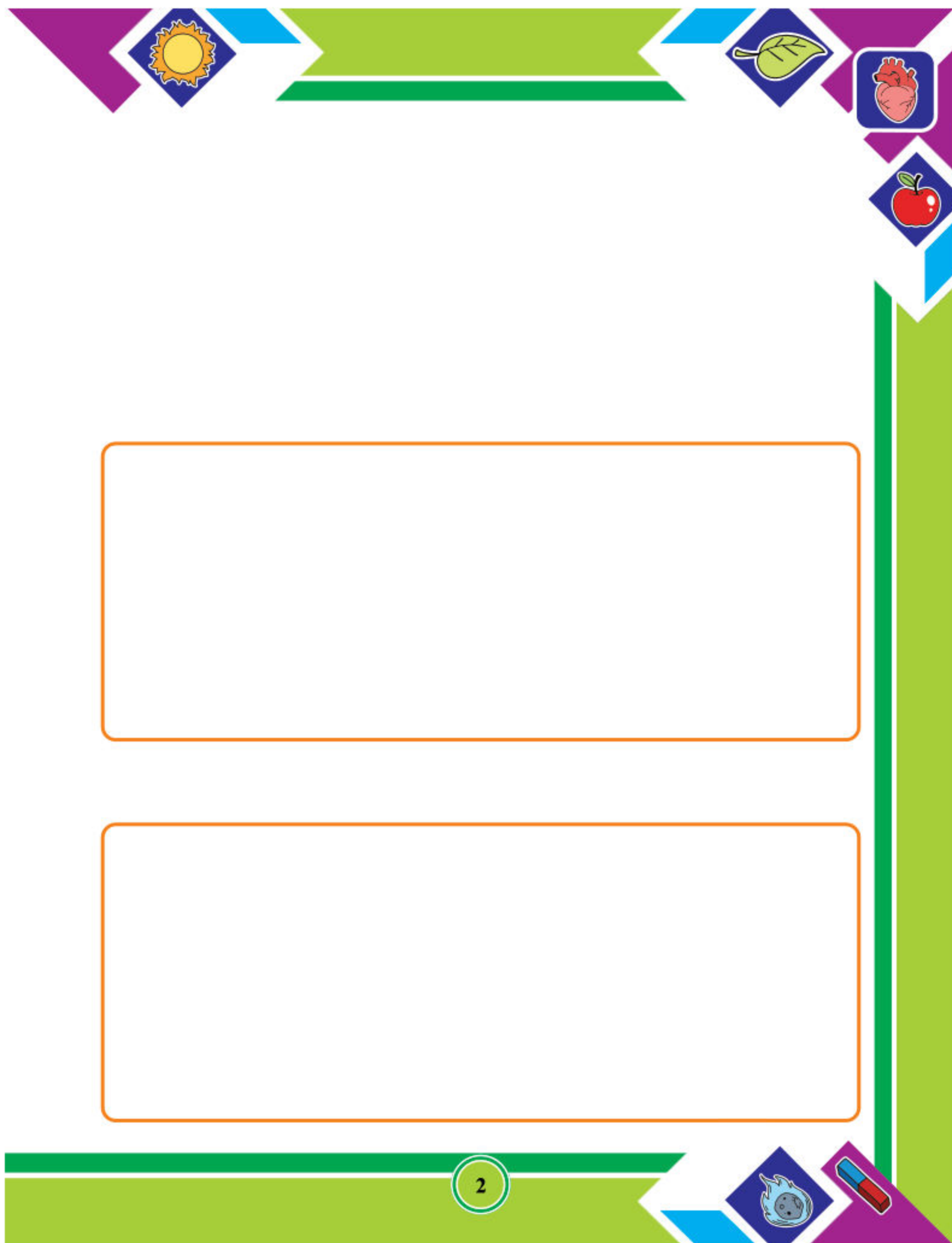
1.1 Intro

6 W X G H Q W / H D U Q I Q J 2 X W F R P H V

\$ I W U H R P S O M Z R Q N K L X Q V V W Q V O N K R X G E H D I D I R

- ◆ Understand that living things grow, take in nutrients, breath, reproduce, eliminate
Z D M D Q G H
- ◆ ' I D F X V W D W O M Q J W Q I V Q H C H Q I N W R J U R Z W E Y H D Q G U P D I Q K H O K \ 3 O Q W J F W K H L U H Q I N I U R P O I K W 3 R R W R M H I V Z K H I Q L I P V J H N W E J \ I U R P H W Q S O Q W D Q R M U D Q I L D V
- ◆ ([S O R U M H U H I I H R H Q V R I S D Q W I R U O L H I D Q G J U R Z R V D L O K W Z D M U Q X W Q V I U R V R I O D Q G U R R W R I E Z

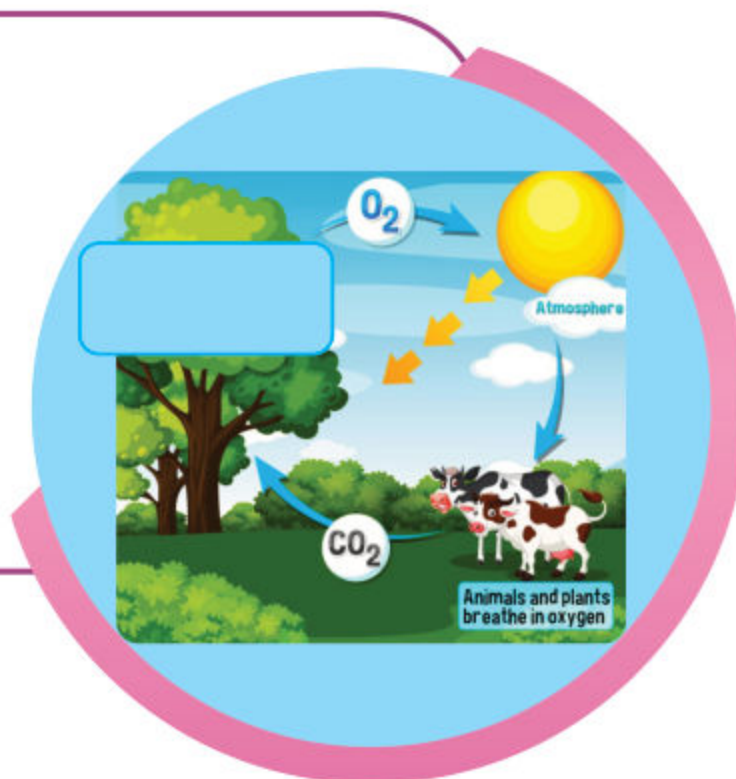


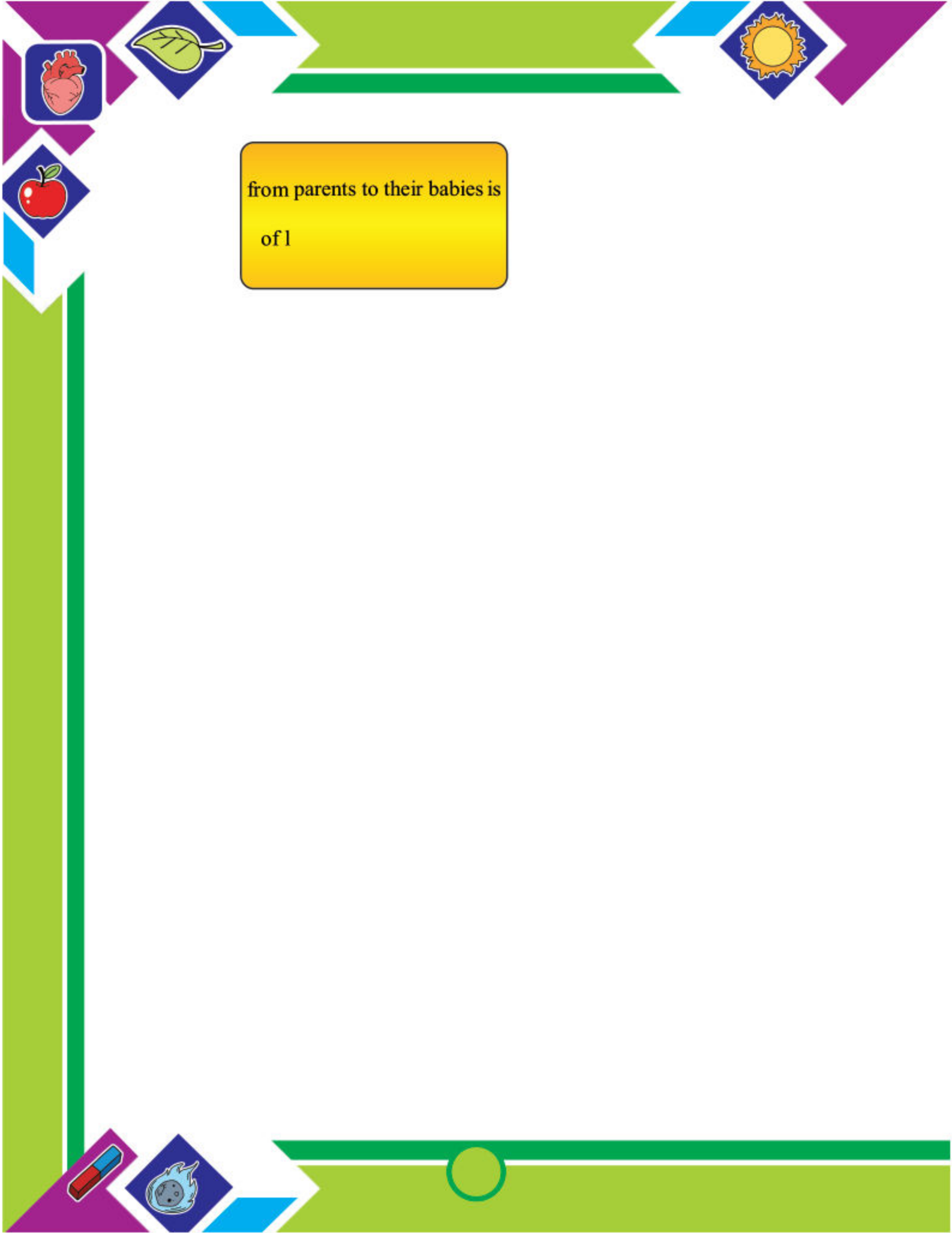




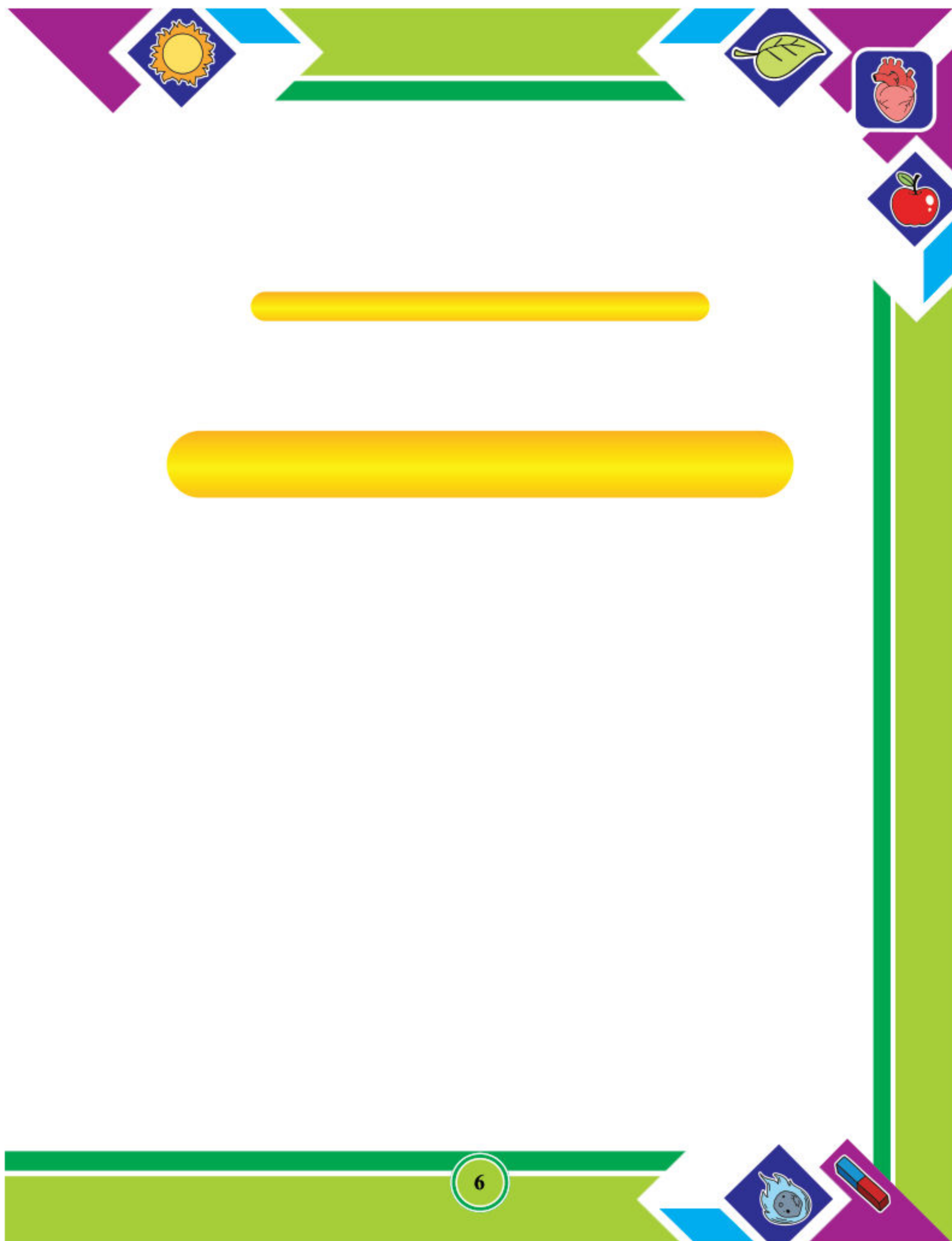
Now we will discuss some characteristics of living thing from the above characters







from parents to their babies is
of 1

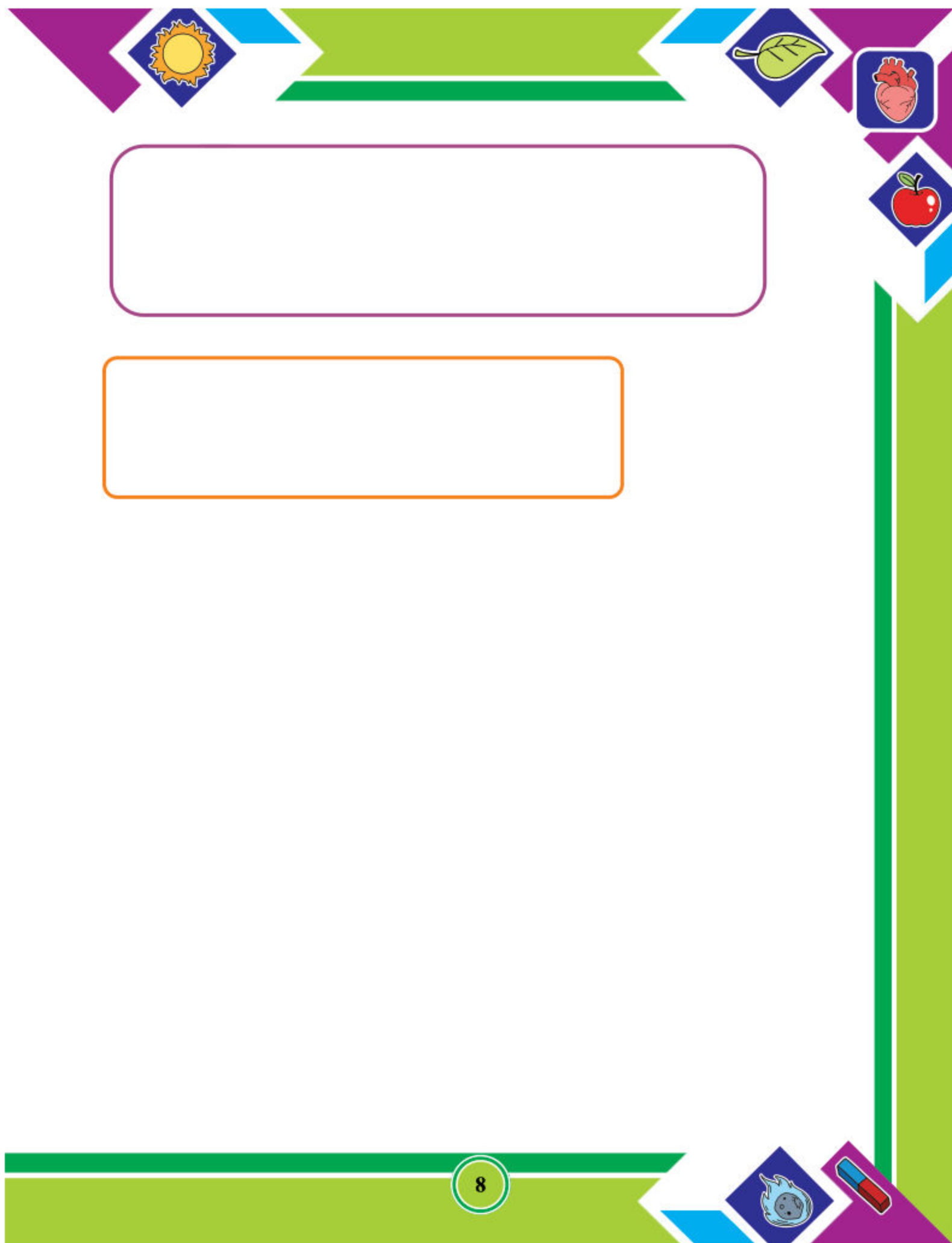




<RX NQRZ WKDW HDFK OLYLQJ WKLQJ UHTX'
IRU OLIH 3ODQWV DUH DOVR OLYLQJ WKLQJ UHTX'
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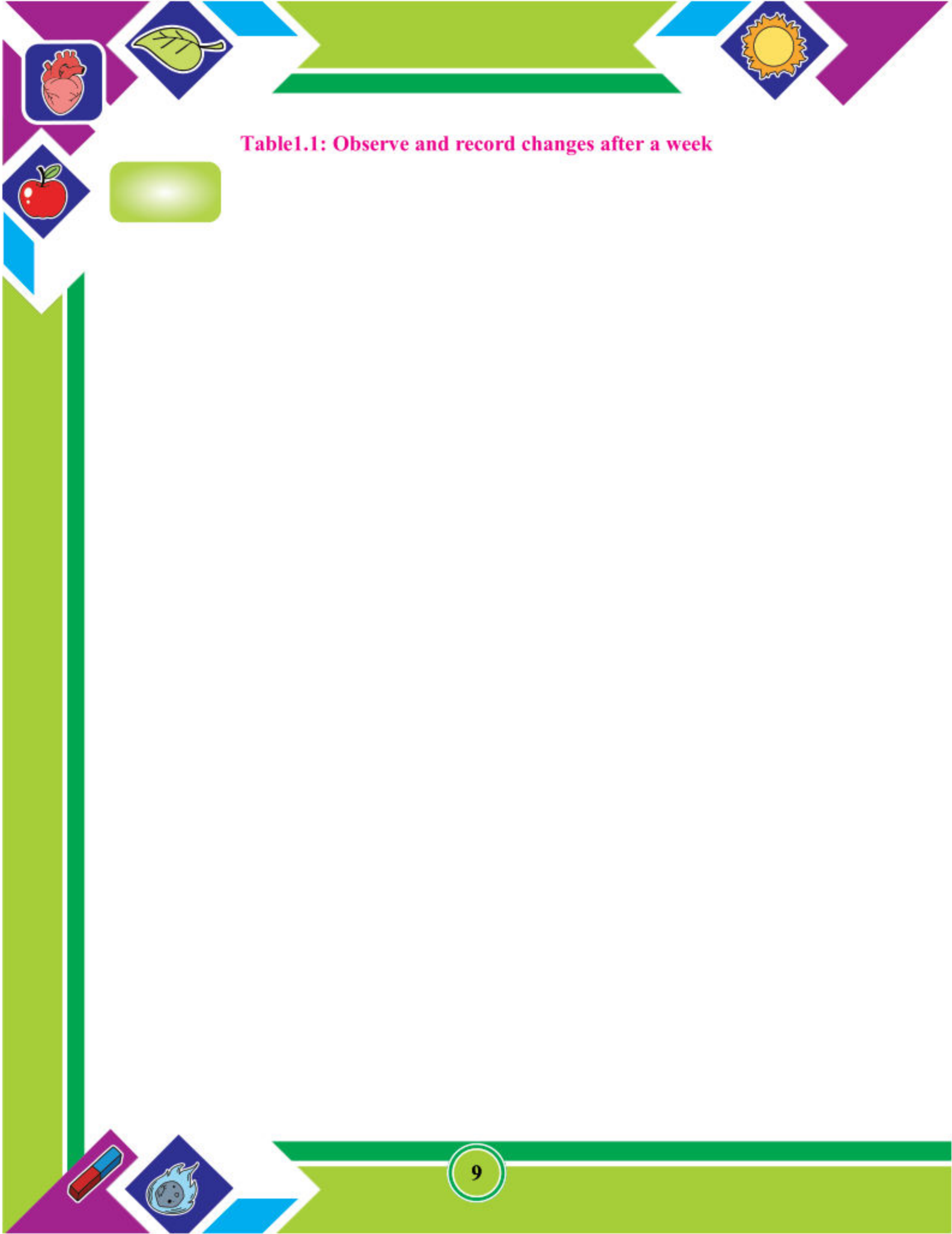
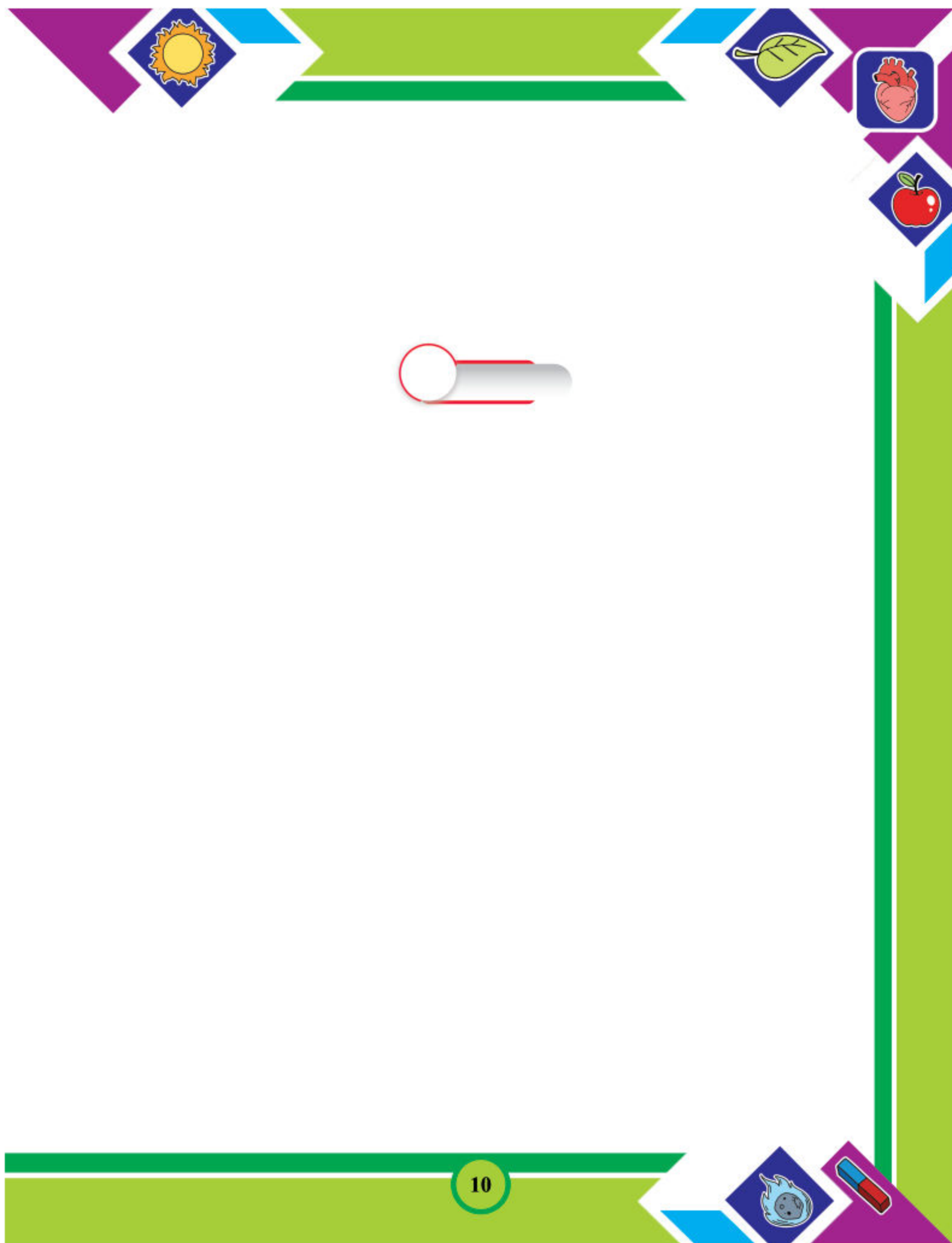


Table1.1: Observe and record changes after a week





v) **The living organisms eliminate some material from their body because this is**

A



Section D: Project:

& K D S



3 O D Q W X F W H X U
D Q G) X Q F W L R Q

10 D M R Q F H S W V

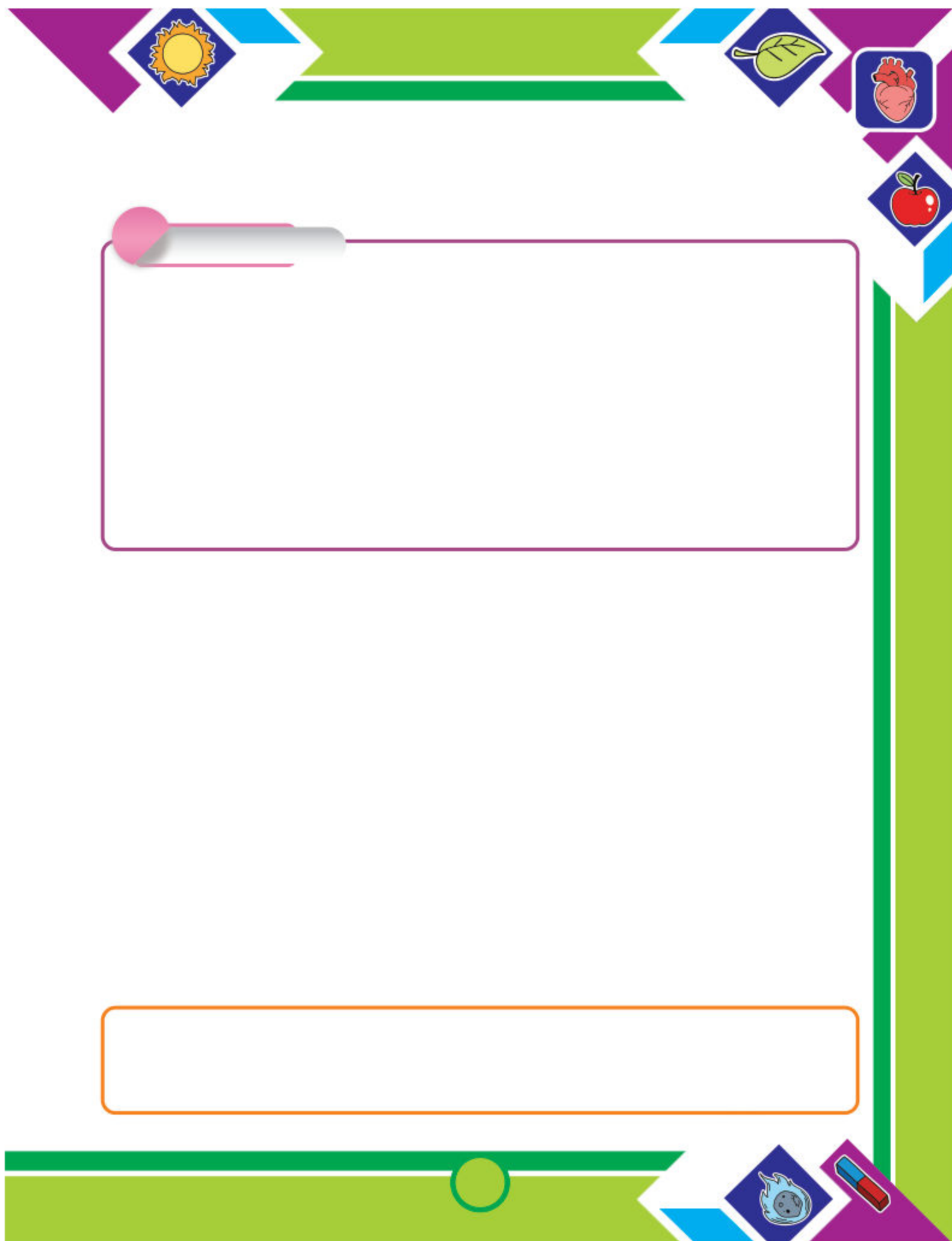
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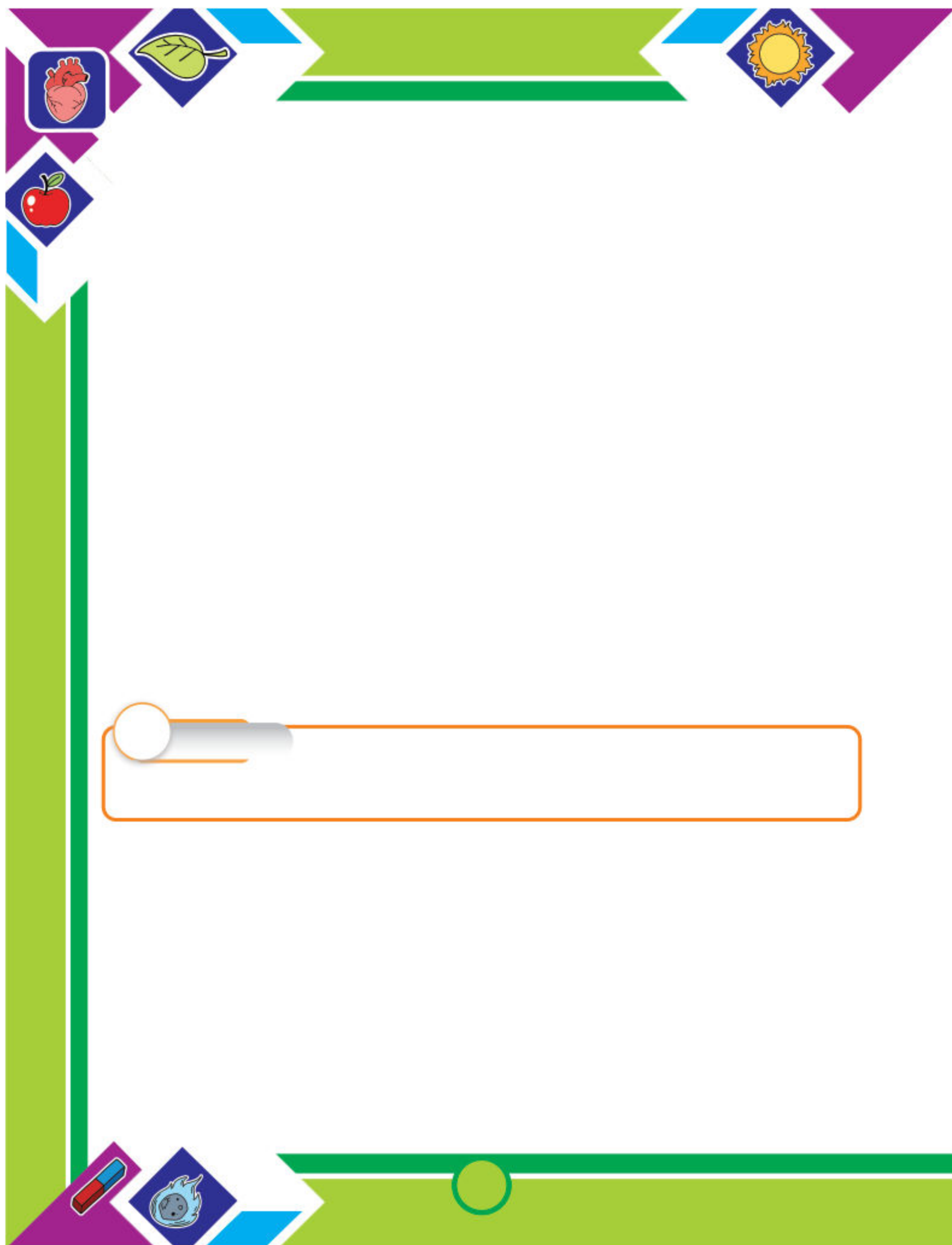
6 W X G H O W / H D U Q L Q I 2 X W F R P H V

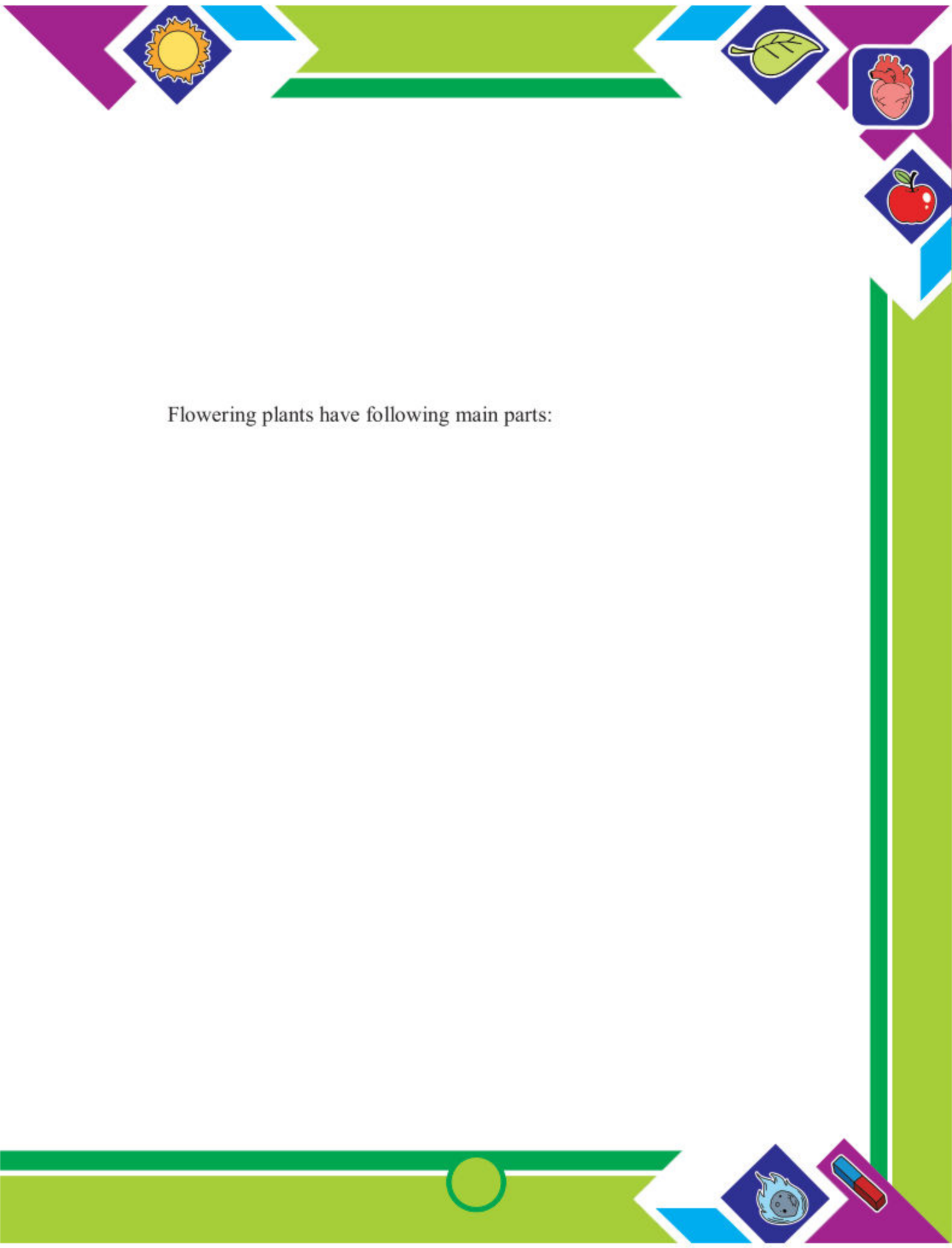
S T W H R P S O M R Q Q K L X Q V V W Q W K R X G O H D I H R

- ◆ Classify the plants into two major groups (flowering, non-flowering), and give examples of each group.
- Describe the functions of different parts of flowering plants: Roots, stem/trunk, leaves and flowers.
- Identify the parts of the plant transport system and describe their functions (Stem, leaf and root).
- Investigate the way in which water is transported within plants.
- Identify the parts of a flower and describe their functions (limited to petals, sepals, anthers, filaments, stamens, stigma, style, carpel, and ovary by diagram).
- Identify stages in the life cycles of common flowering plants.
- Explore the role of flowers in the life cycle of flowering plants, including pollination, fruit and seed formation and seed dispersal.
- Describe seed germination and know that seeds require water and an appropriate temperature to germinate.
- Relate that why plants are vital for sustaining life on earth
- Identify various professions associated with this unit of science.
farmers, gardeners, florists, etc.

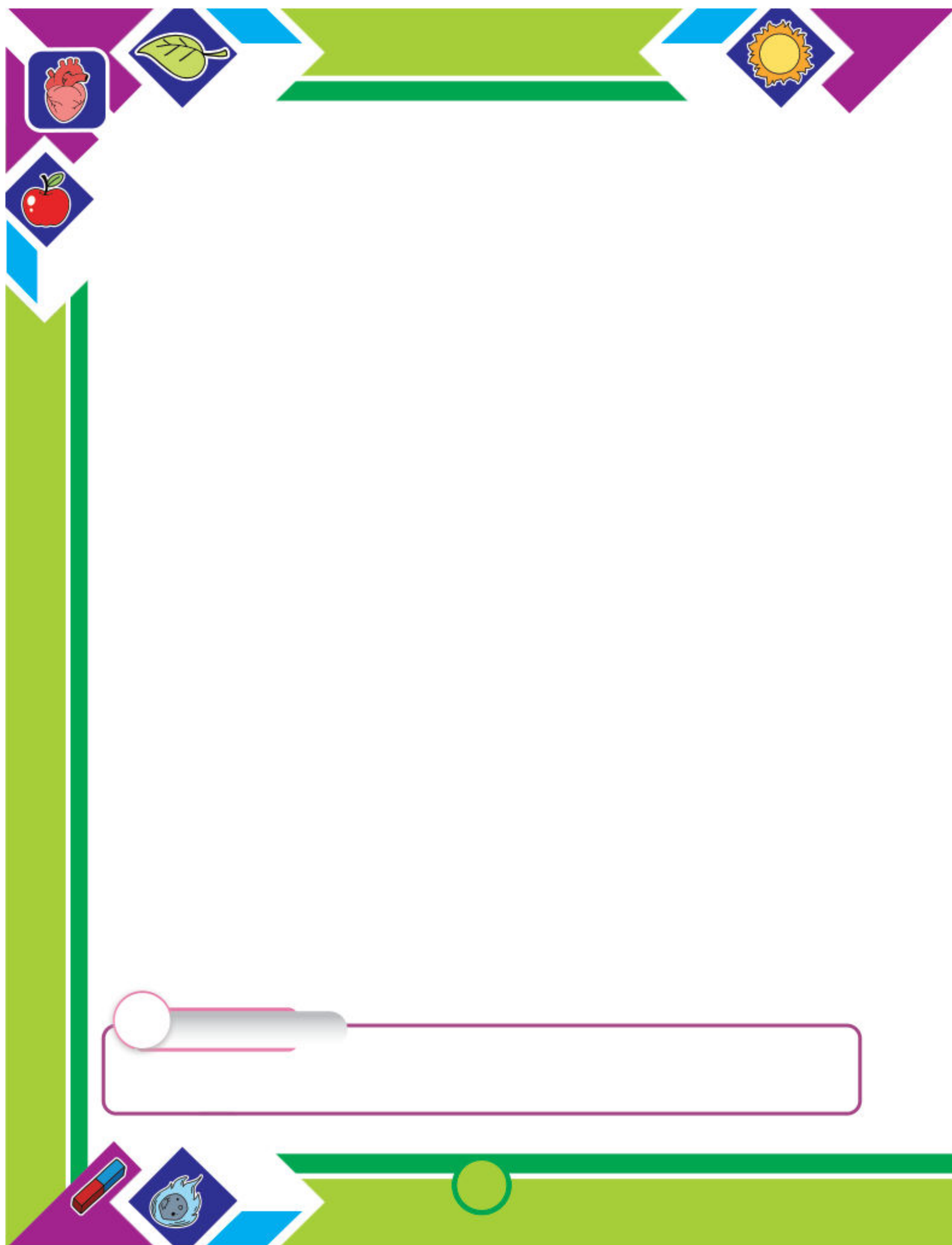


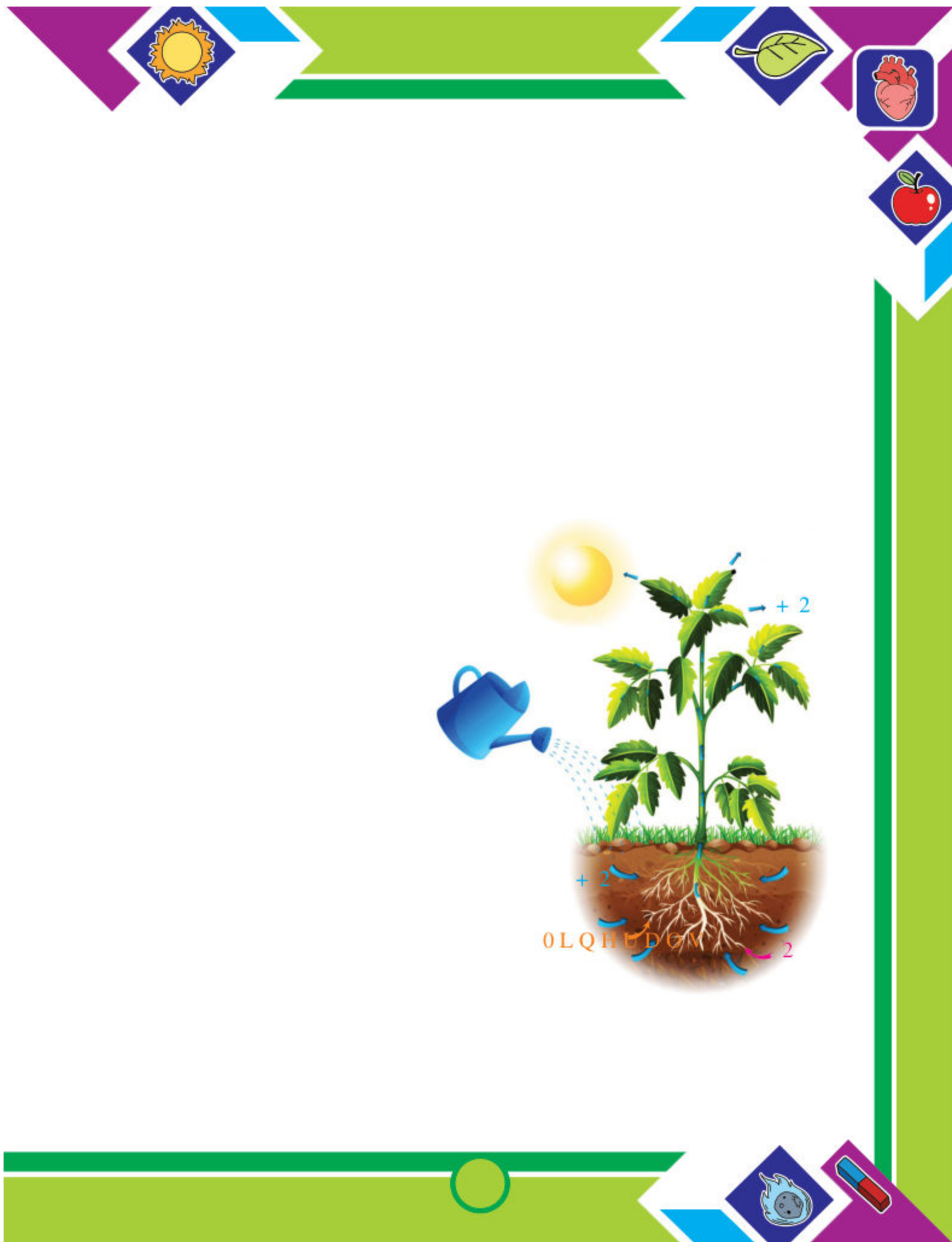


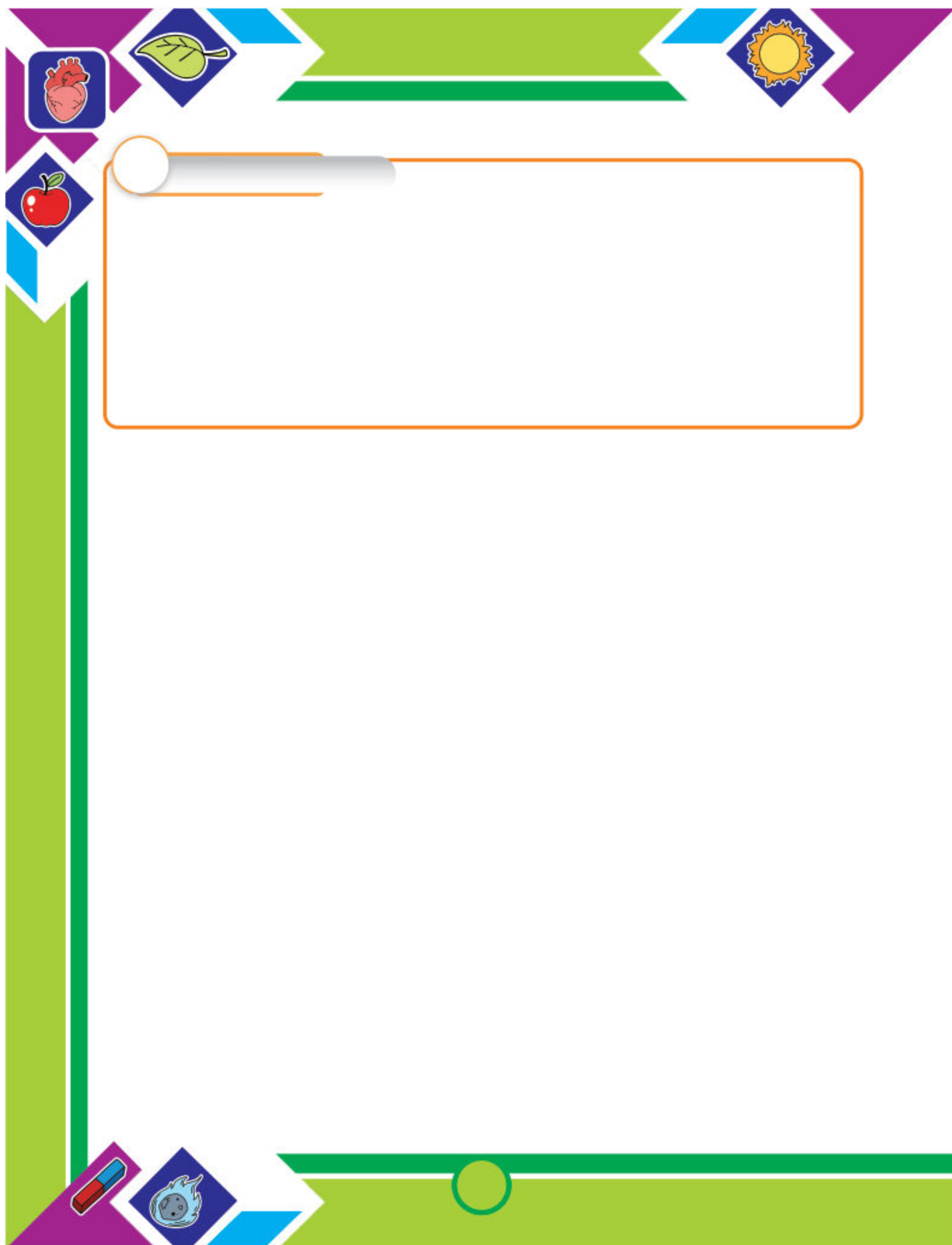


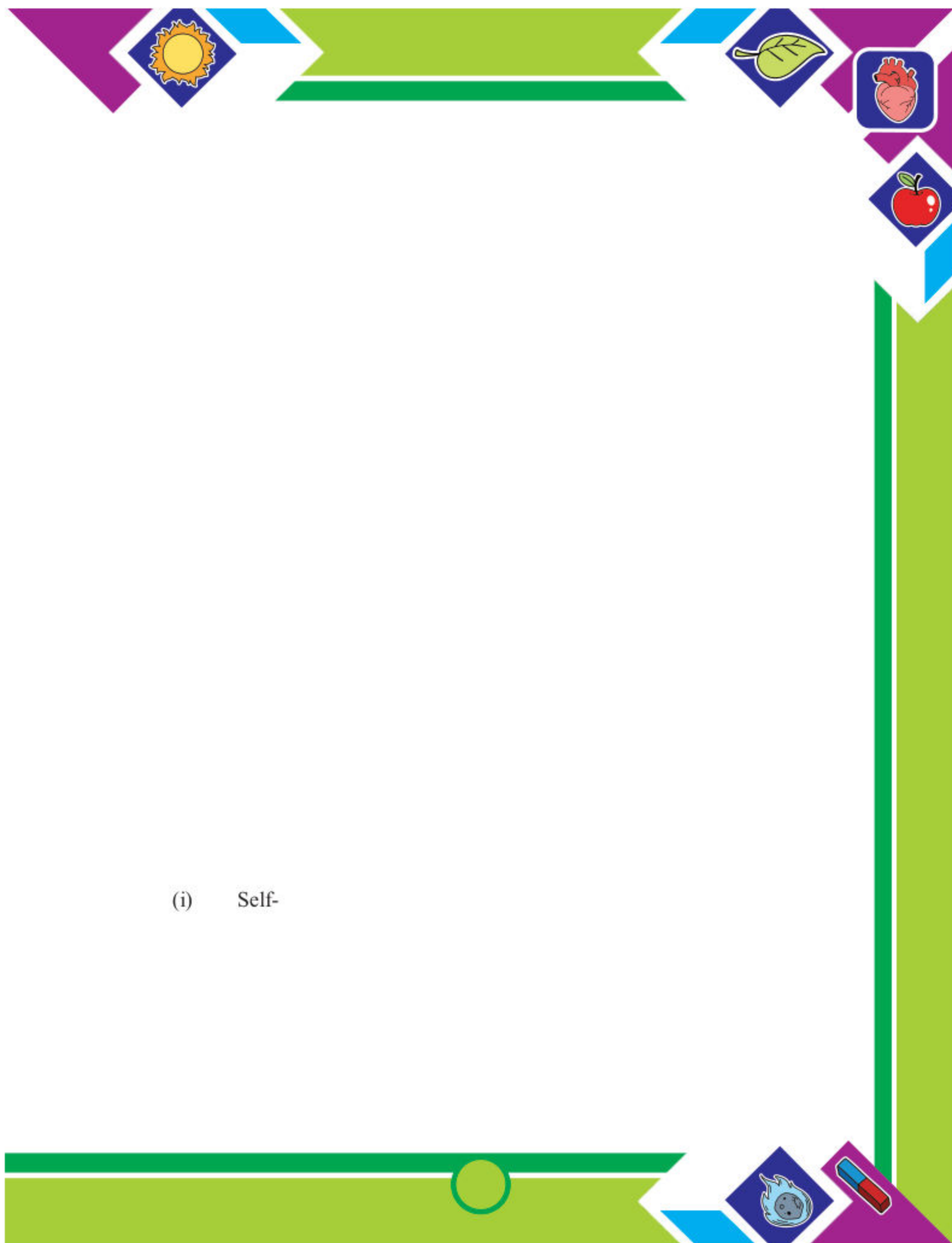


Flowering plants have following main parts:

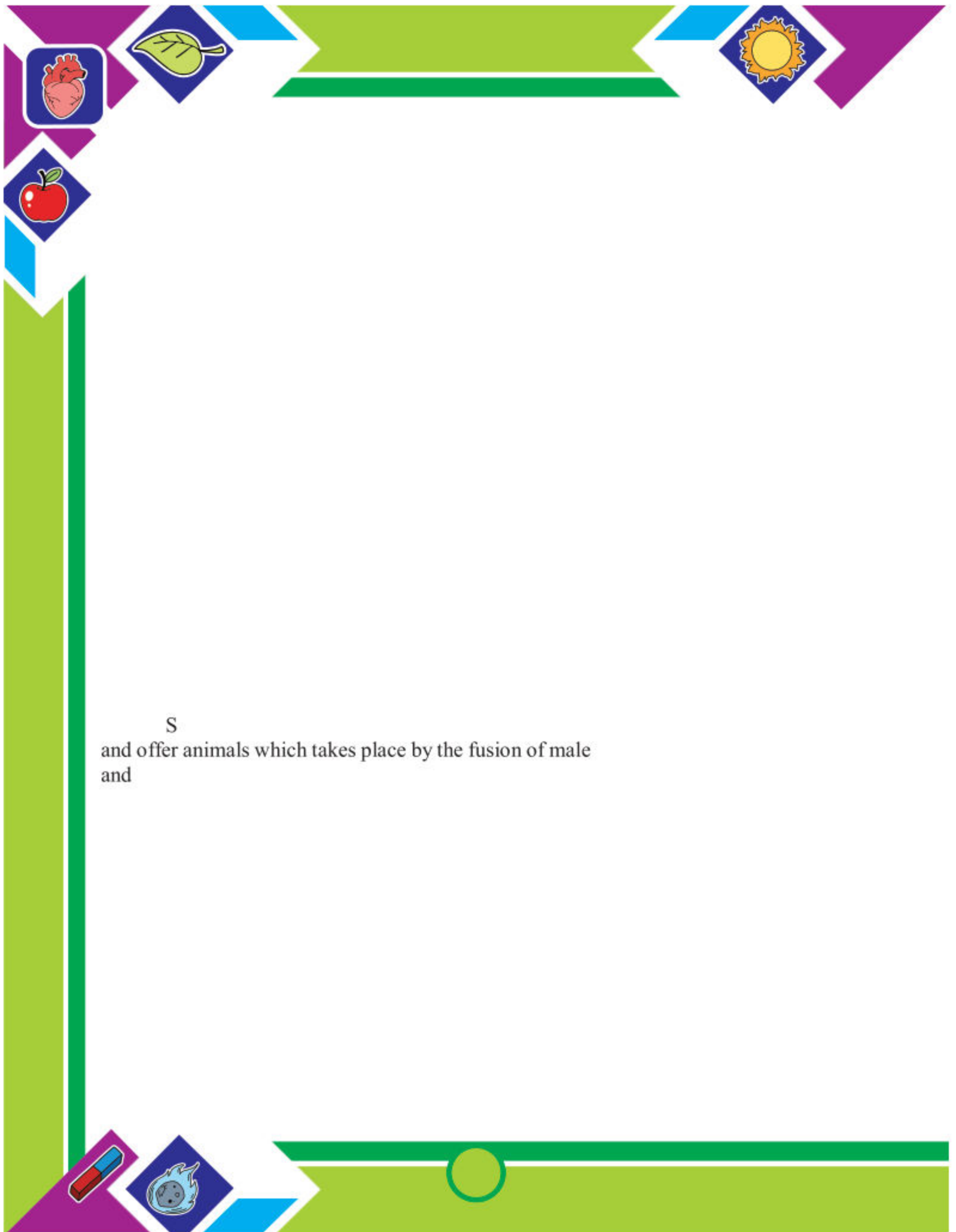




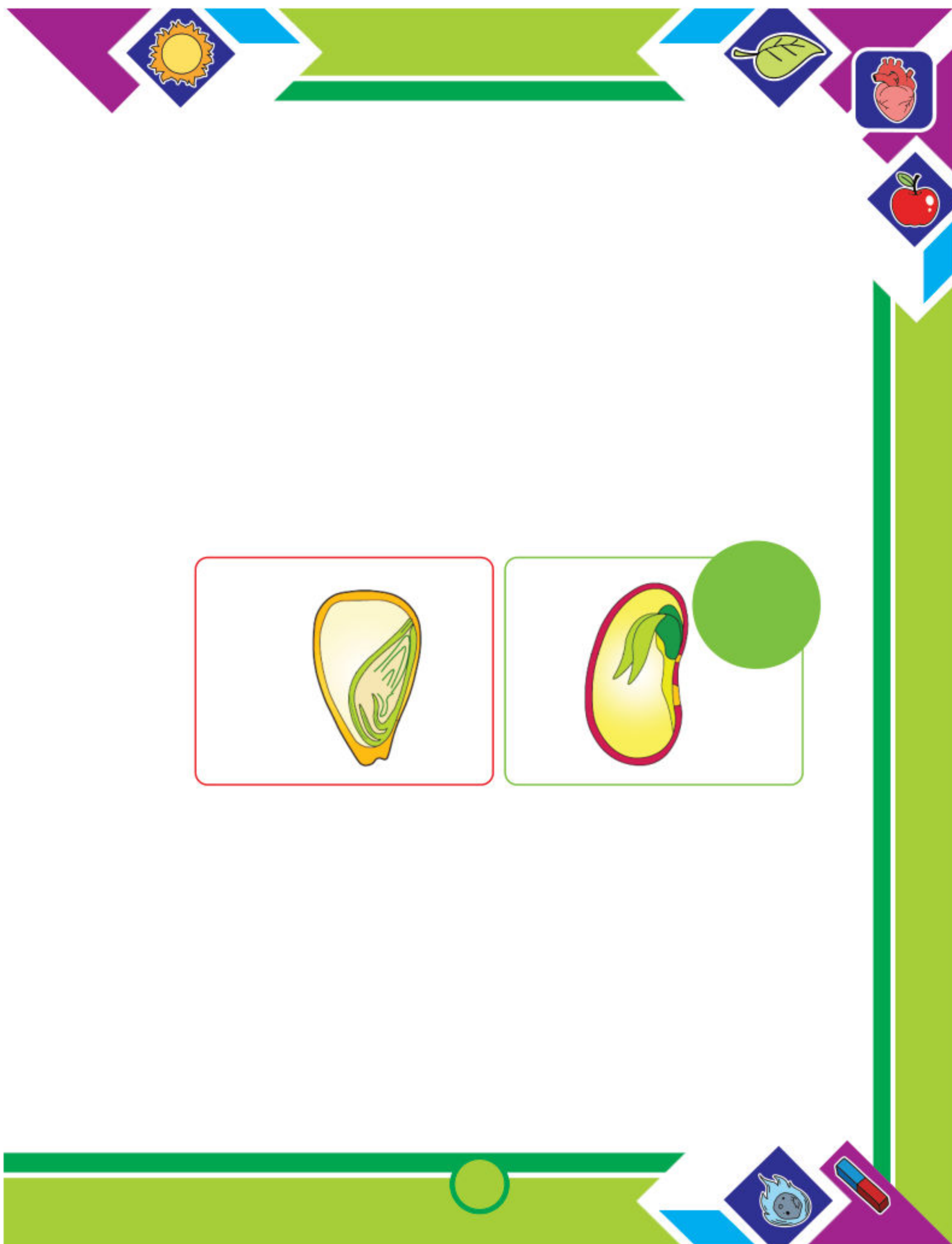




(i) Self-



S
and offer animals which takes place by the fusion of male
and

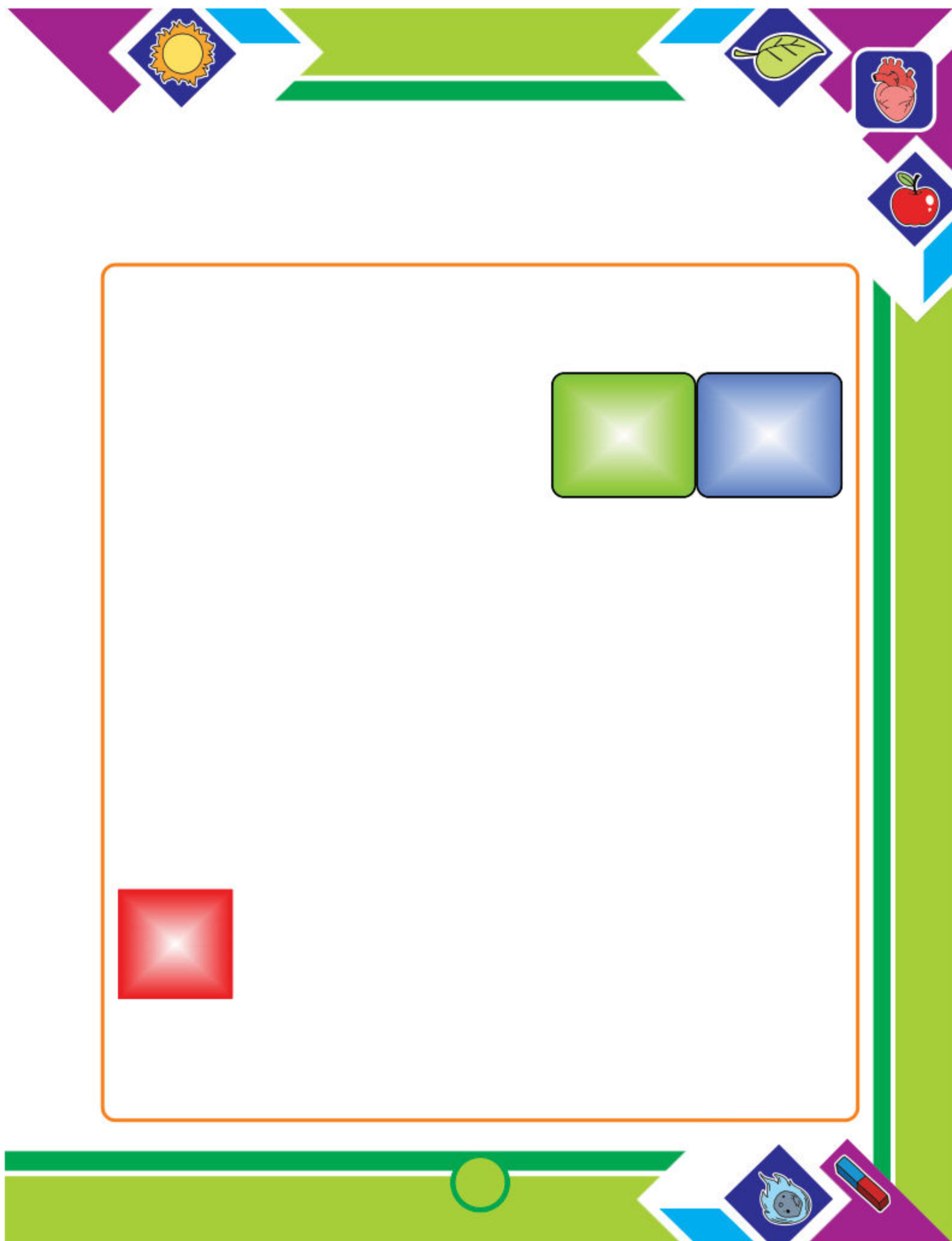


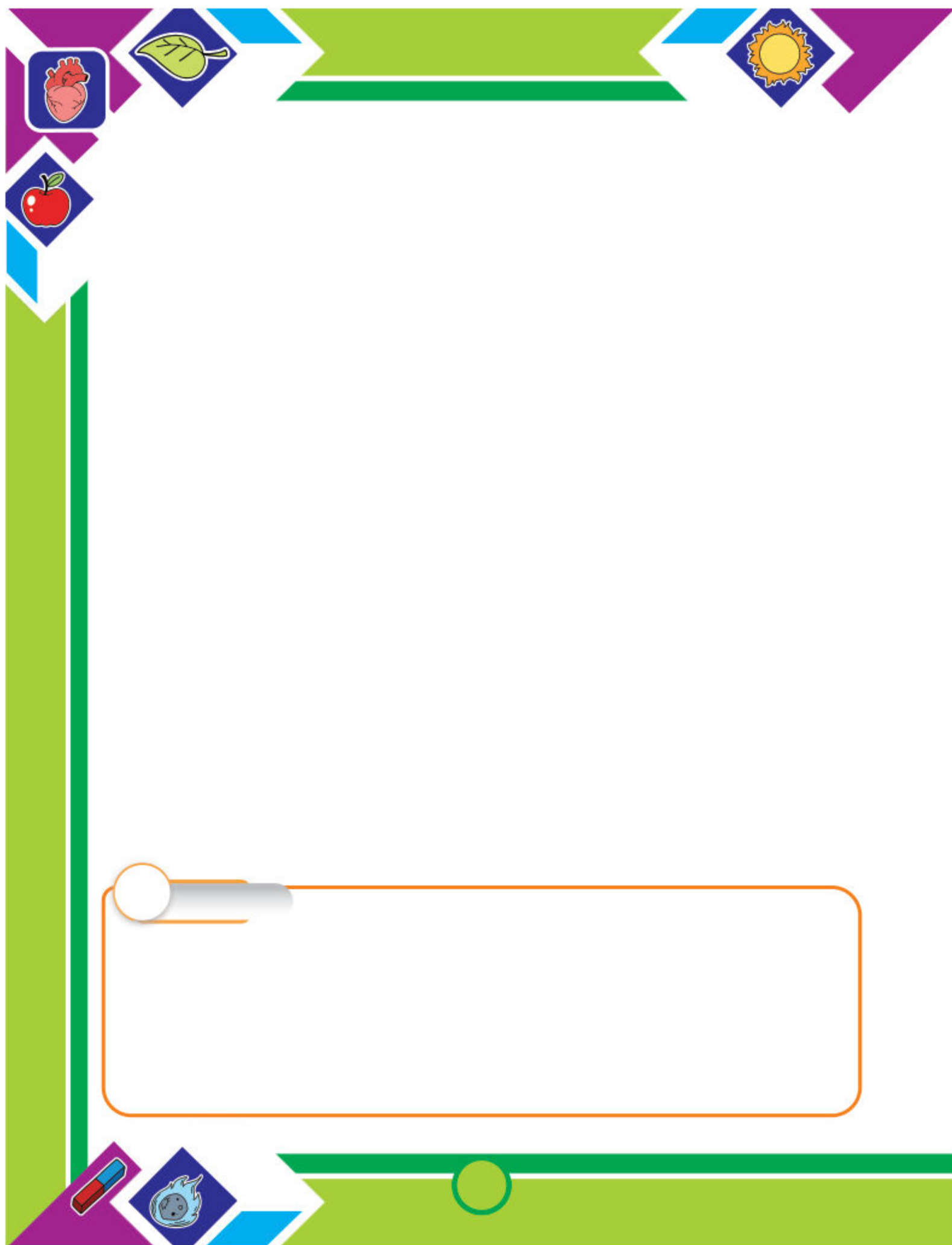


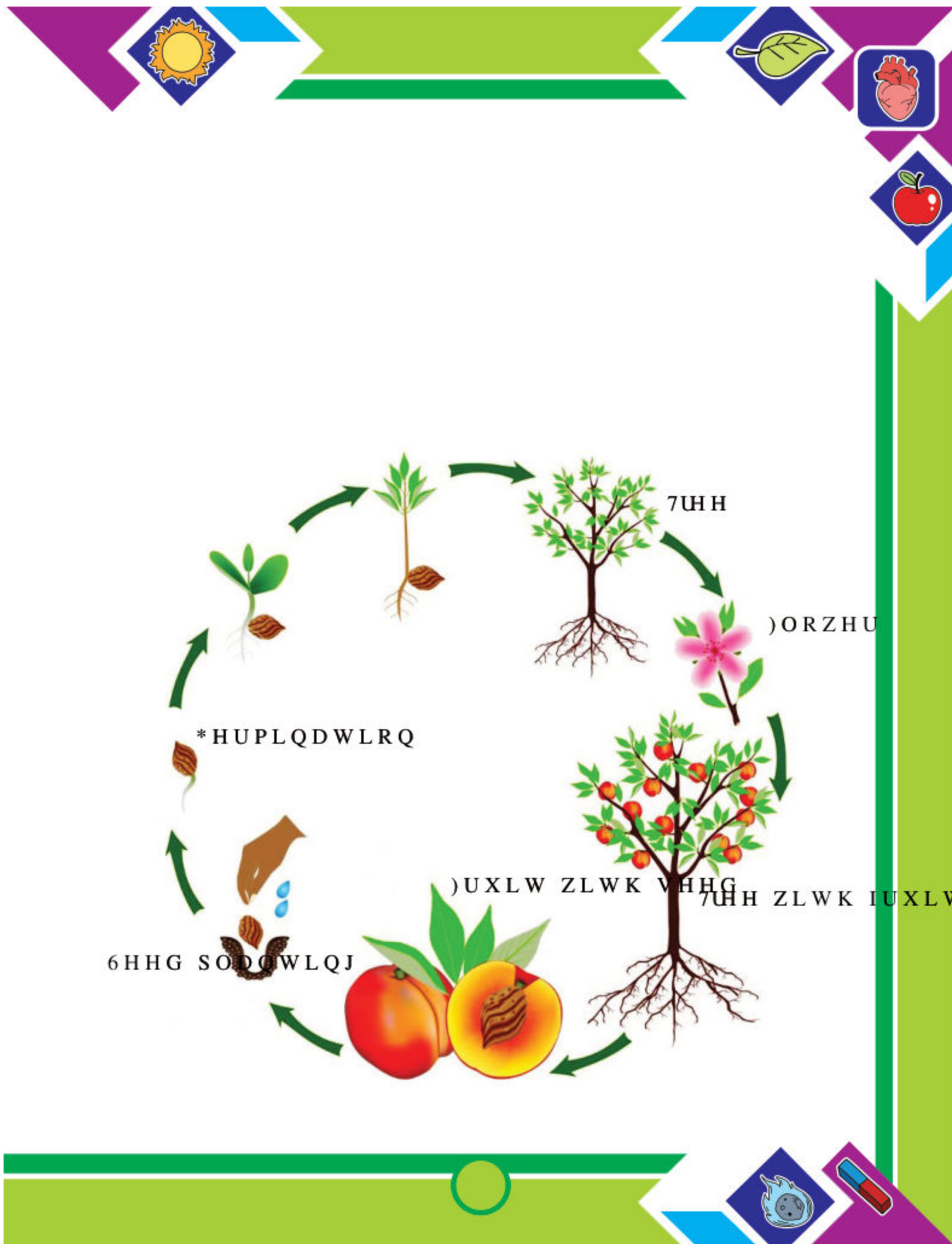
Dispersal by wind

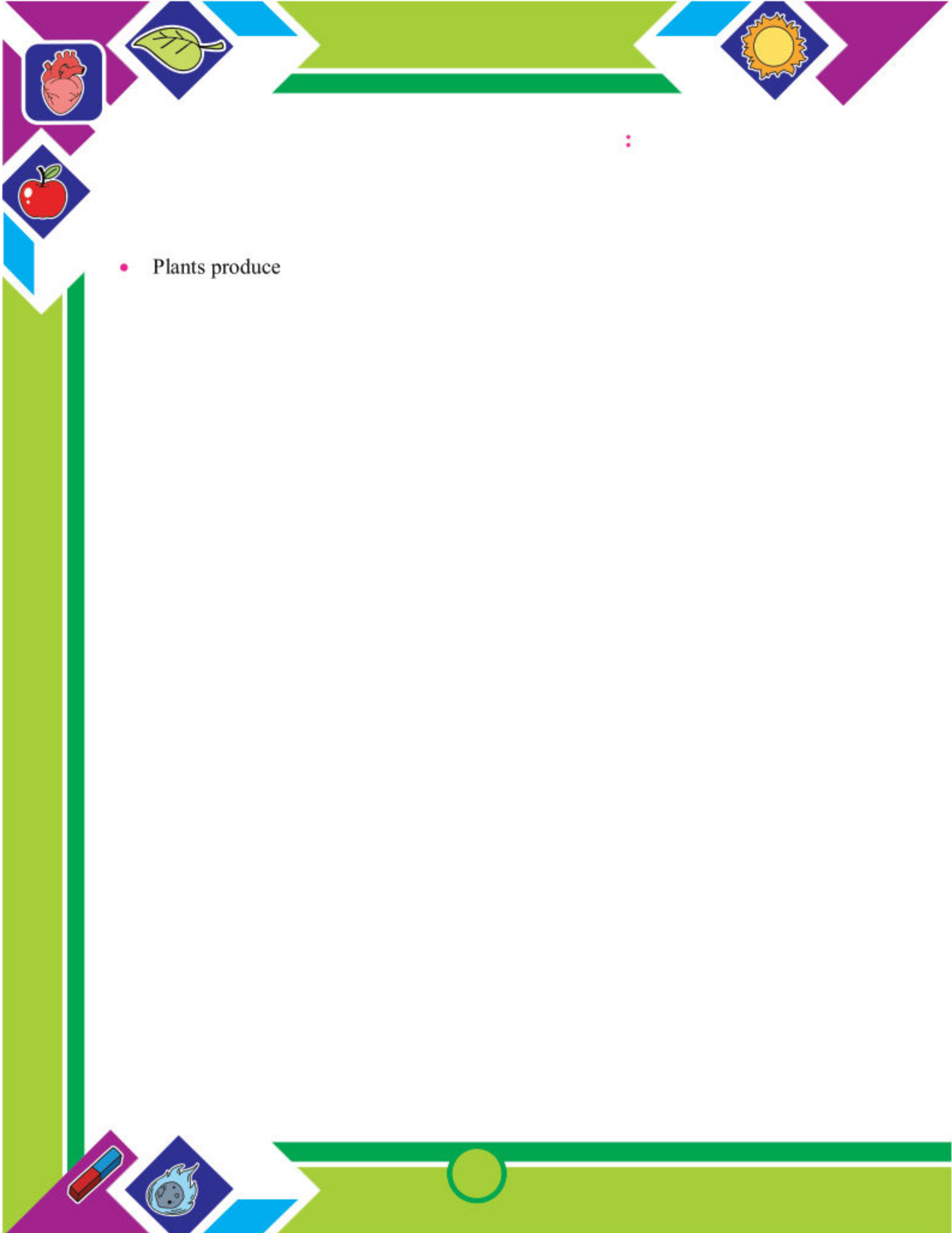
Seeds which are very small, light weight or having wings like structures are



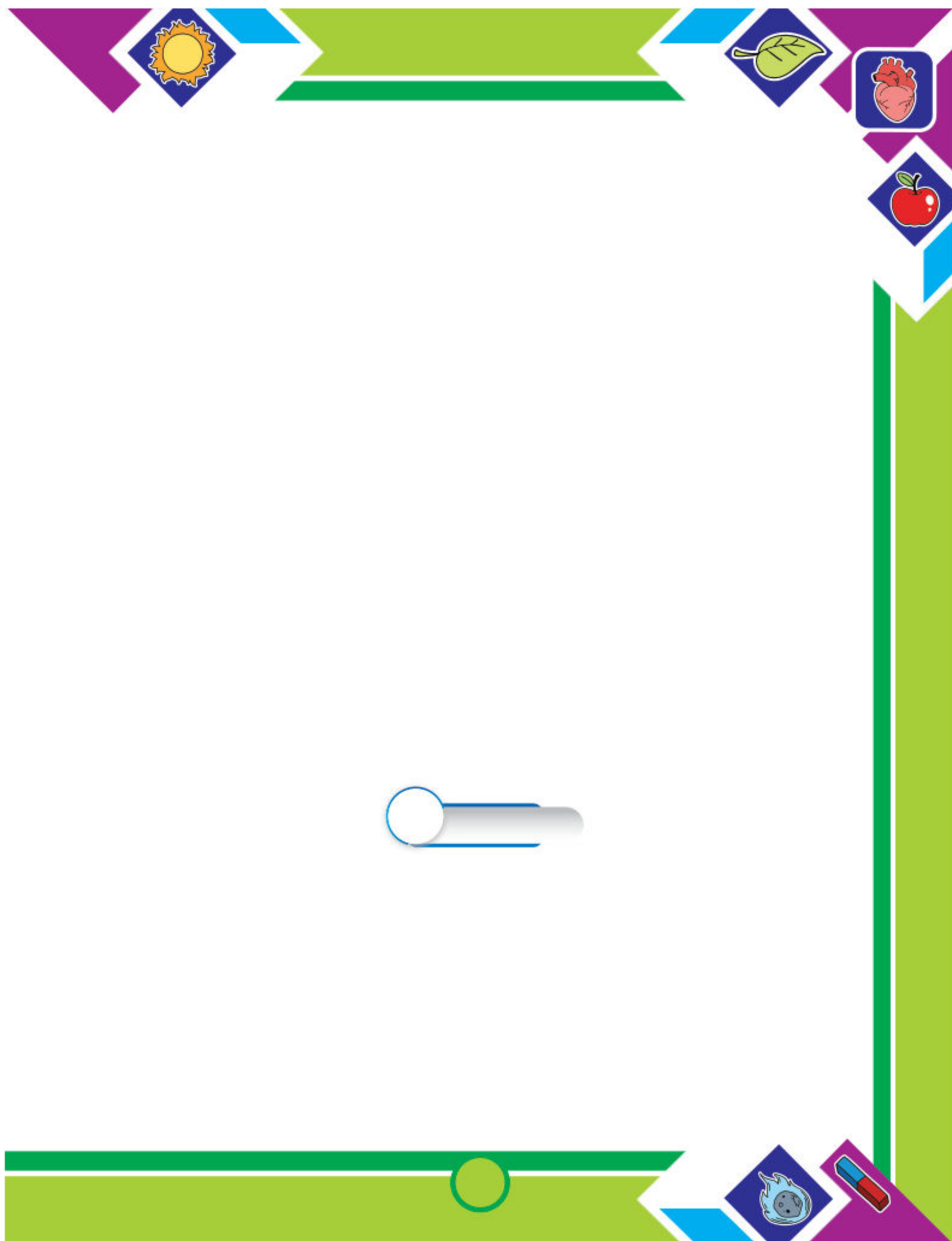


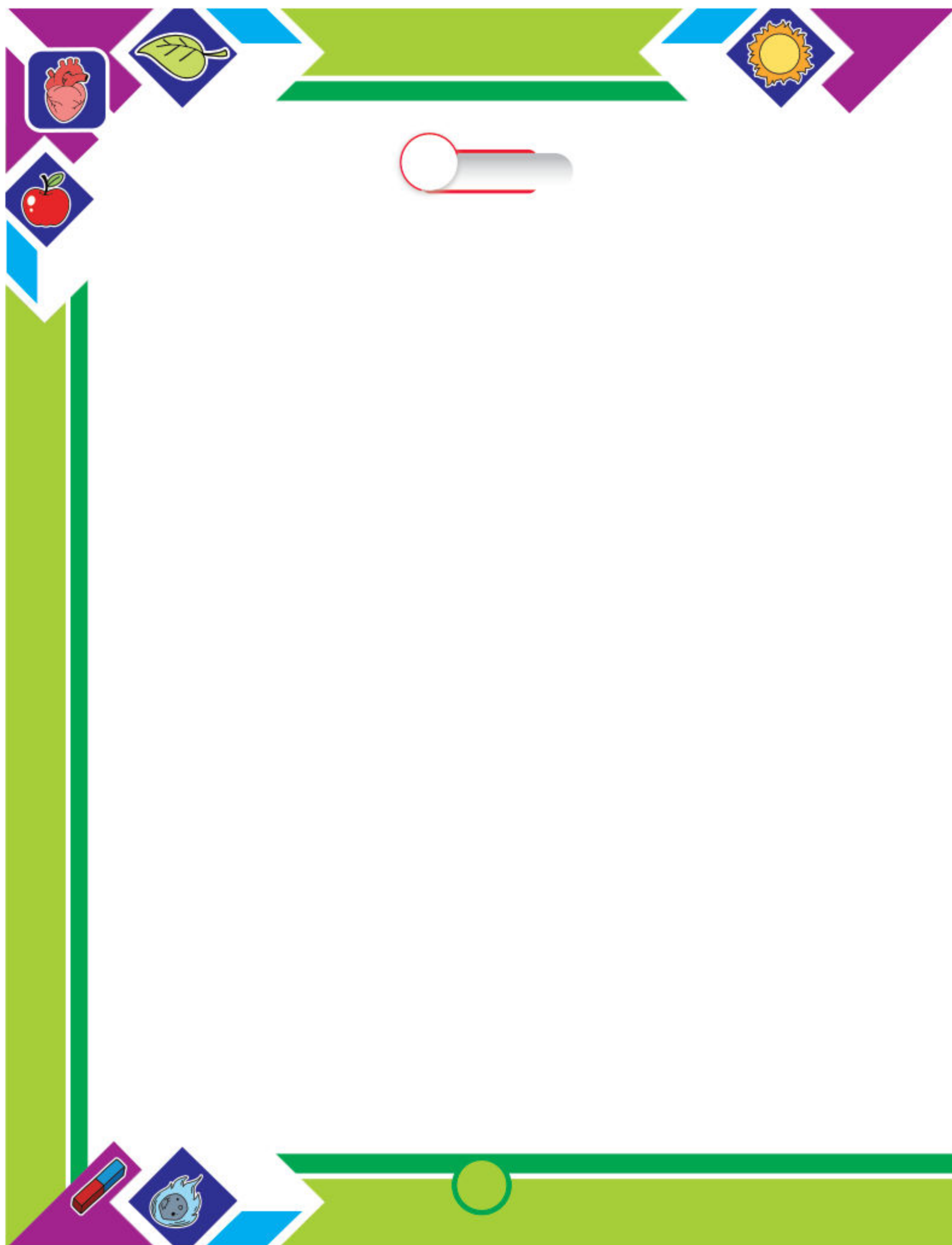


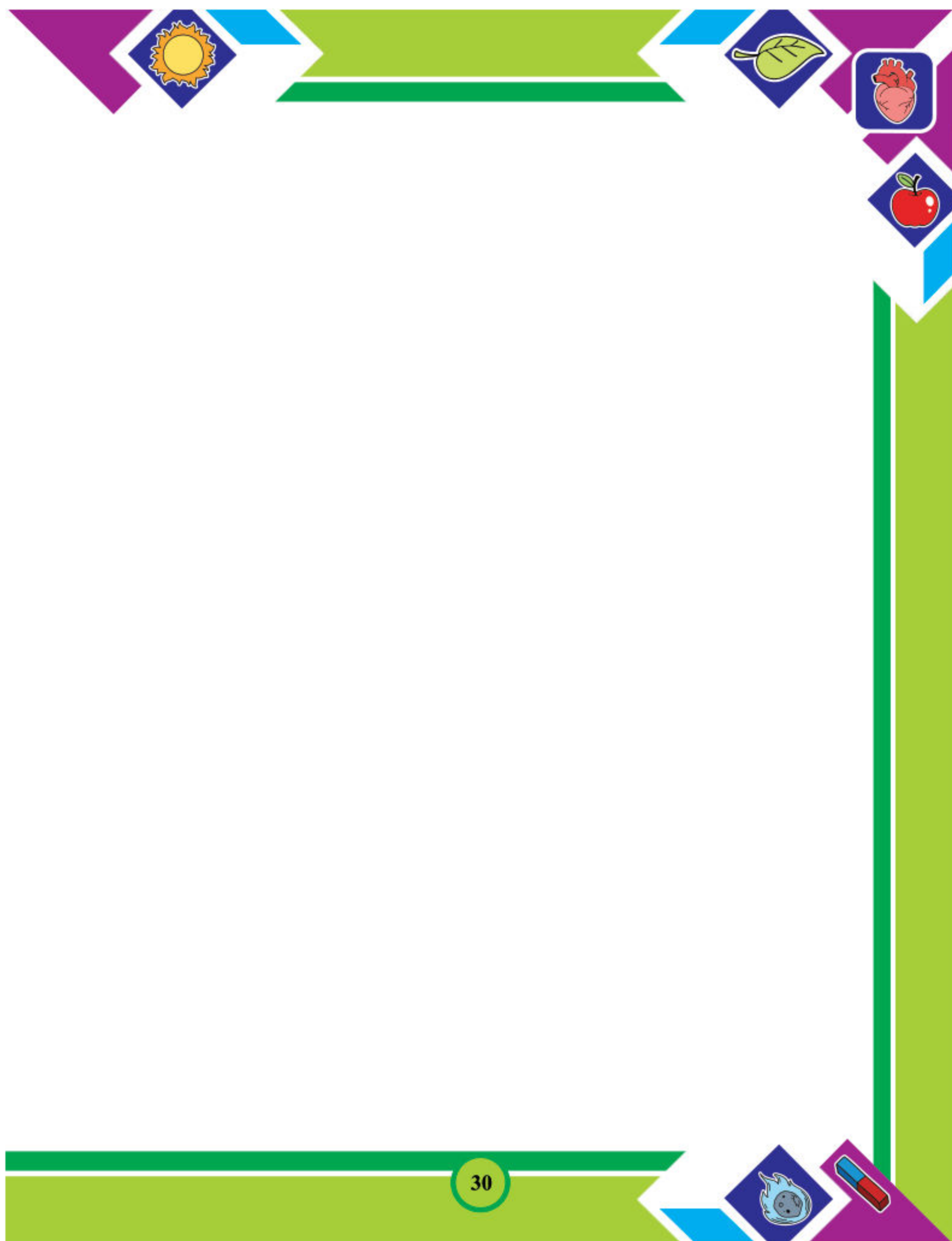




• Plants produce







& K D S



\$ Q L P D W X U
D Q G) X Q F W L R Q

10 D M R Q F H S W V

, Q V L X Q W H Z L O O I Q D E R X W

6 W X G H Q W / H D U Q I 2 X W F R P H V

S I W U I R I S O M E R Q I K L X Q V V W I V O K R X G E H D I D I R

- ◆ Distinguish between major groups of animals with backbones (vertebrates: fish, amphibians, reptiles, birds and mammals) and without backbones (invertebrates: insects, snails, earthworm, the basis of their characteristics).

Identify that some animals (spider, crab, beetles) have an exoskeleton.

Describe some of the important functions of the skeleton.

Describe the human digestive system including the simple functions of the organs involved (mouth, esophagus, stomach, small and large intestine).

Recognize that humans have different types of teeth (molar, premolar, incisors, canines) and know their functions in digestion of food.

Investigate the causes and prevention of tooth decay and gum diseases.

Identify that many vertebrates (Cat and Goat) have a digestive system similar to humans.

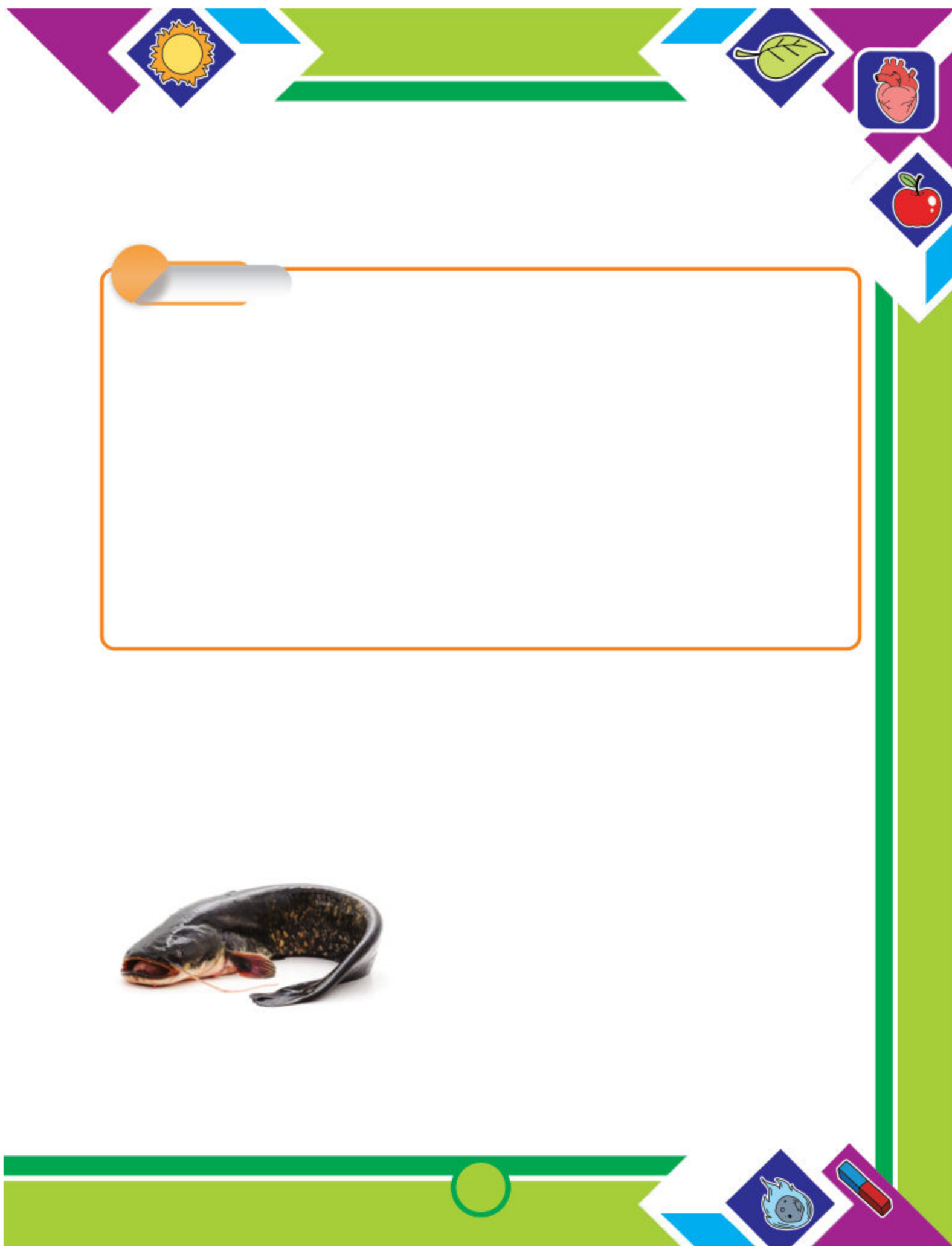
Describe the handling and storage of food and water.

Recall main food groups with examples.

Identify the food pyramid.

Differentiate between balanced and unbalanced diet.







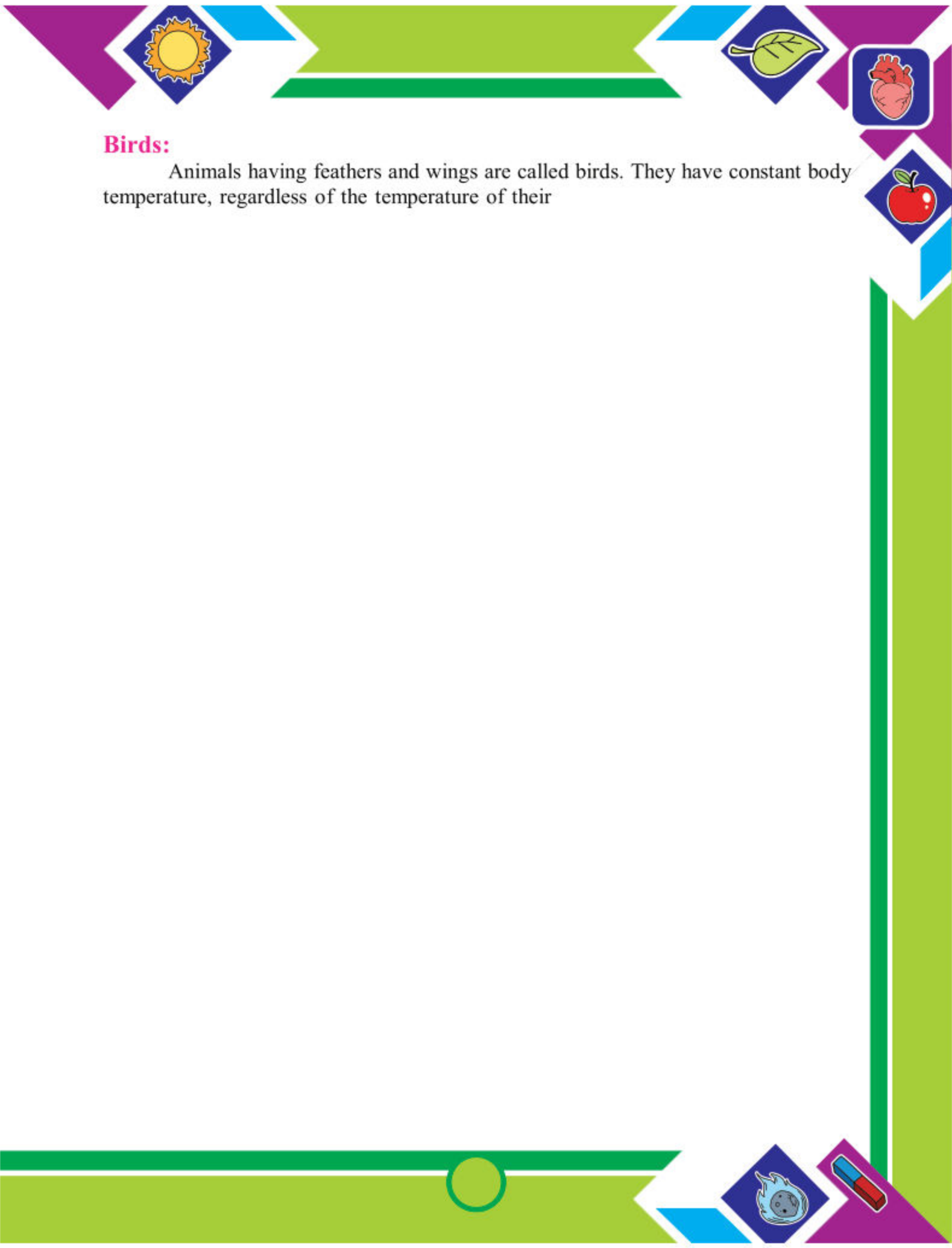
Amphibians:



Reptiles:

Tortoise





Birds:

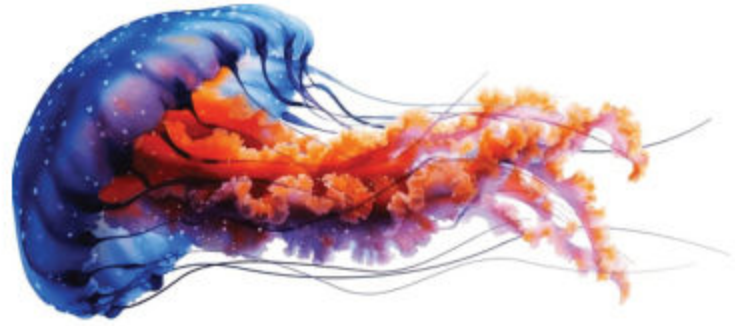
Animals having feathers and wings are called birds. They have constant body temperature, regardless of the temperature of their

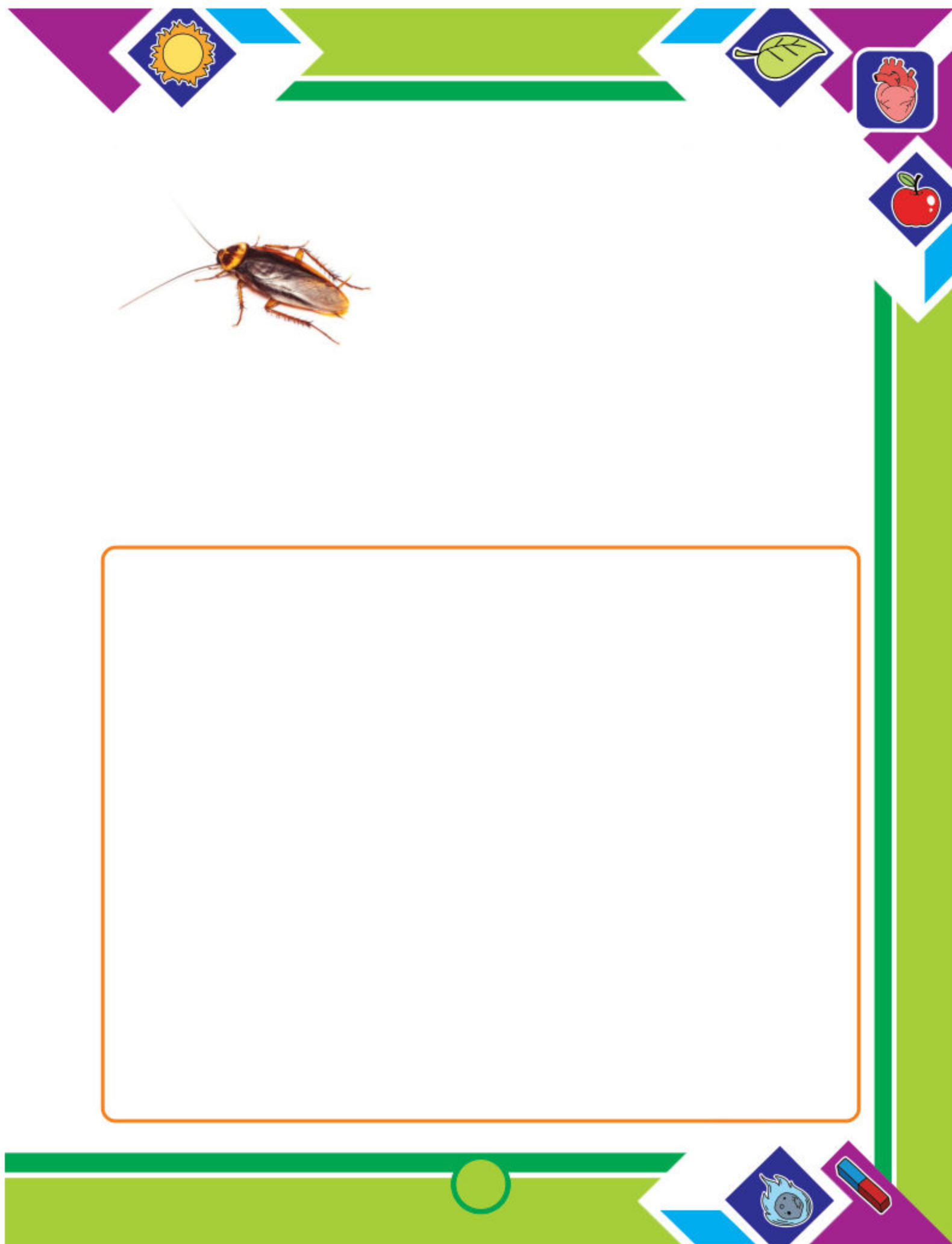
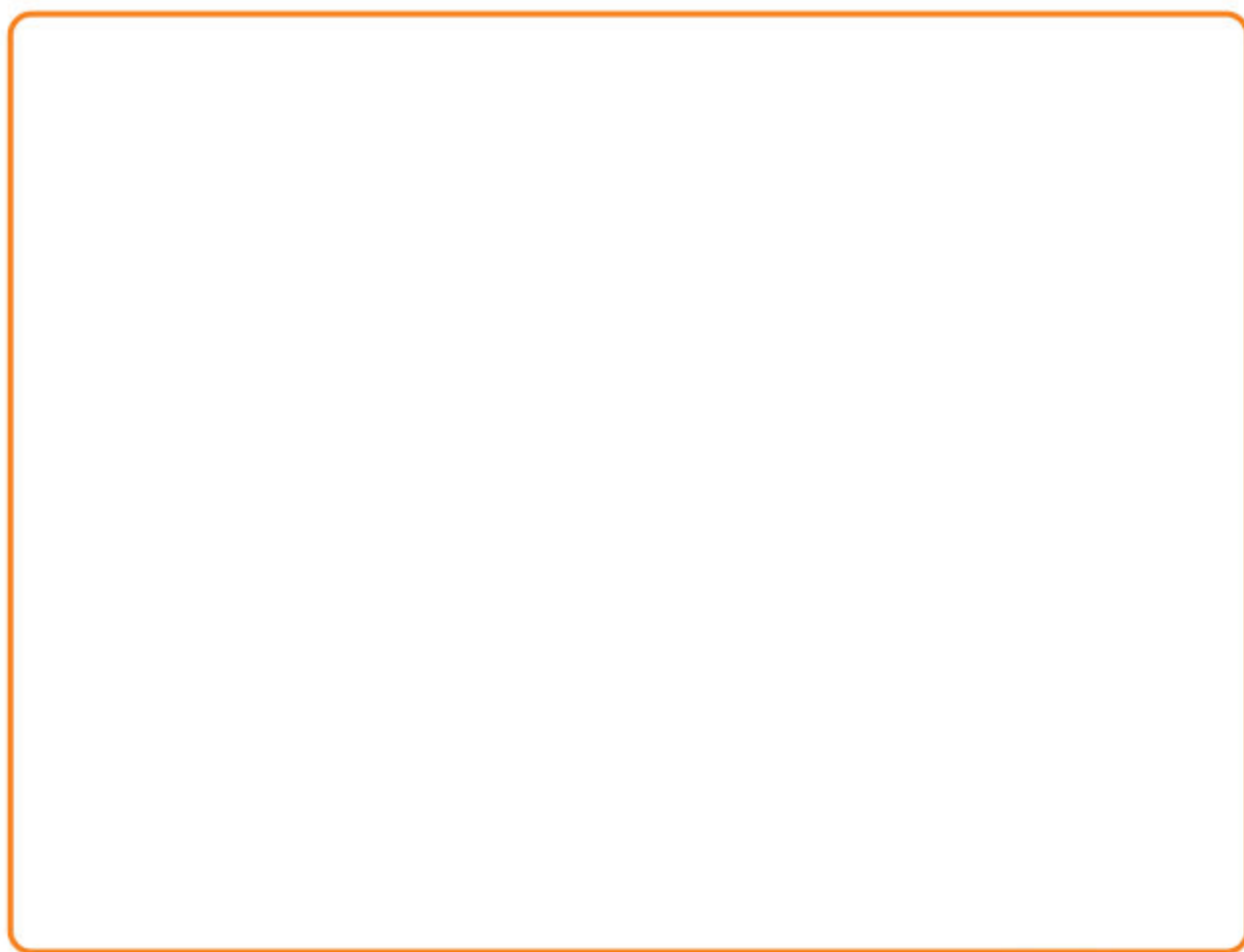


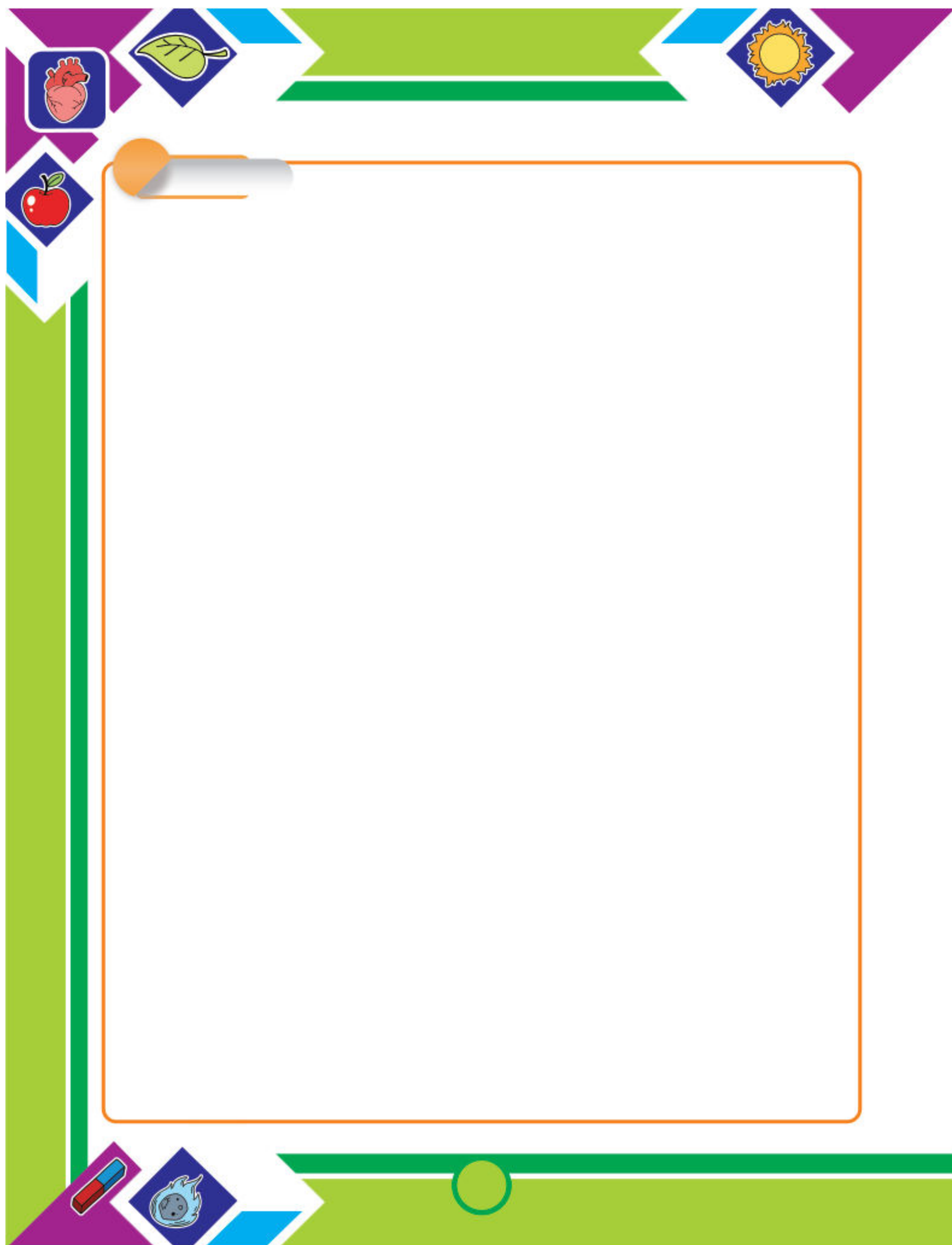
3.3 GROUPS OF INVERTEBRATES

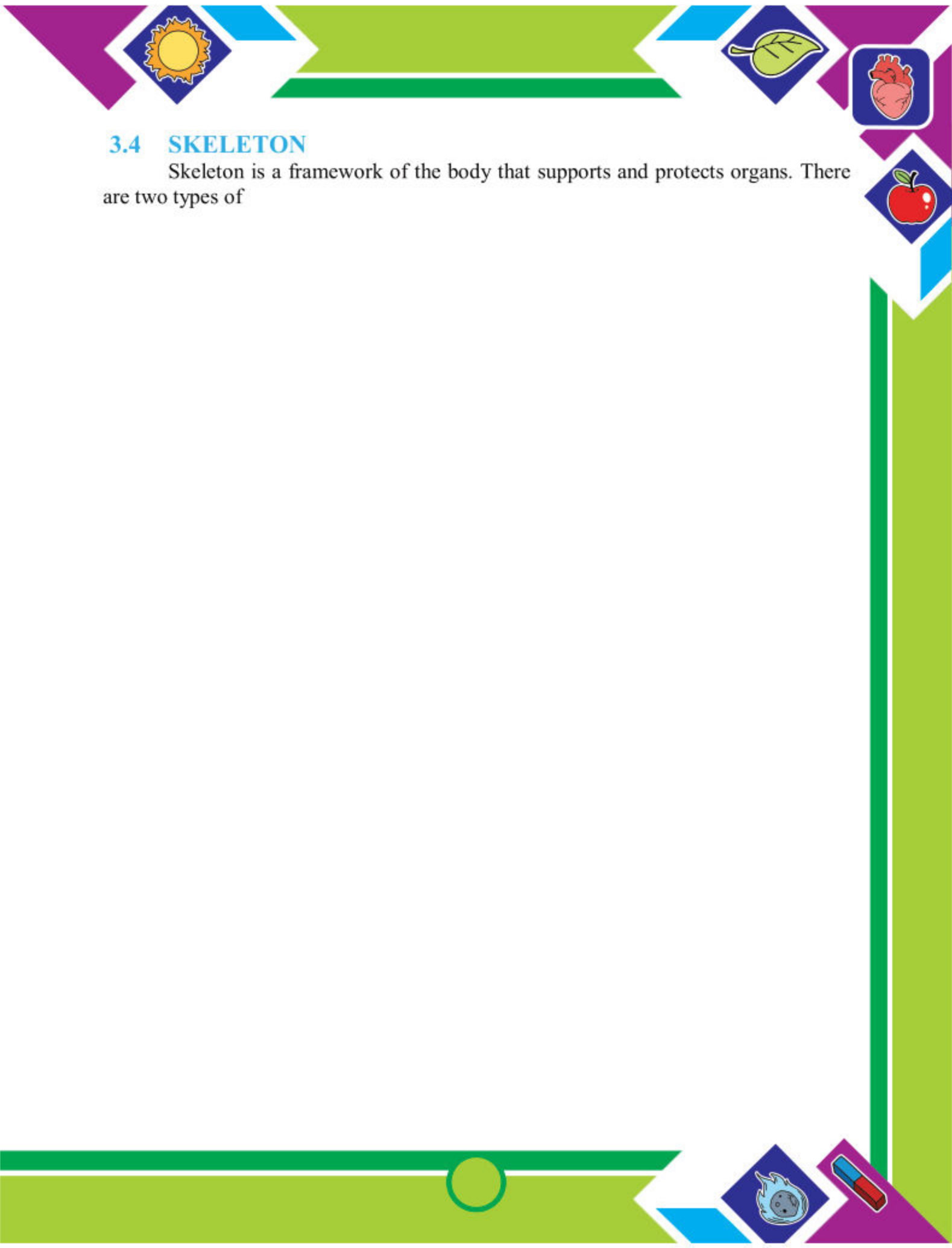


Jelly fish:



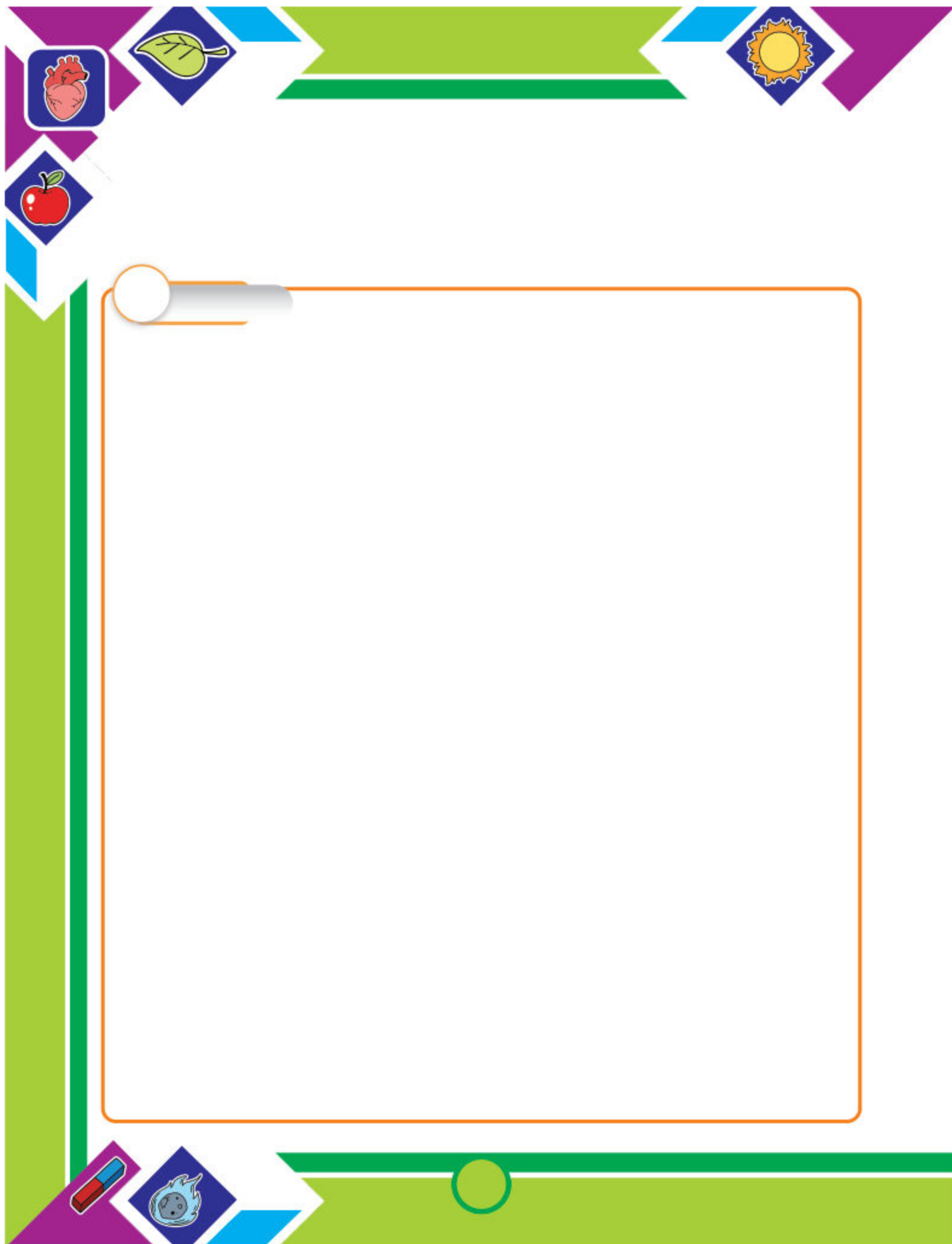






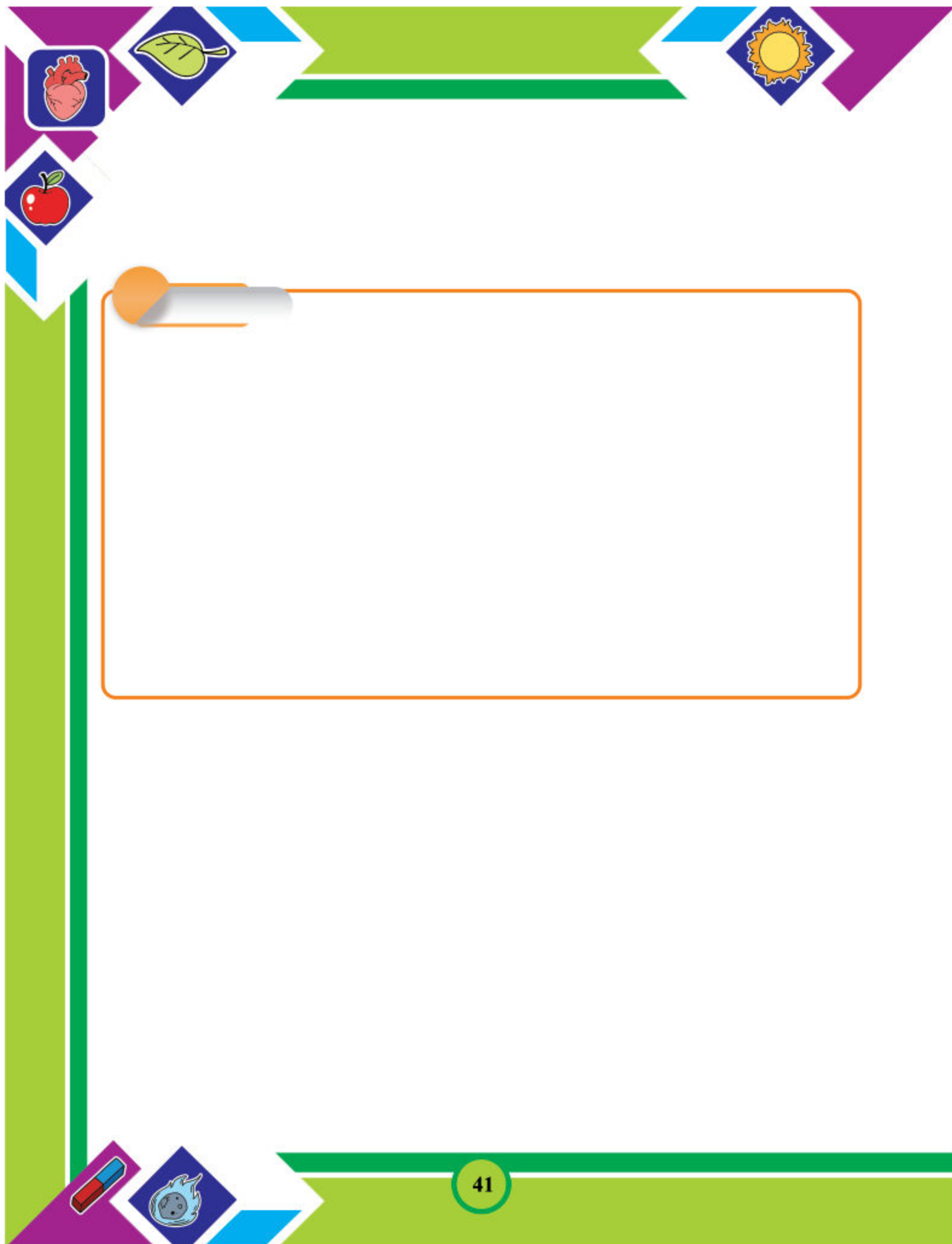
3.4 SKELETON

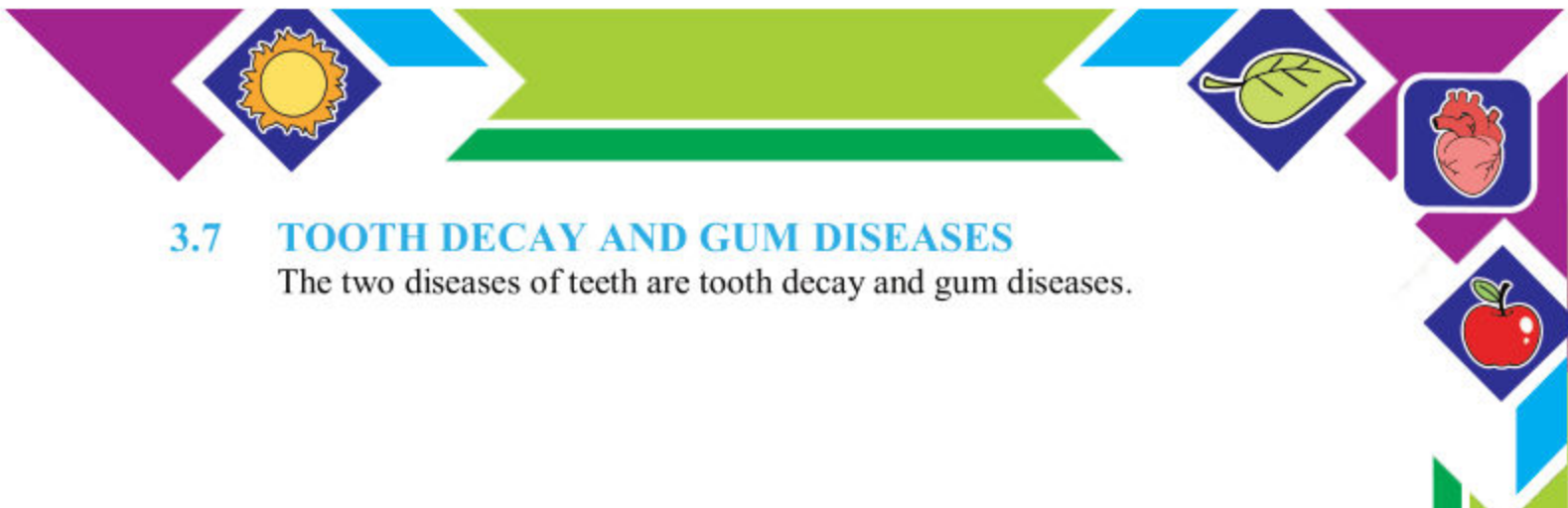
Skeleton is a framework of the body that supports and protects organs. There are two types of



3.6 HUMAN DIGESTIVE SYSTEM

Mouth:

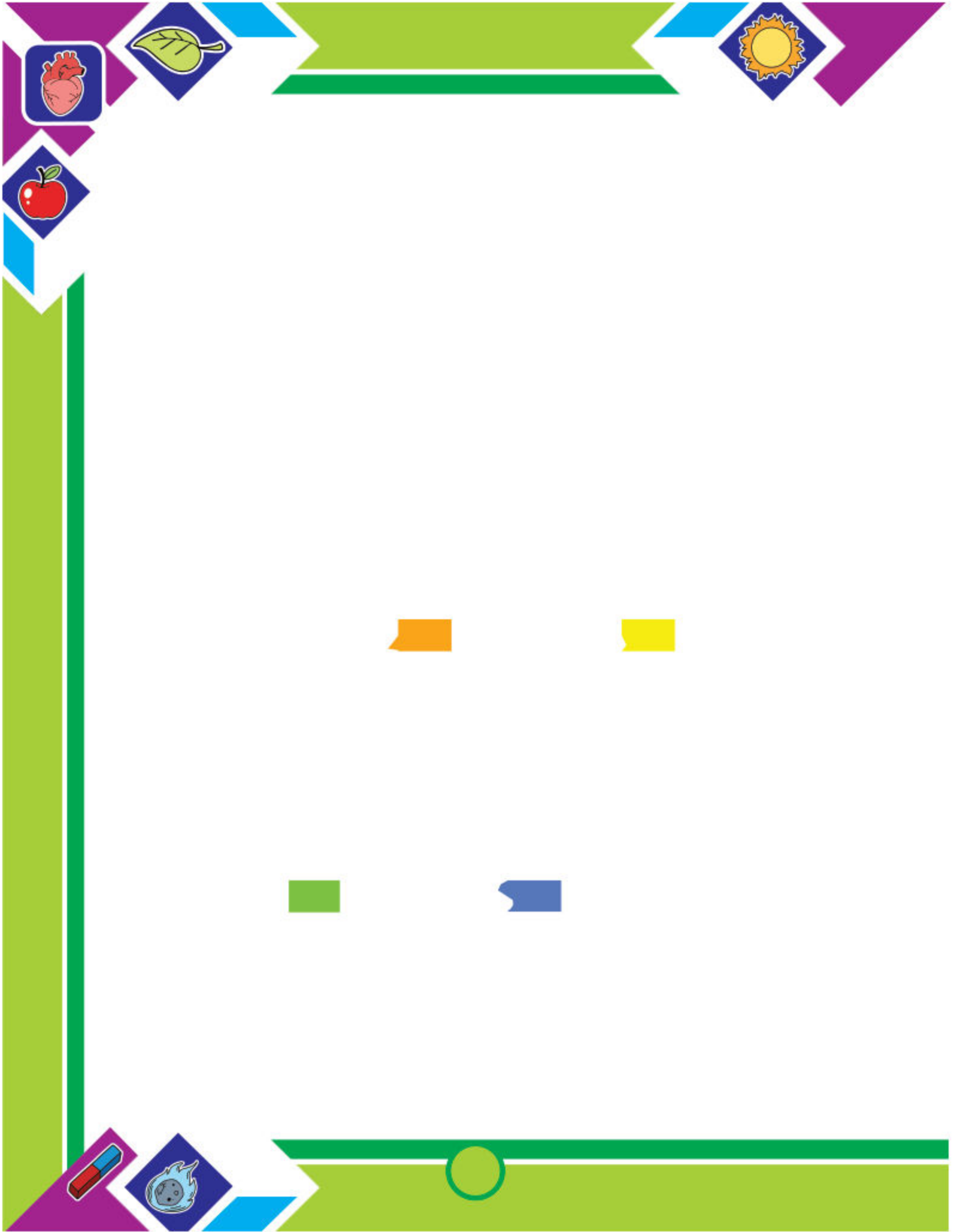


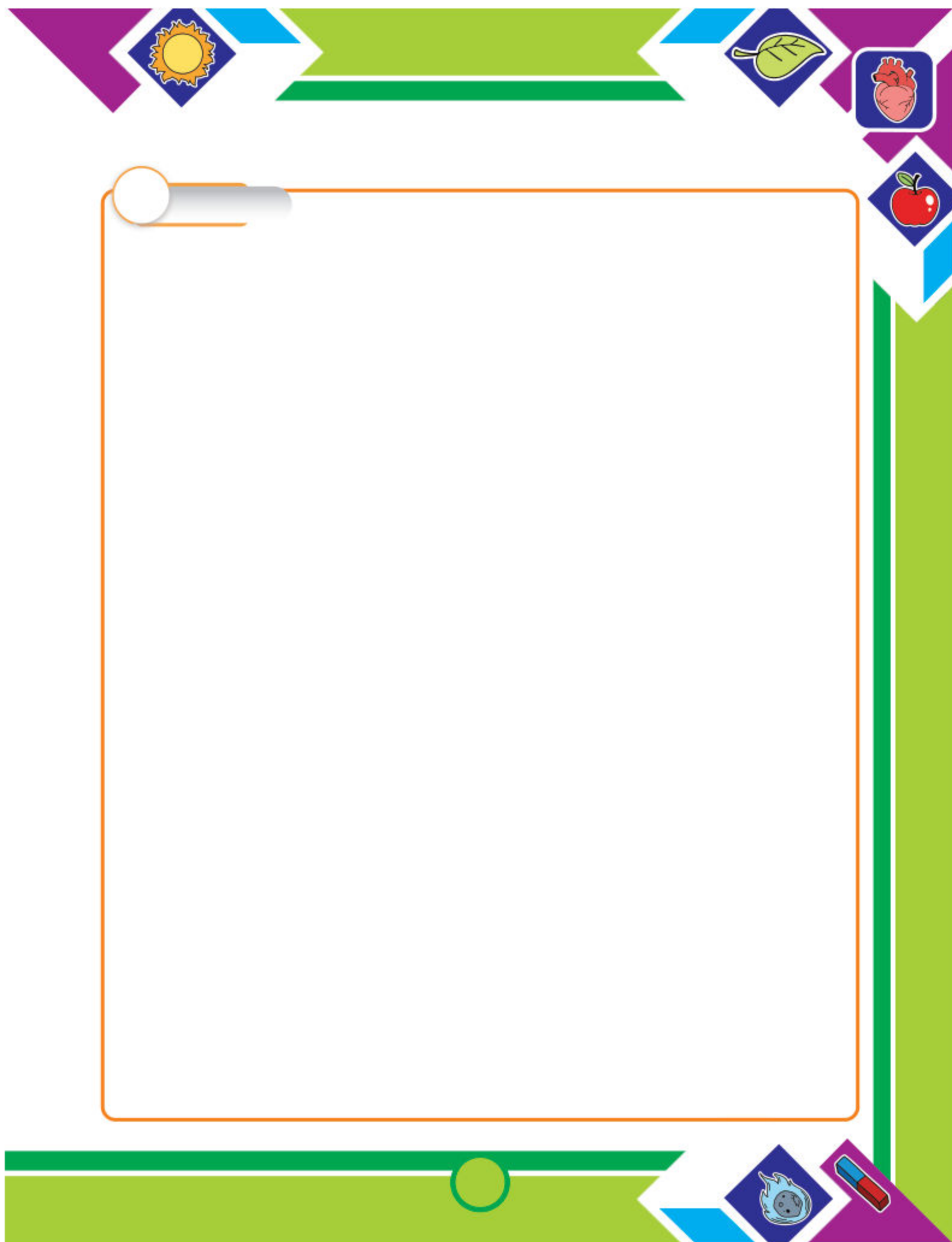


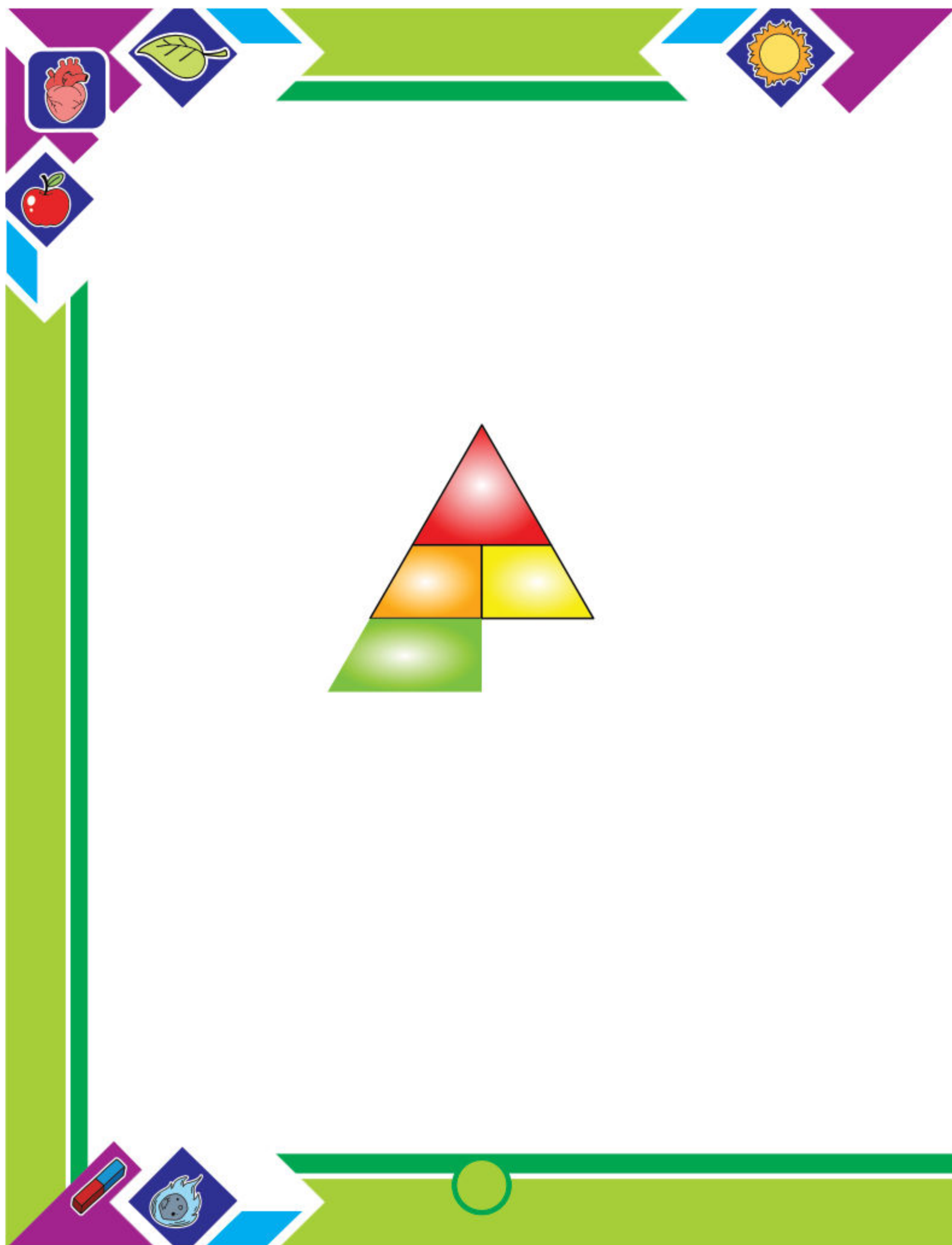
3.7 TOOTH DECAY AND GUM DISEASES

The two diseases of teeth are tooth decay and gum diseases.





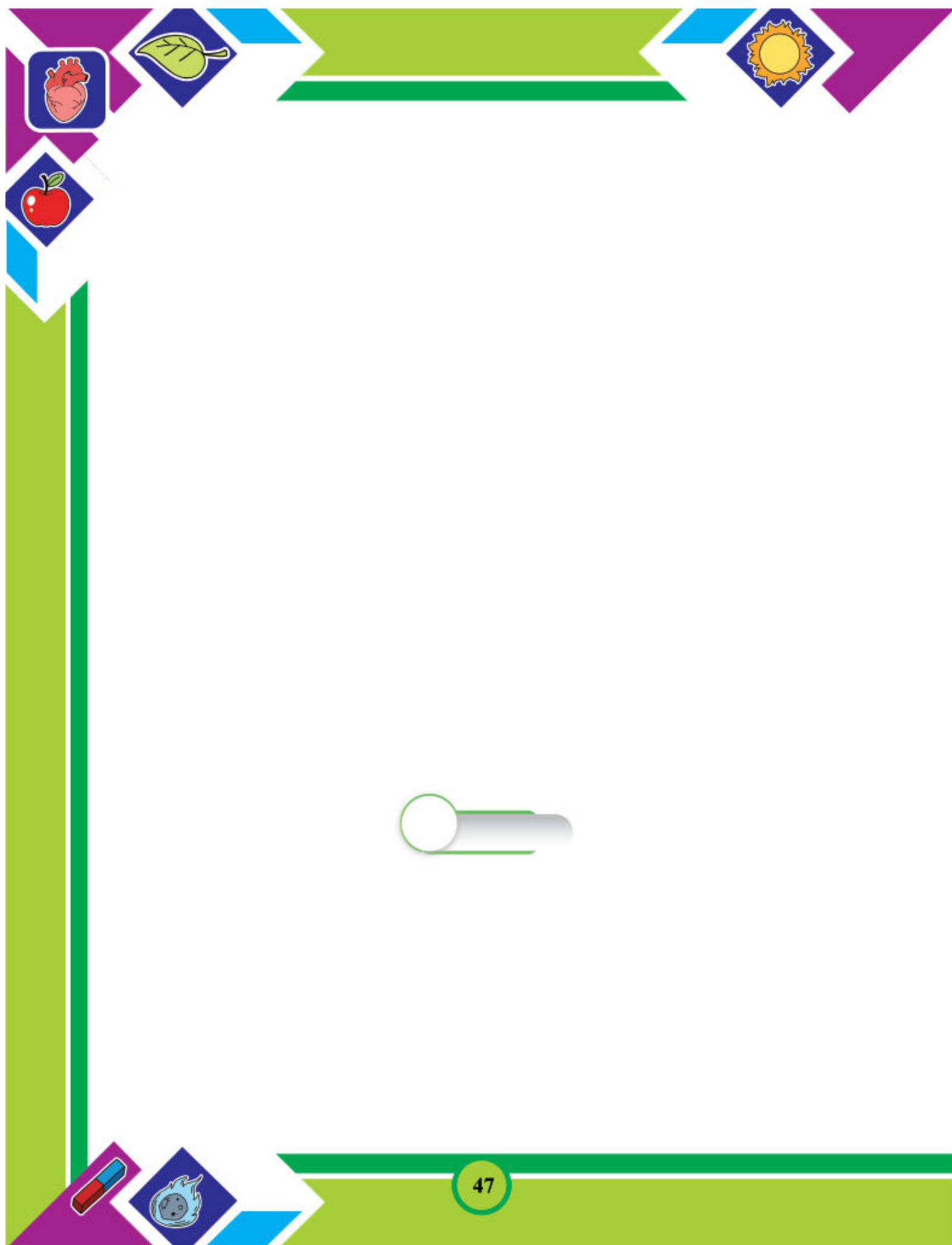


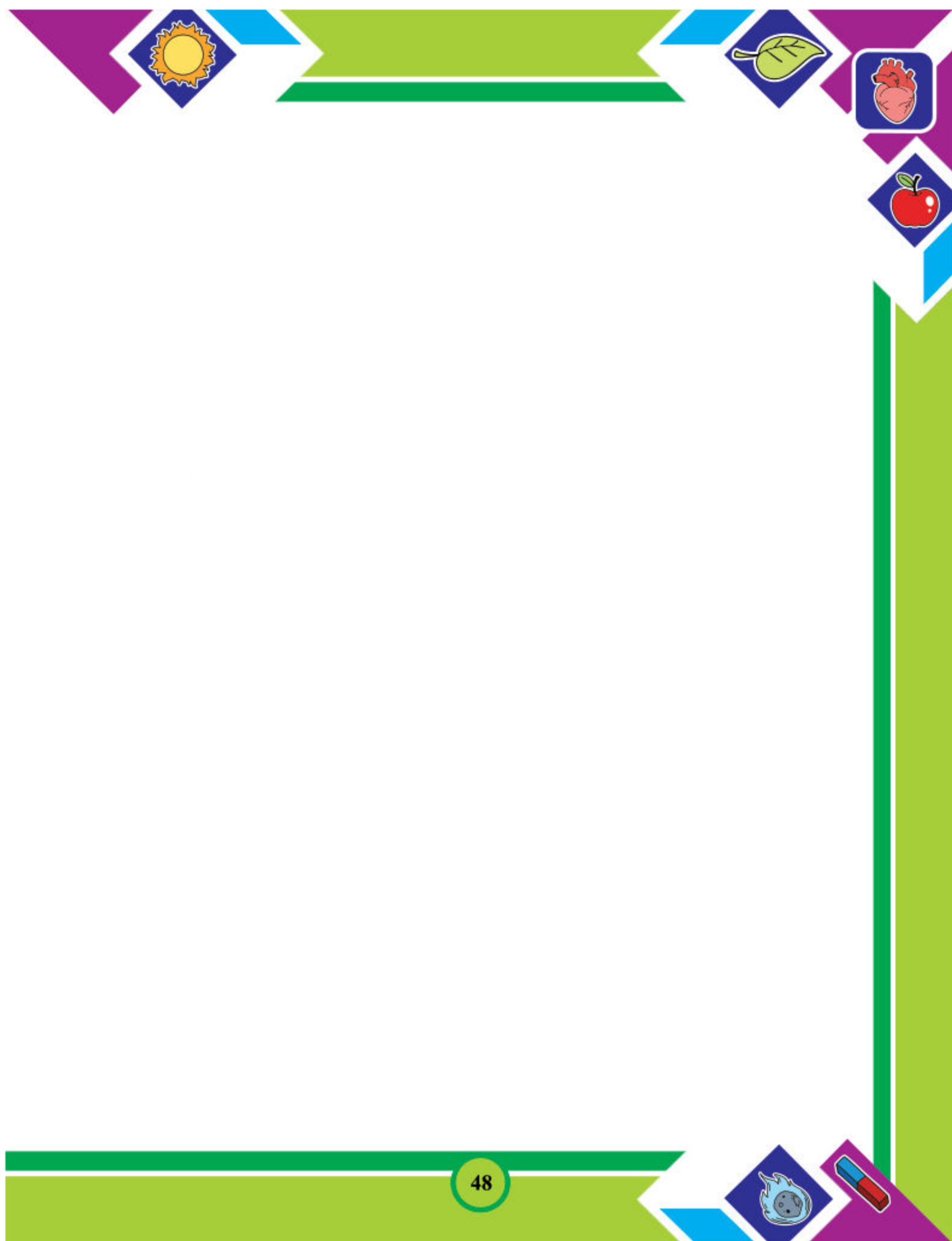


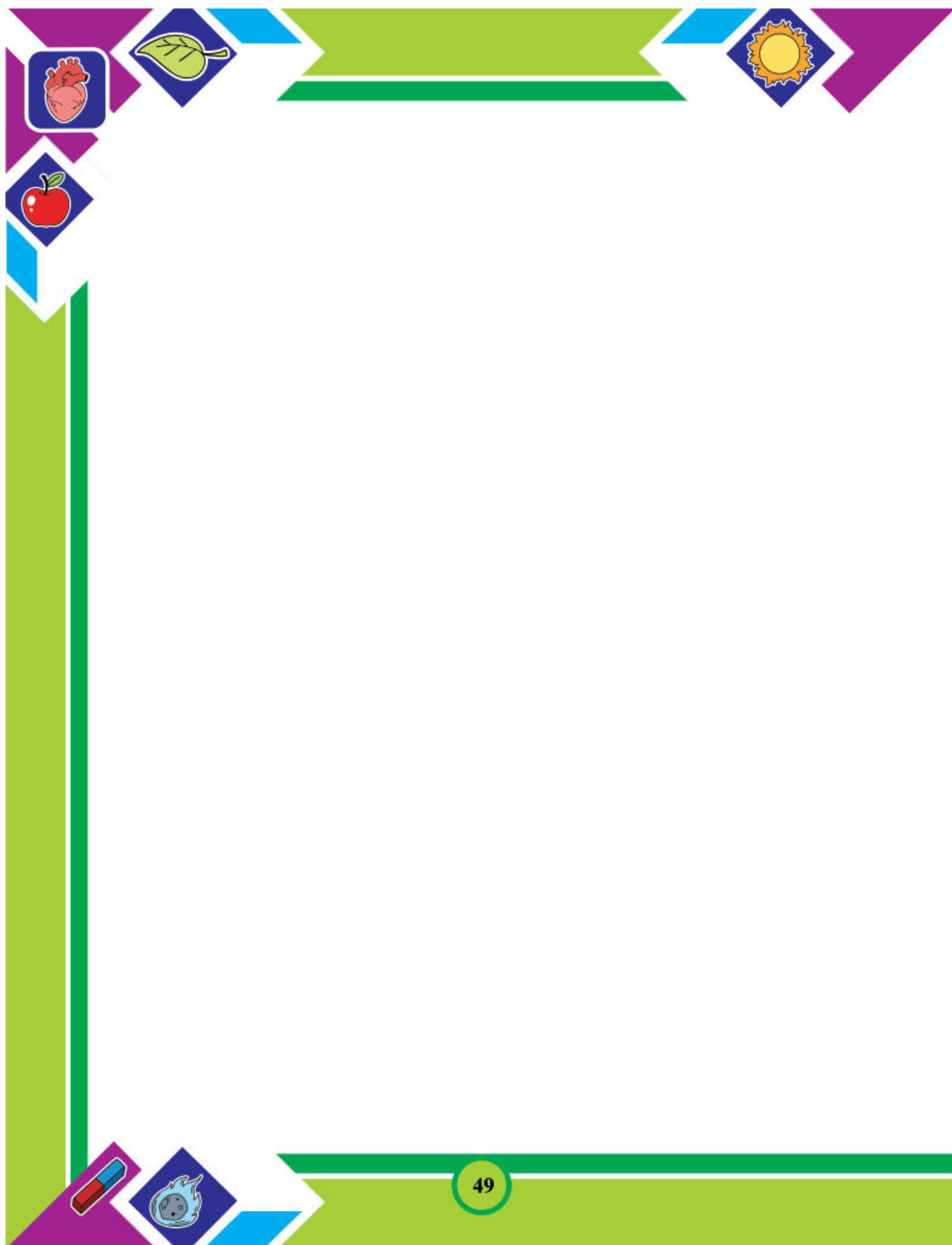


Differentiate between balanced and unbalanced diet:

Balanced Diet	







- i) Why animals are classified as vertebrate and invertebrate?

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\$ G D S W L R Q V

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, Q V L X W Z H Z L O O I Q D E R X W

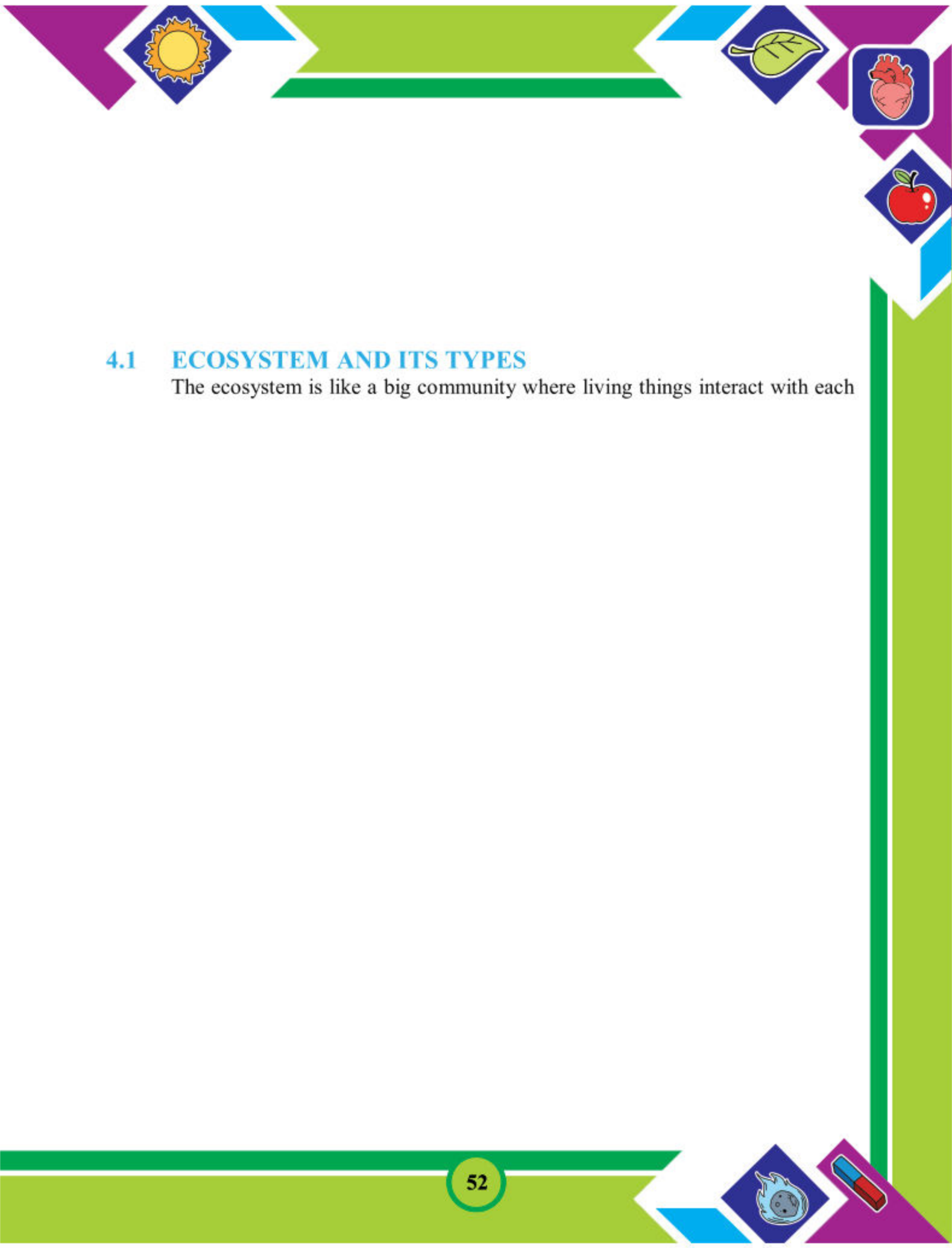
- 4.1 Ecosystem and its types
- 4.2 Response of living thing to environment
- 4.3 Adaptation of plant and animals to environment
- 4.4 Effects of human activities on habitat
- 4.5 How changes in habitat effect on living organisms

6 W X G H O W H D U O I Q J 2 X W F R P H V

\$ I W U H R P S O M R Q N K L X Q V V W Q W K R X G E H D I W K

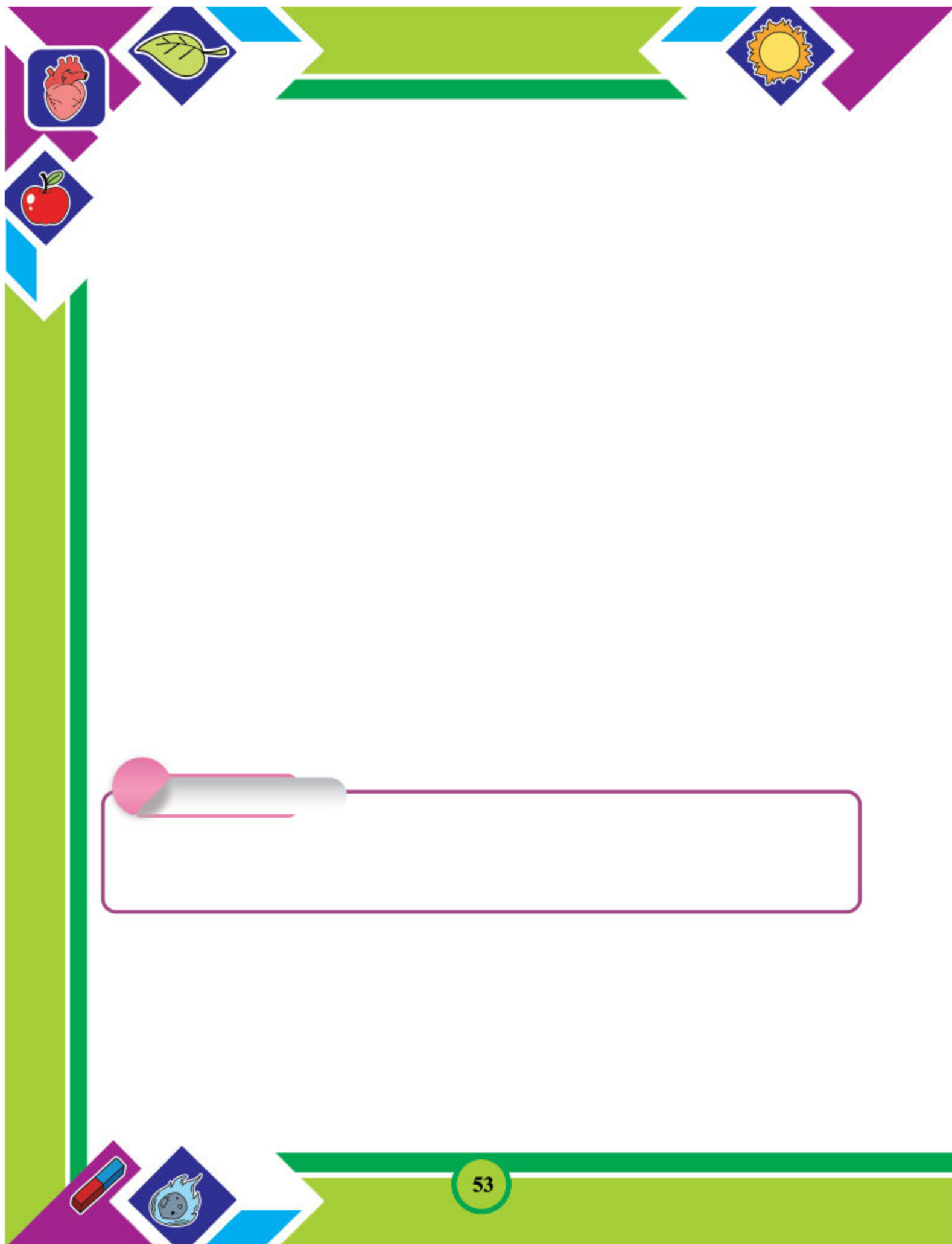
- ◆ Recognize that ecosystems (e.g., forests, ponds, rivers, grasslands and deserts) consist of habitats that provide living things with what they need it. Recognize and explain that living things respond to environmental conditions. Describe how many plants and animals adapt to environments that are hot, cold, wet and or dry and describe common physical adaptations of plants (e.g., a thick stem, a waxy coating helps it survive with less water) and animals (e.g., colours of animals help in camouflage). Associate behavior of animals with the environments in which they live, and



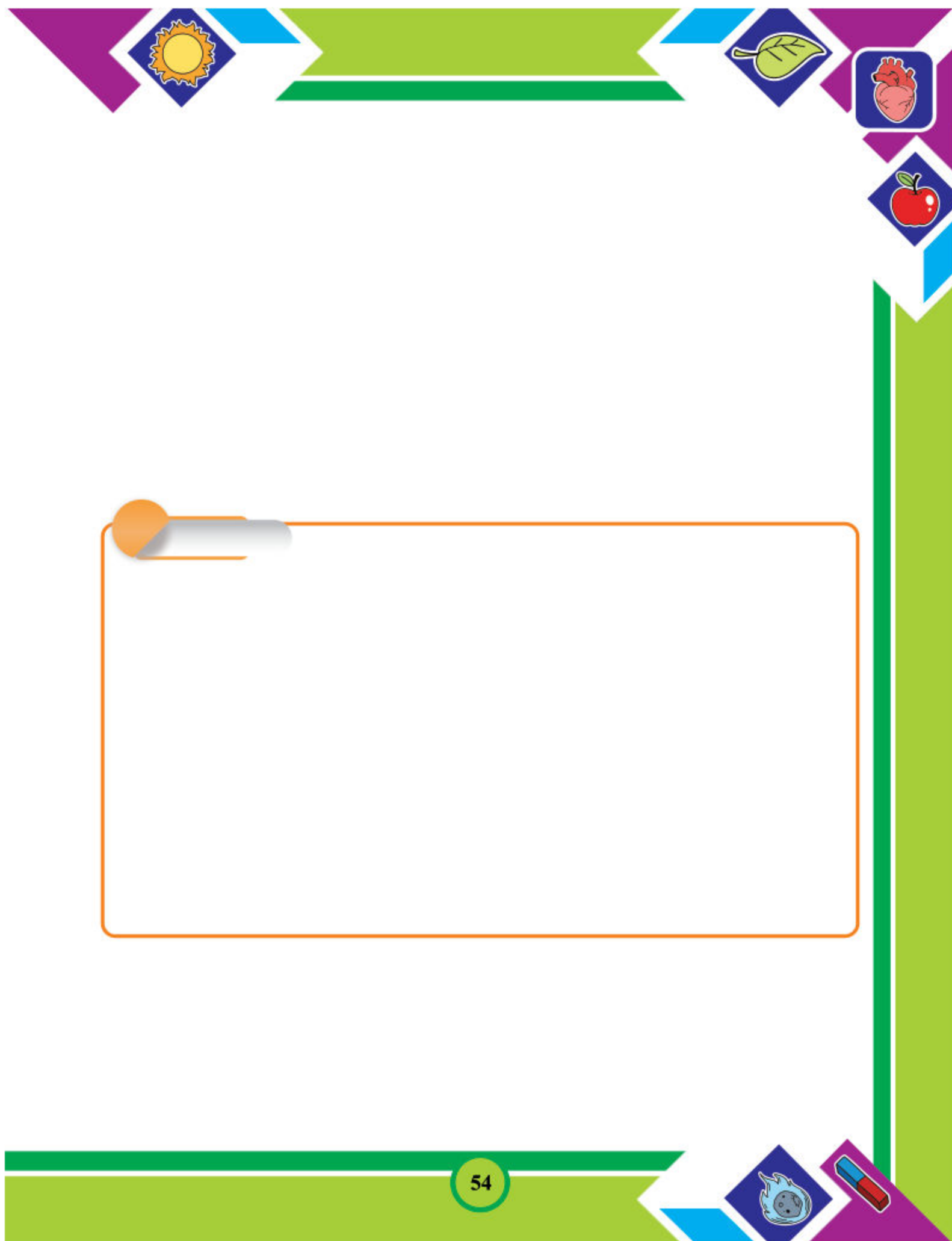


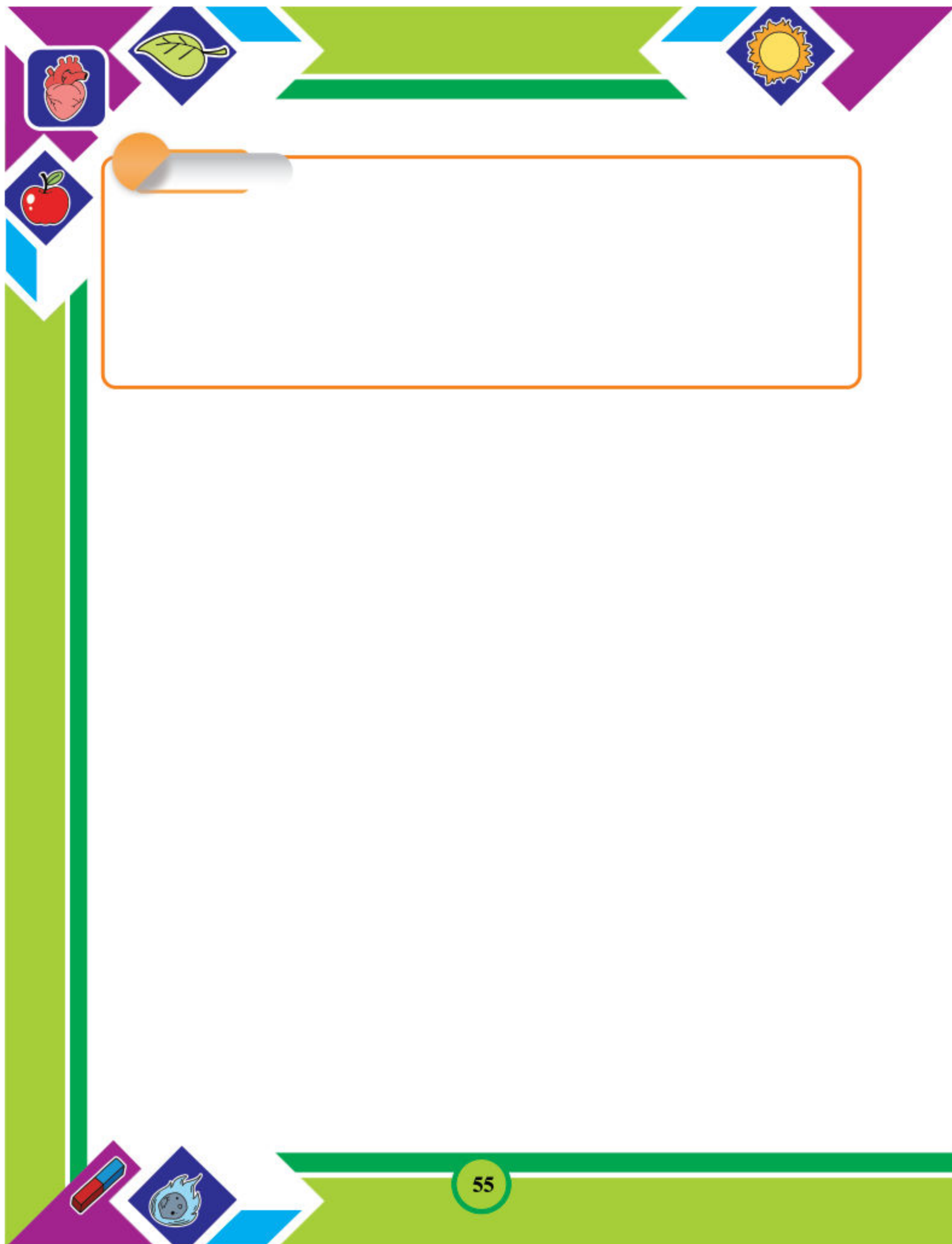
4.1 ECOSYSTEM AND ITS TYPES

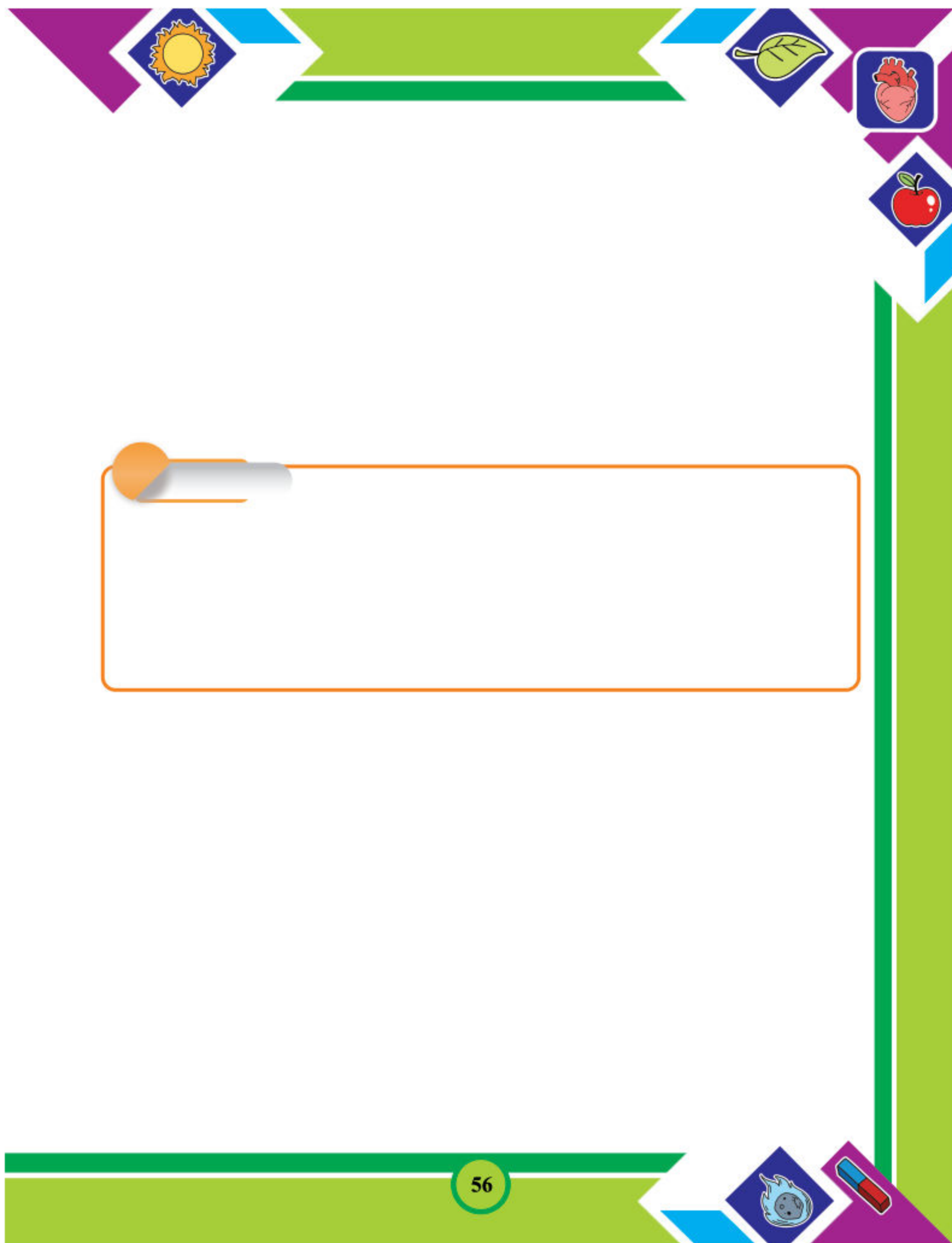
The ecosystem is like a big community where living things interact with each

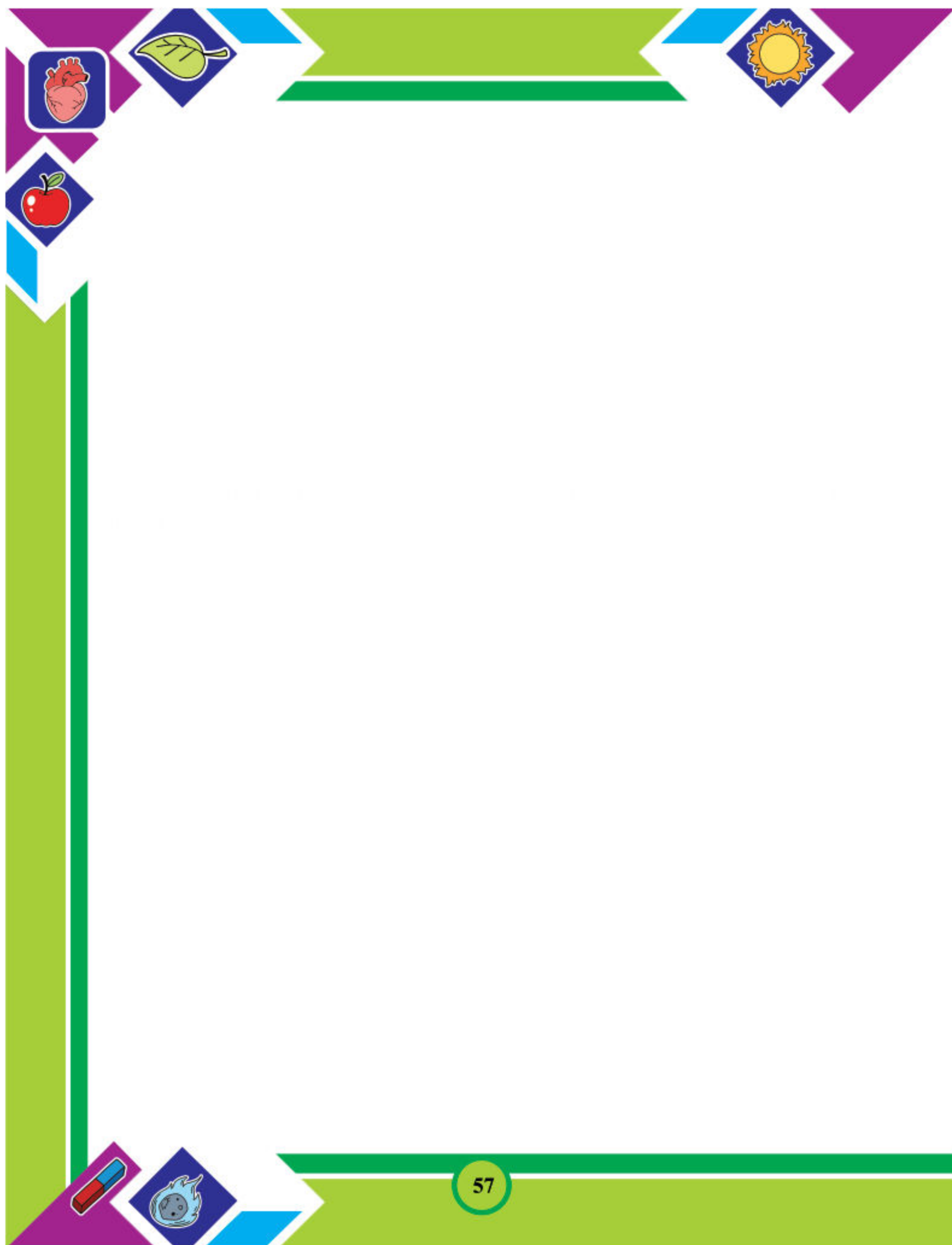


A rectangular box with a purple border and a pink tab on the left side, intended for text entry.





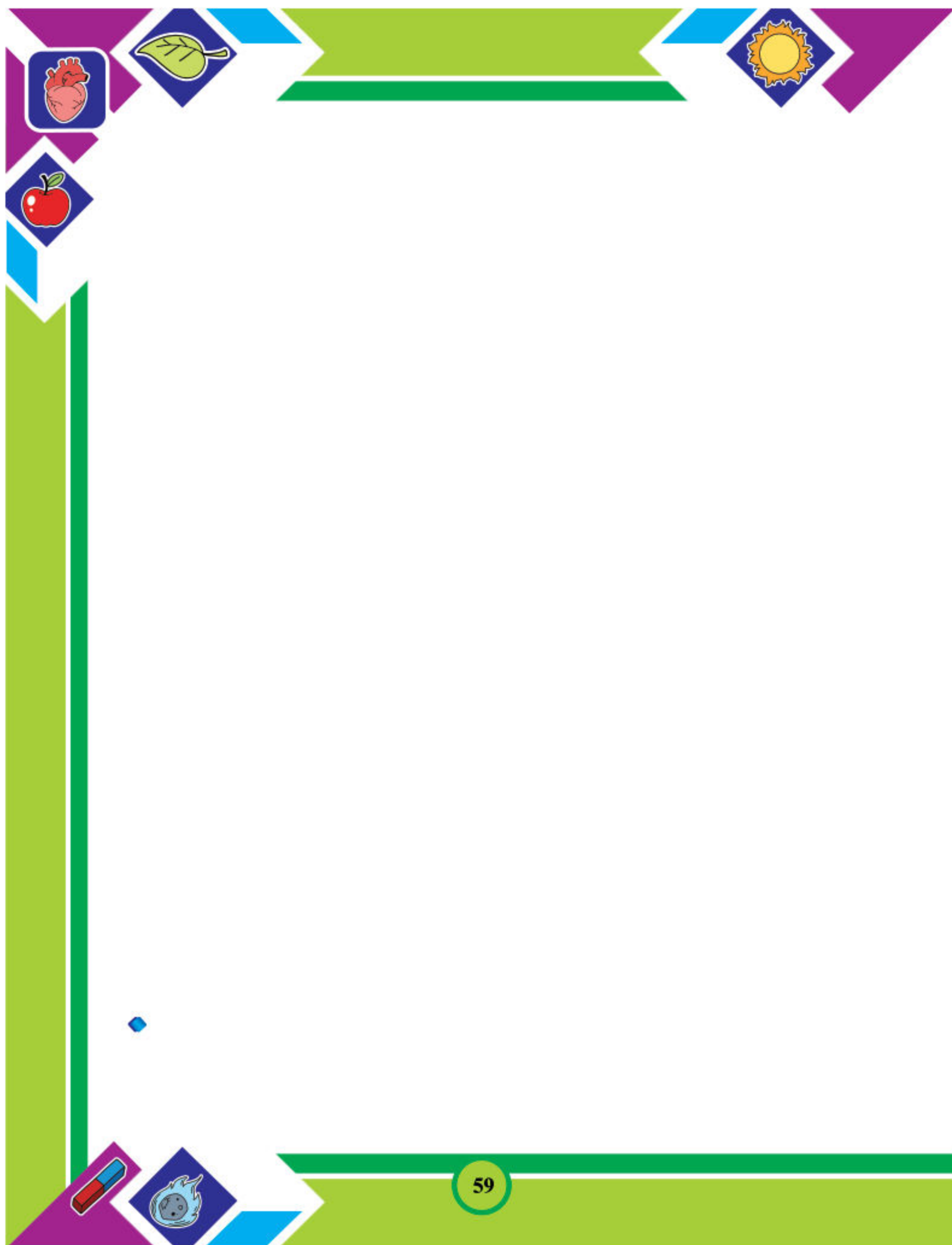


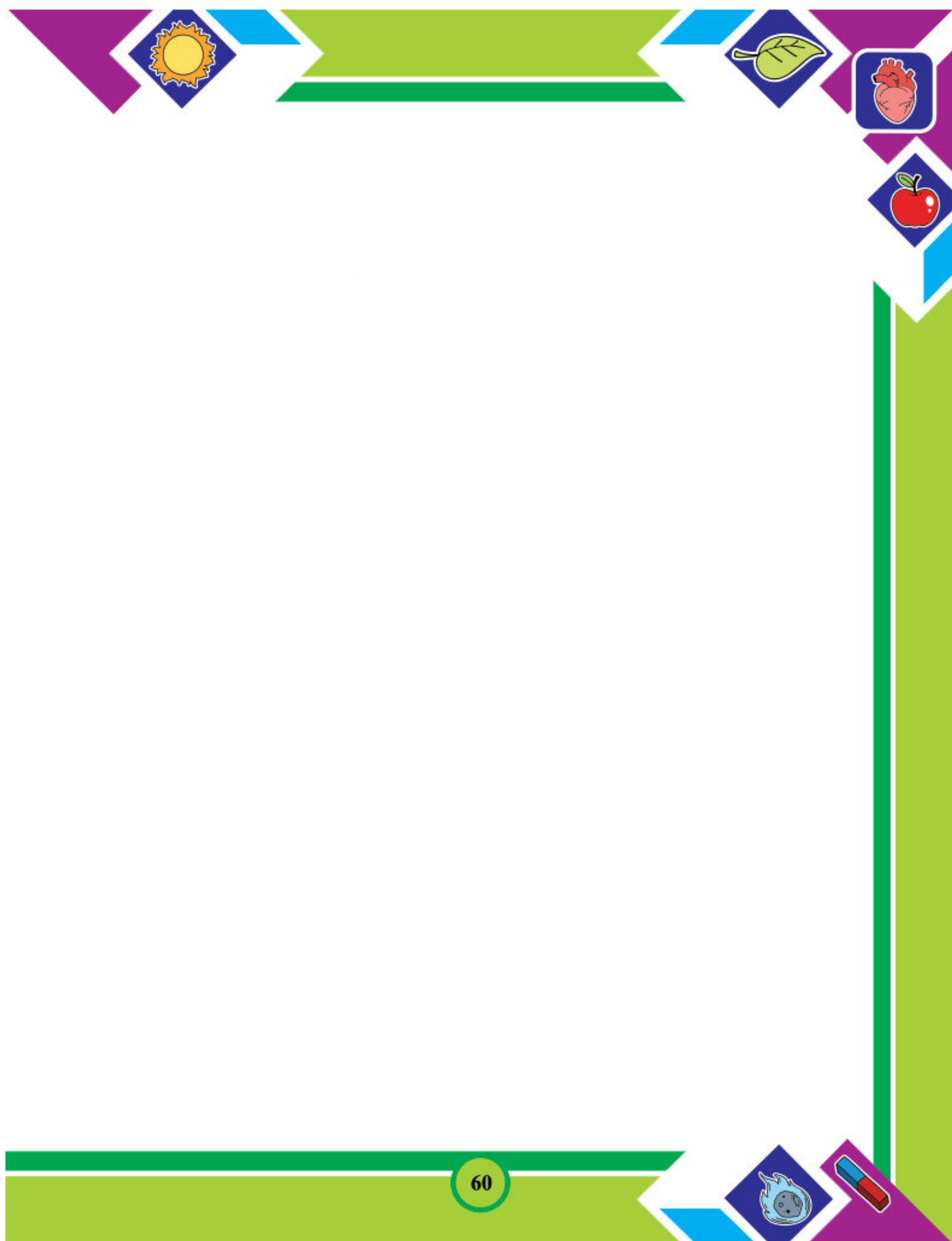


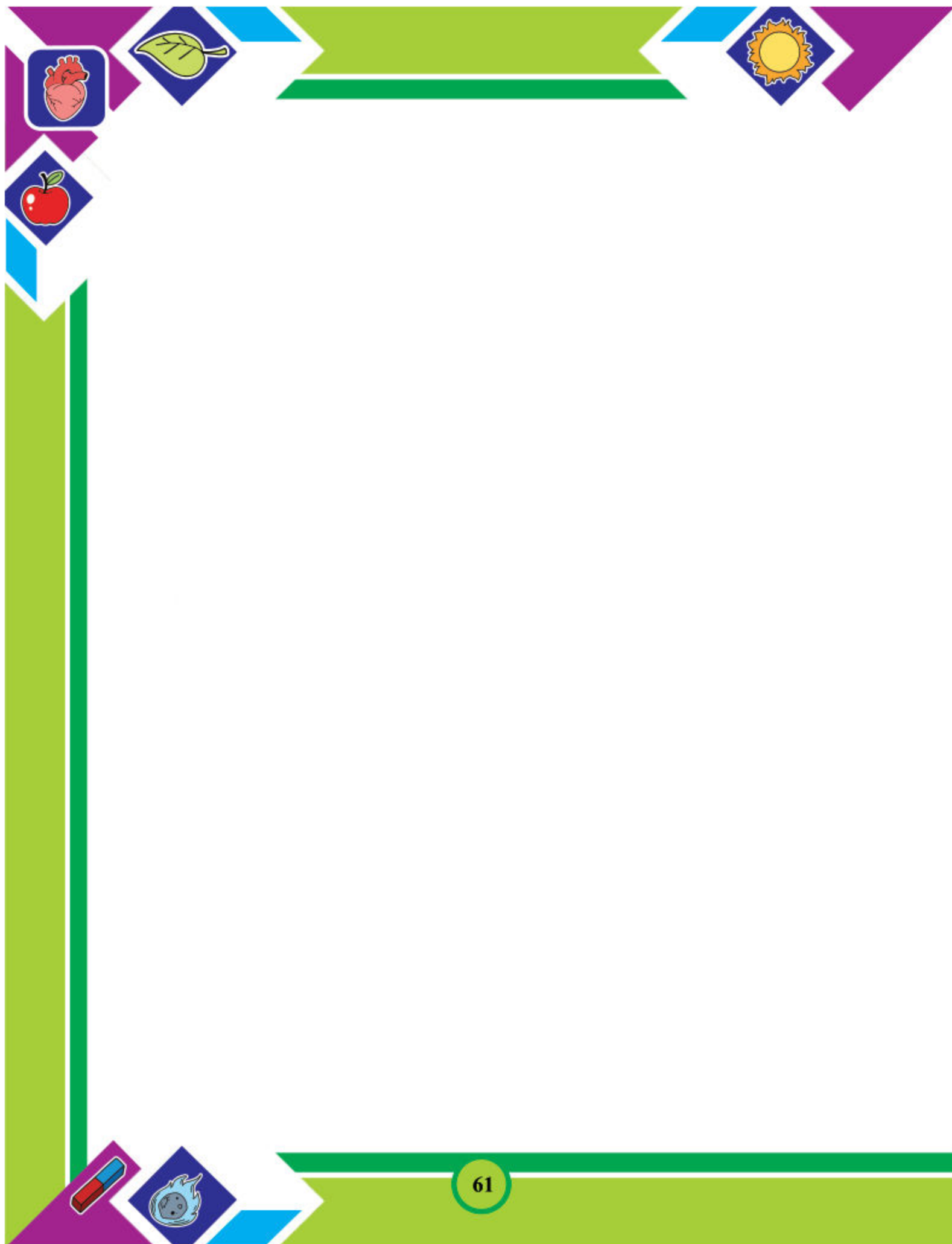
Hibernation:

4.4 EFFECTS OF HUMAN ACTIVITIES ON HABITAT









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0 D W H W D Q G L W V
& K D U H D F W W L F V

10 D M R Q F H S W V

, Q V K L X Q W H Z L O O I Q D E R X W

- 5.1 States of Matter & its Characteristics
- 5.2 Physical properties of matter
- 5.3 Physical Properties of Metal
- 5.4 Conditions cause matter to change states (melting, freezing, boiling)

6 W X G H Q W H D U Q I Q J 2 X W F R P H V

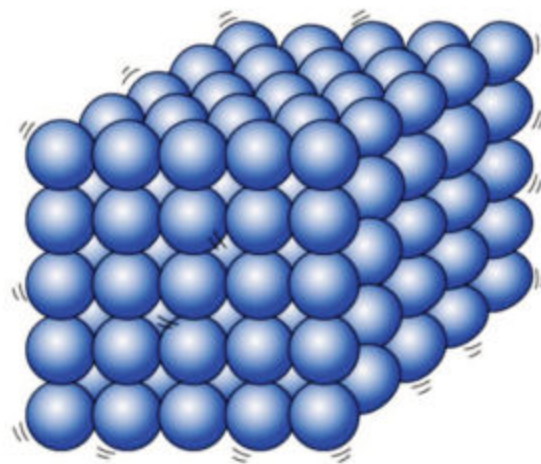
\$ I W U H R P S O M R Q V K L X Q V V W Q V K R X G E H D I D V K

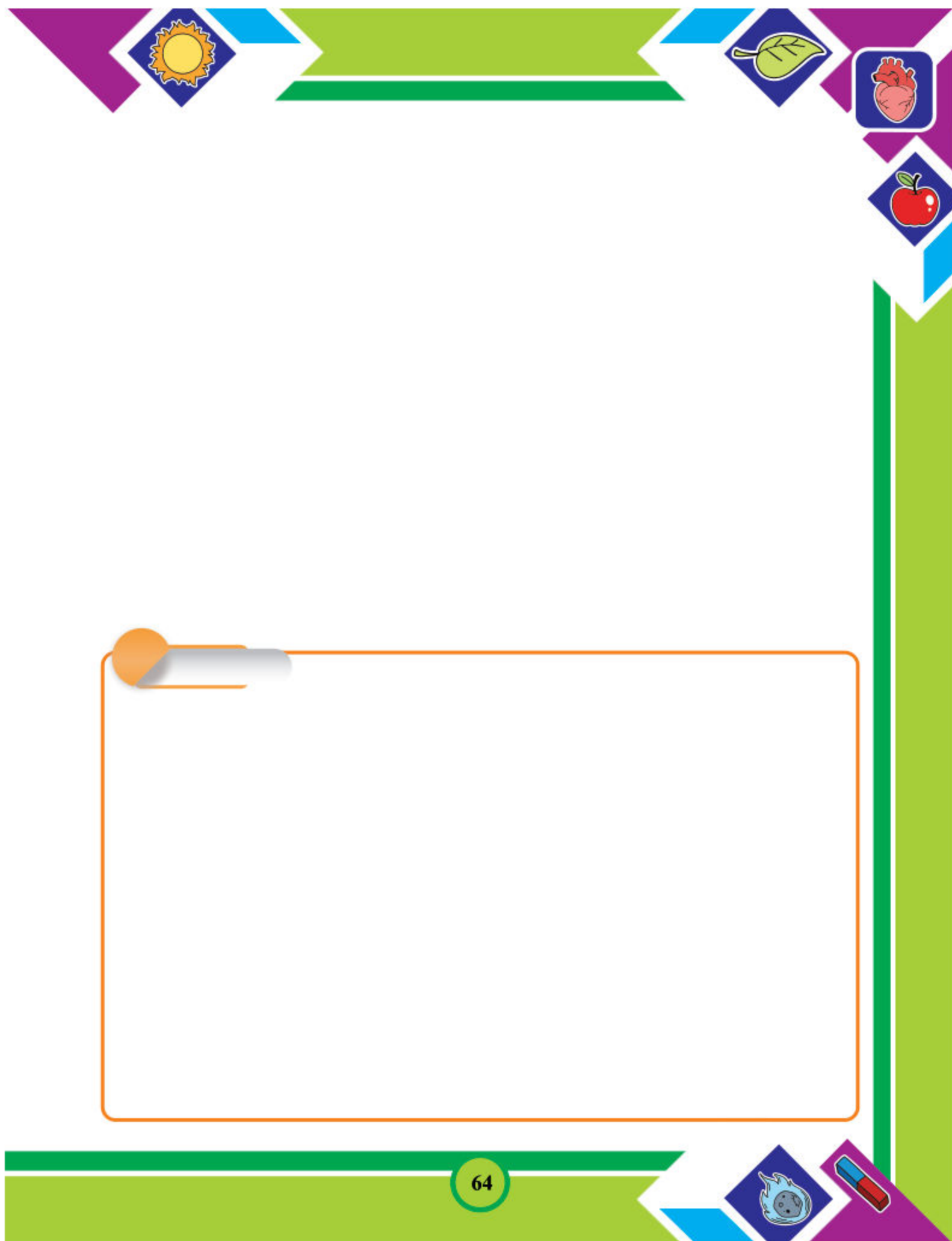


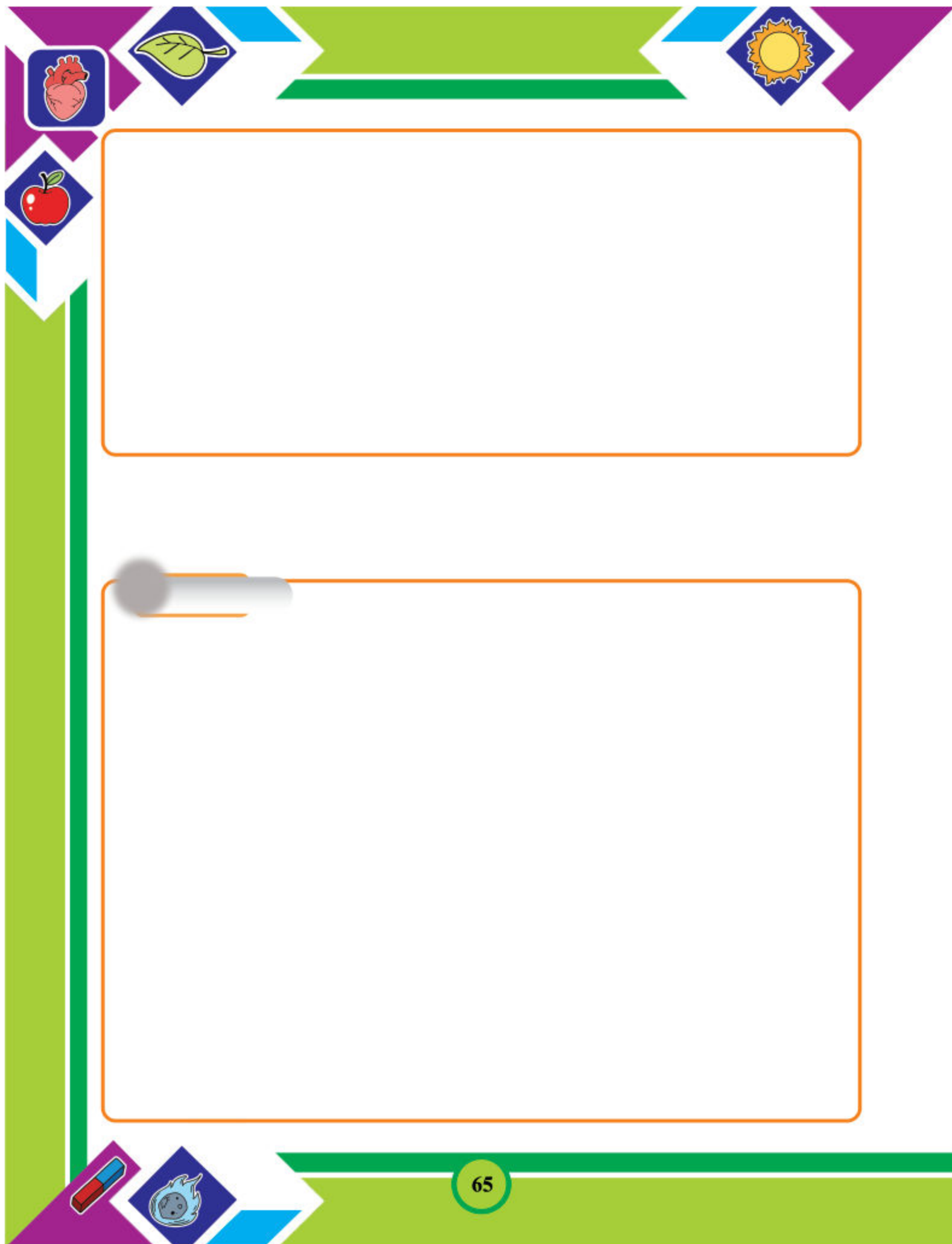


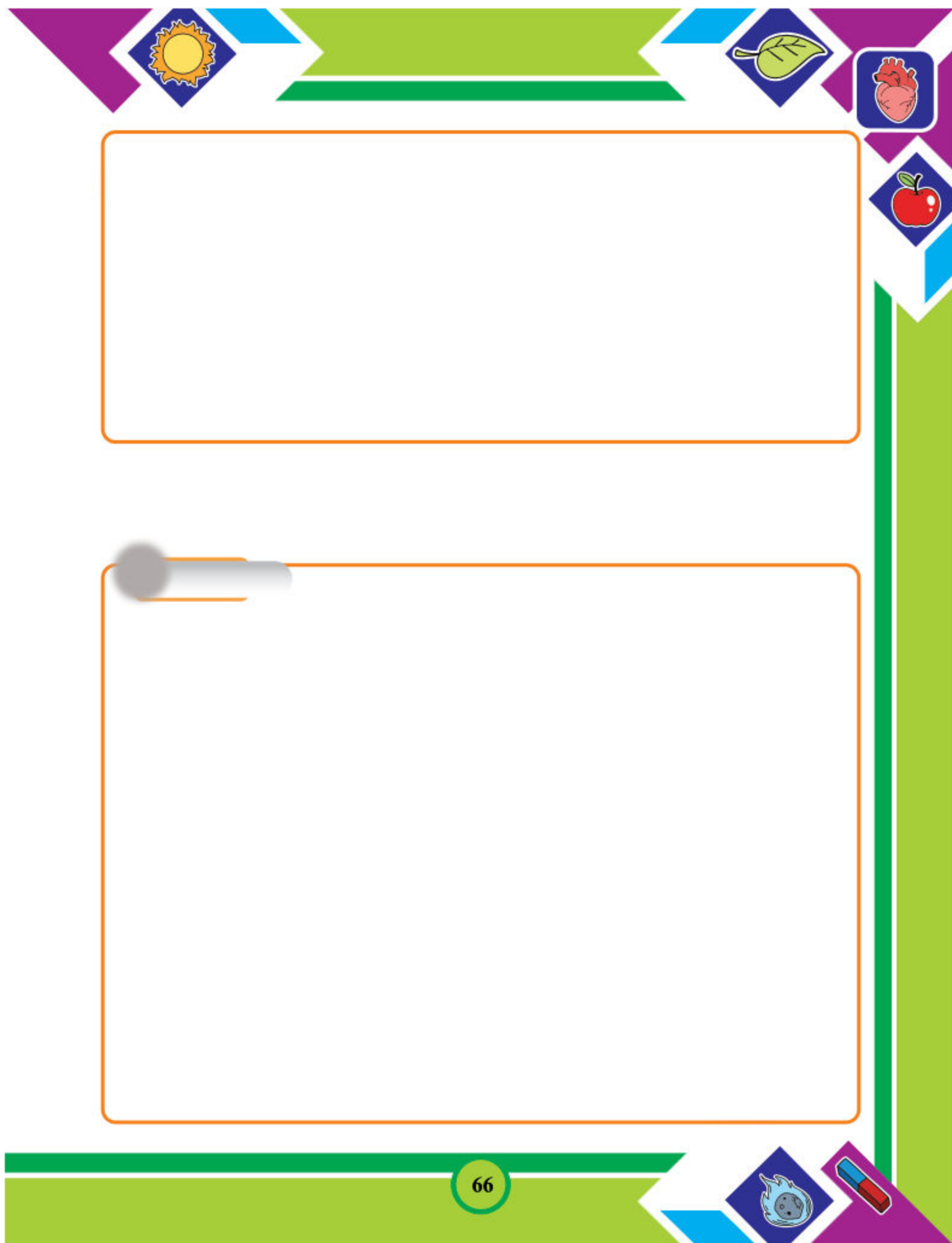
Introduction:

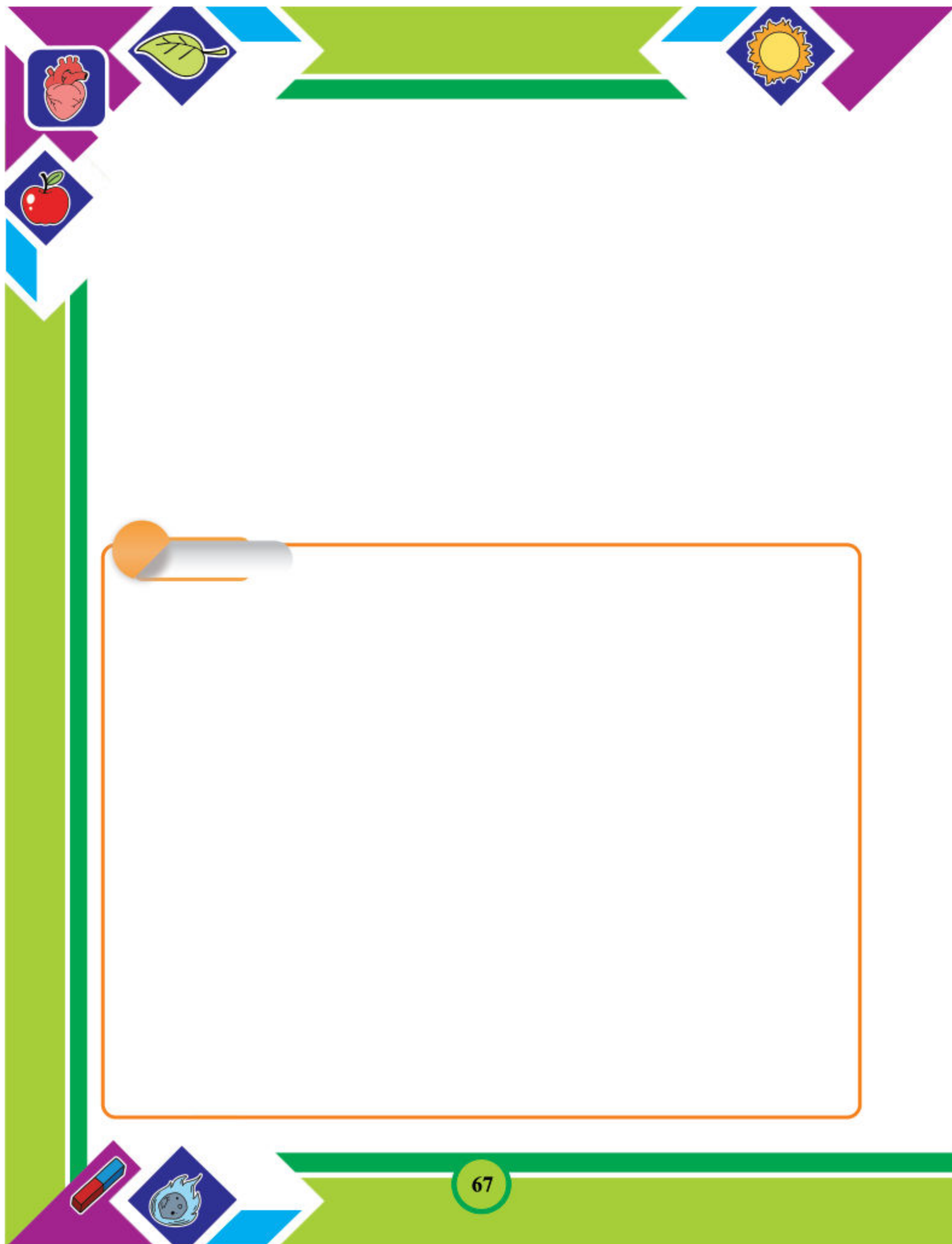
As we know the world is made up of different materials. Have you e

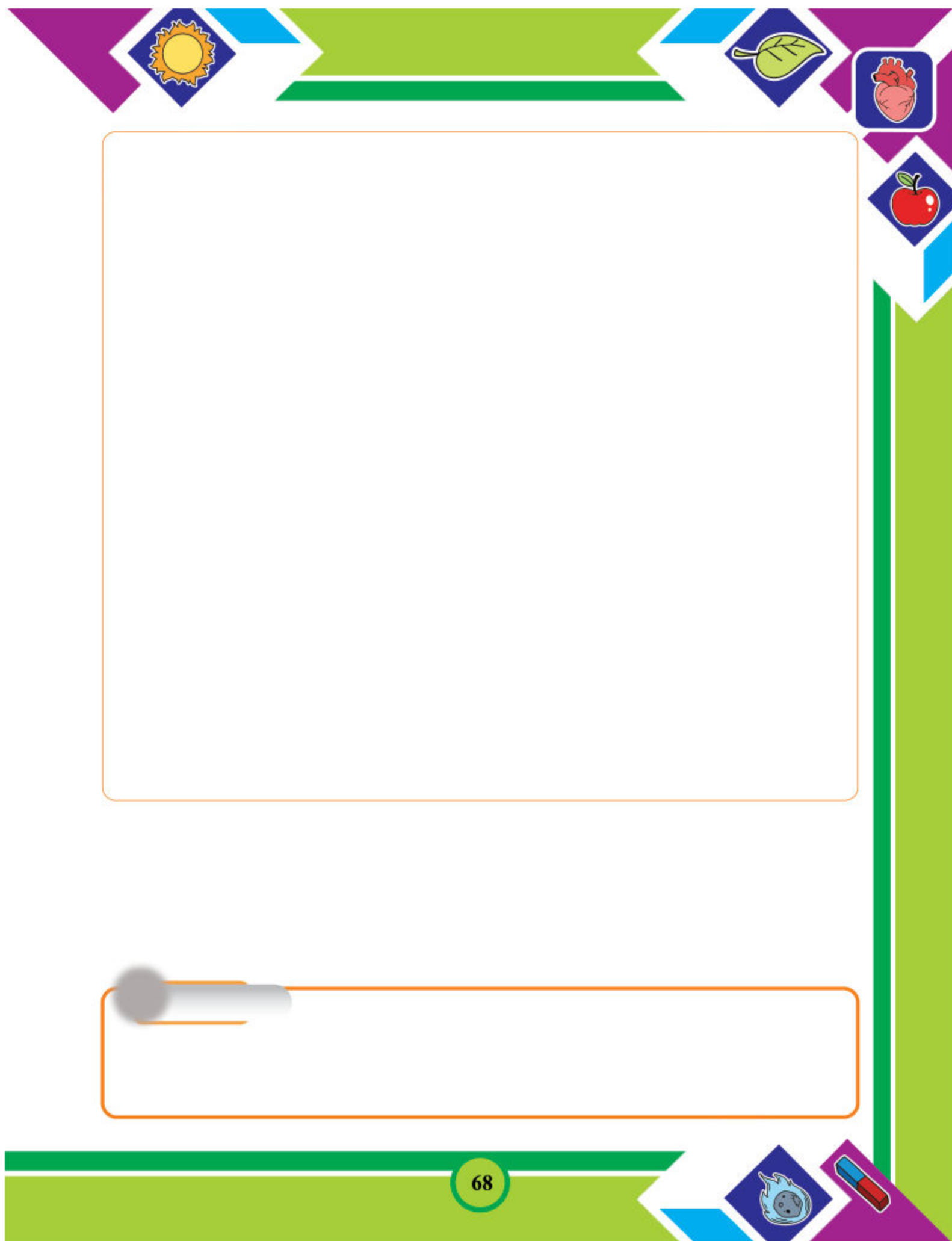












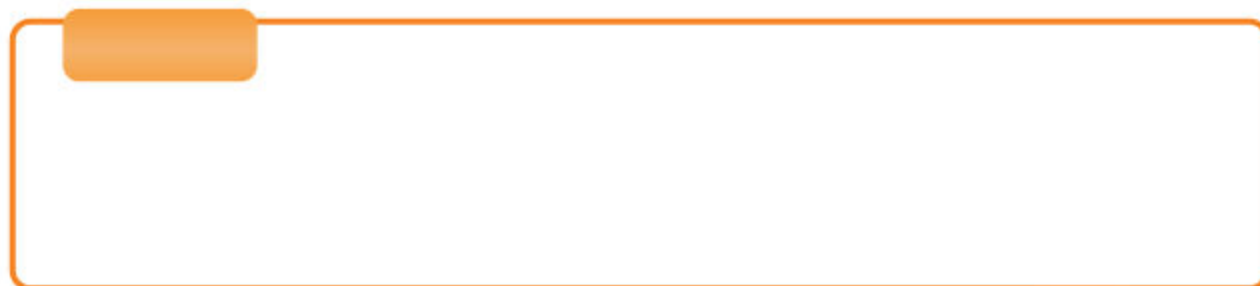


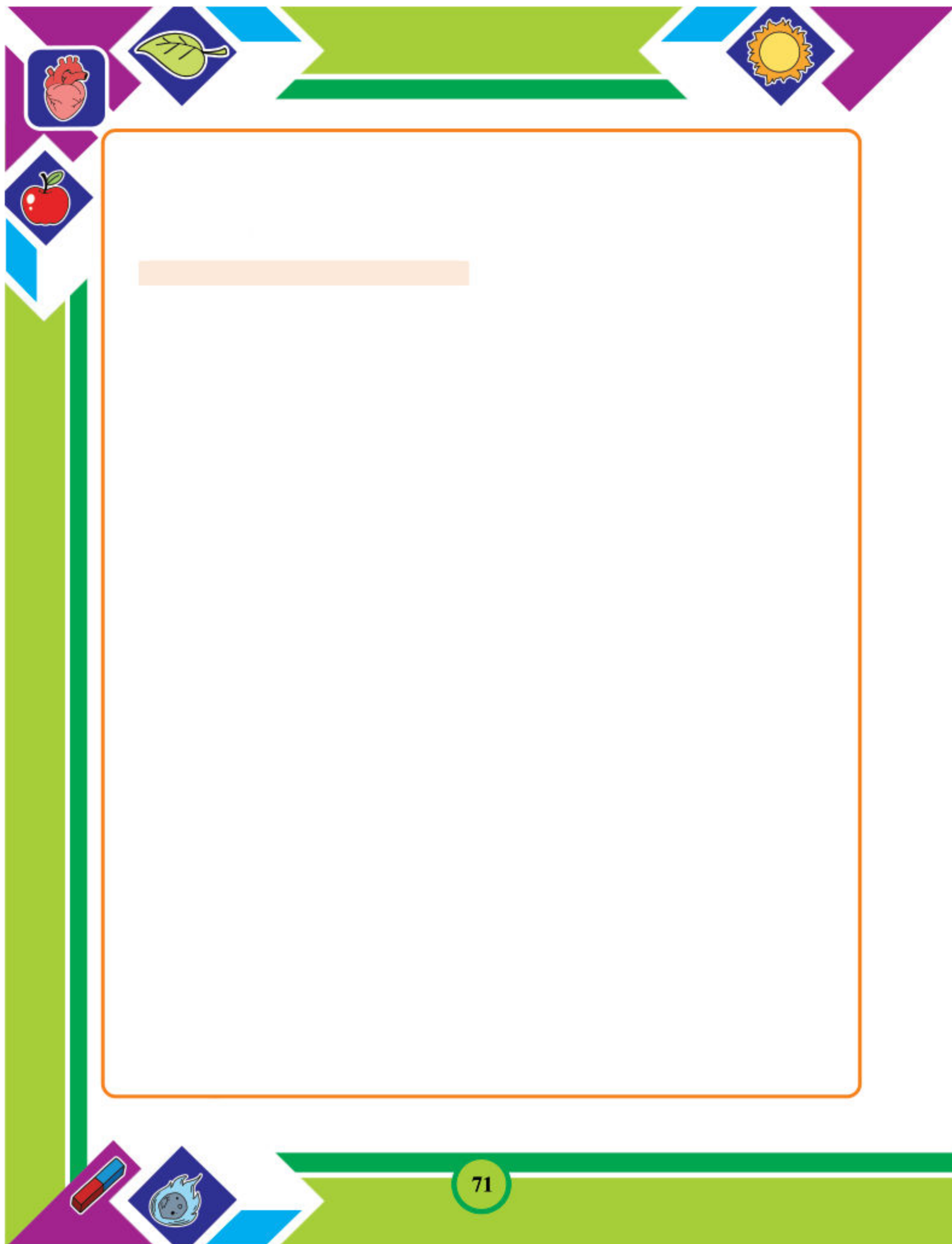
- Put the given objects on the weighing scale one by one and note down their mass in the



(2) Volume

(3) Density





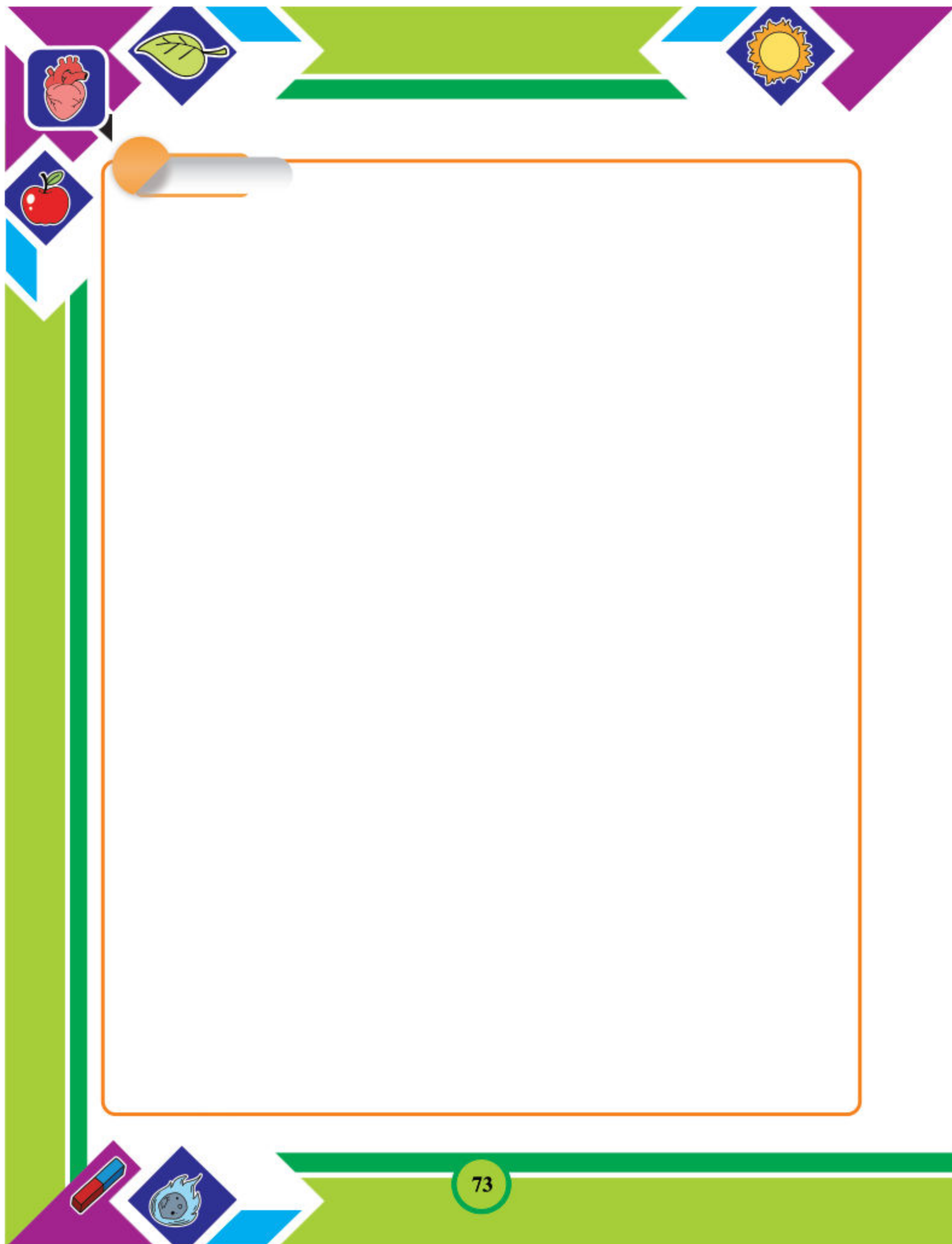


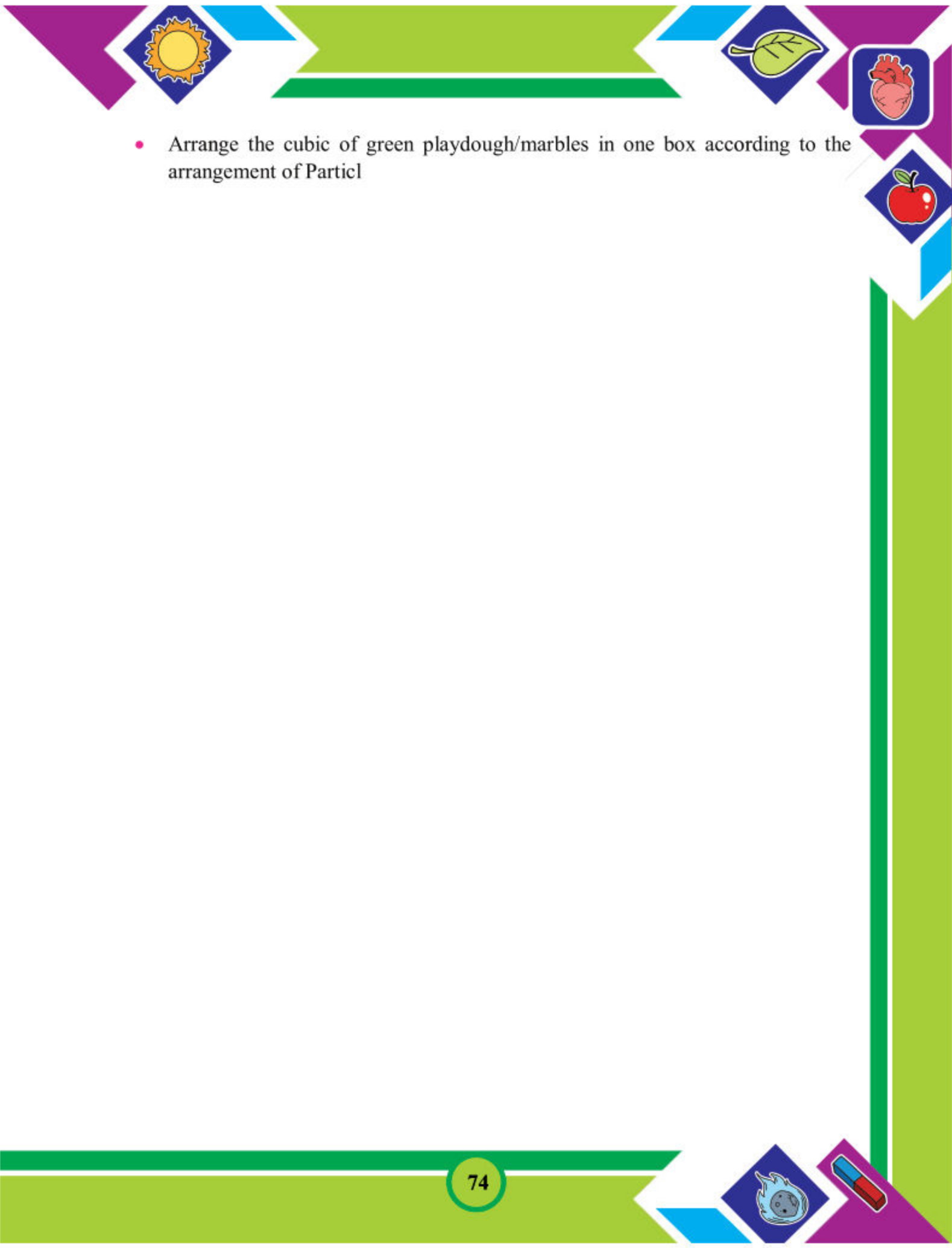
6ROLG

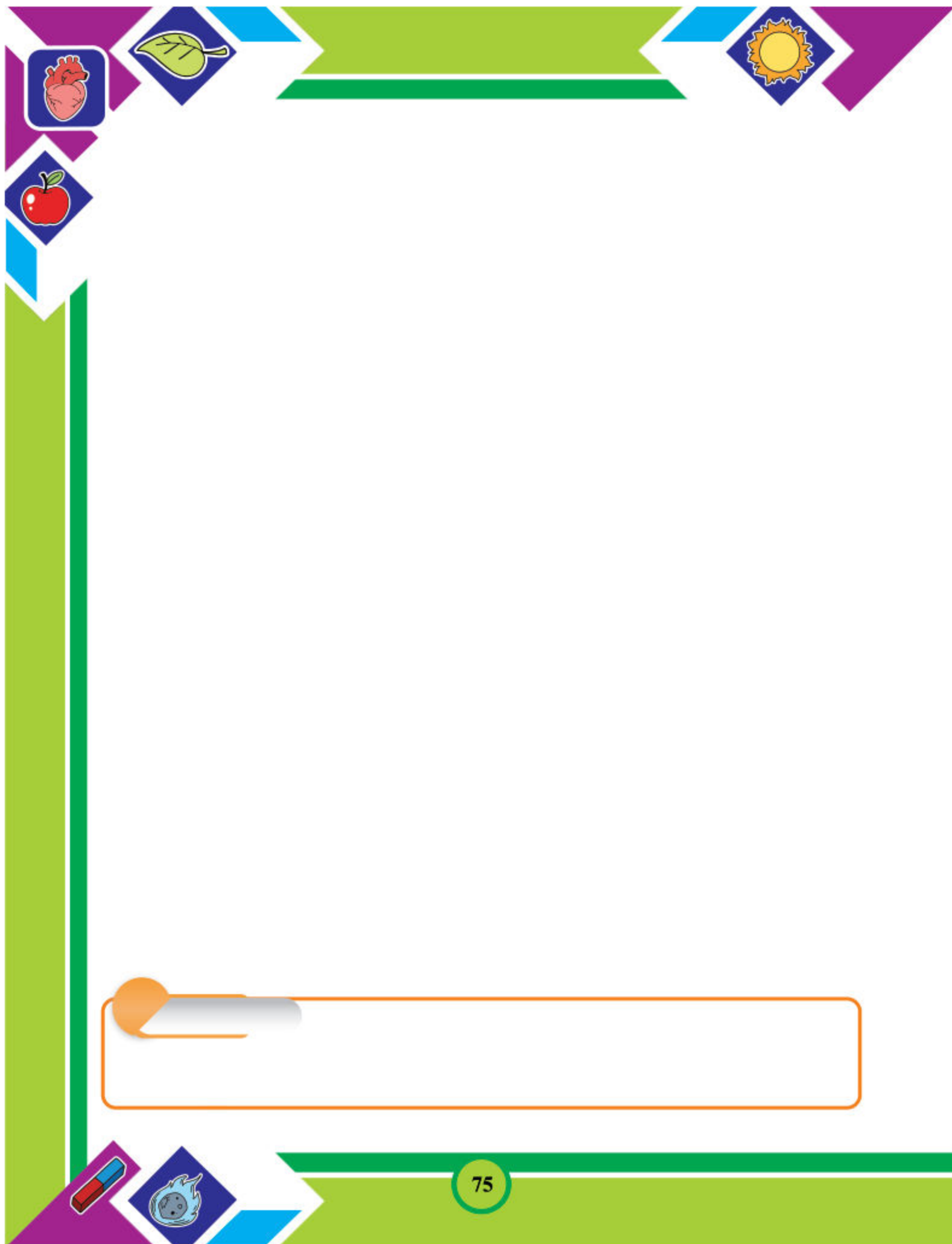
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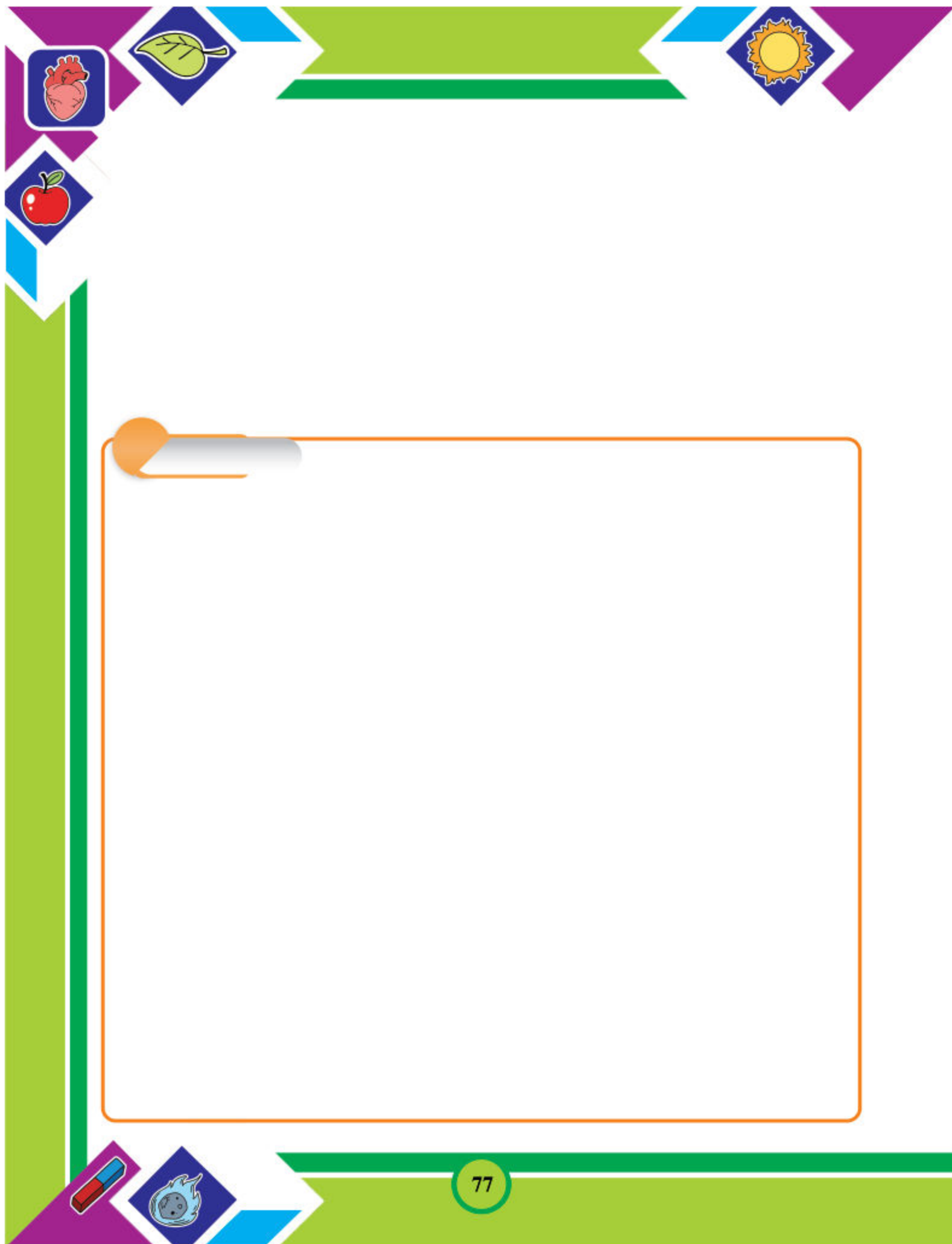


- 
- Arrange the cubic of green playdough/marbles in one box according to the arrangement of Particl



A large, empty rectangular box with an orange border, intended for writing or drawing.

- Observe carefully and note your observations in the appropriate columns of Table 5.5.



- Tick marks your observations in the appropriate column of given table 5.6.



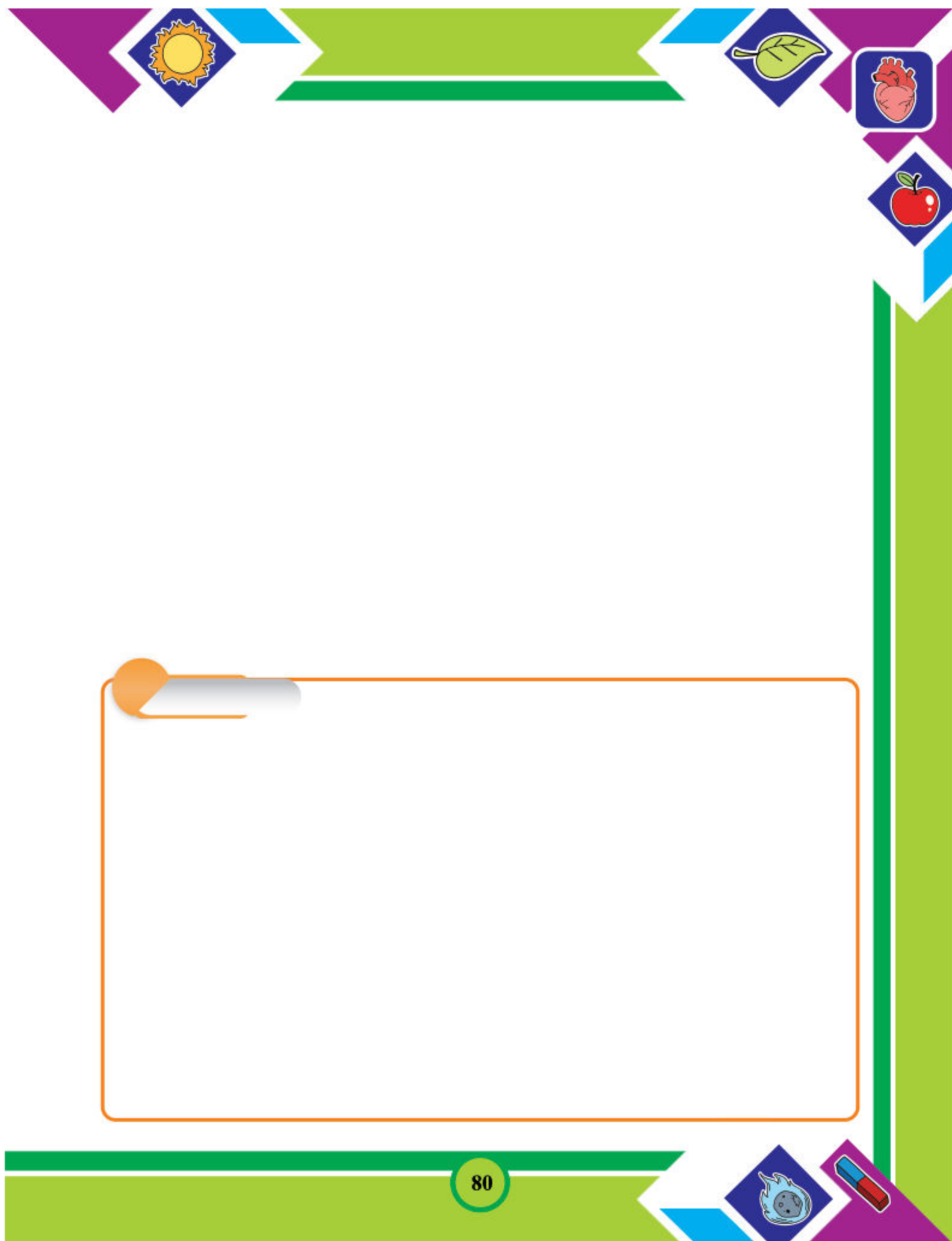


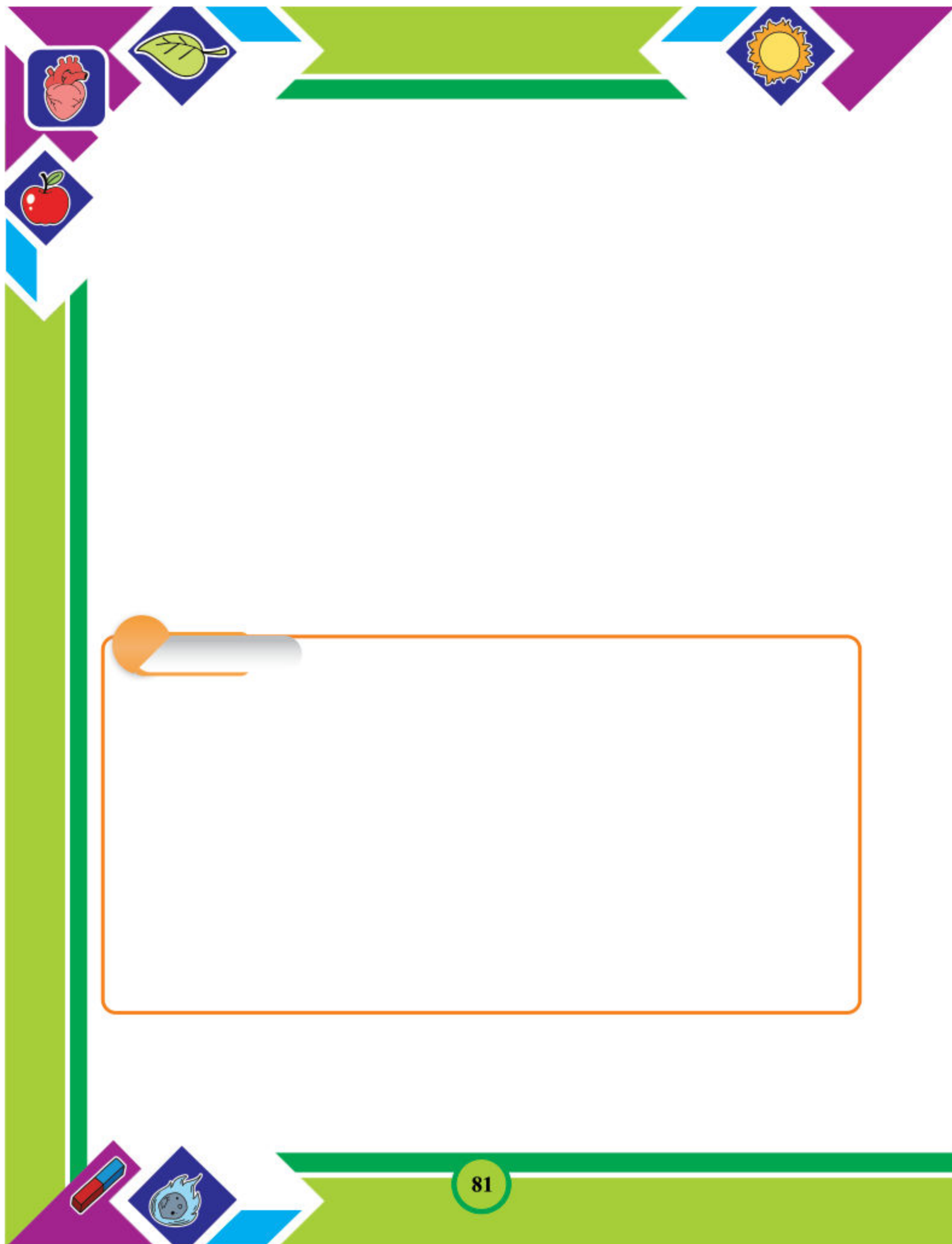
What did you conclude?

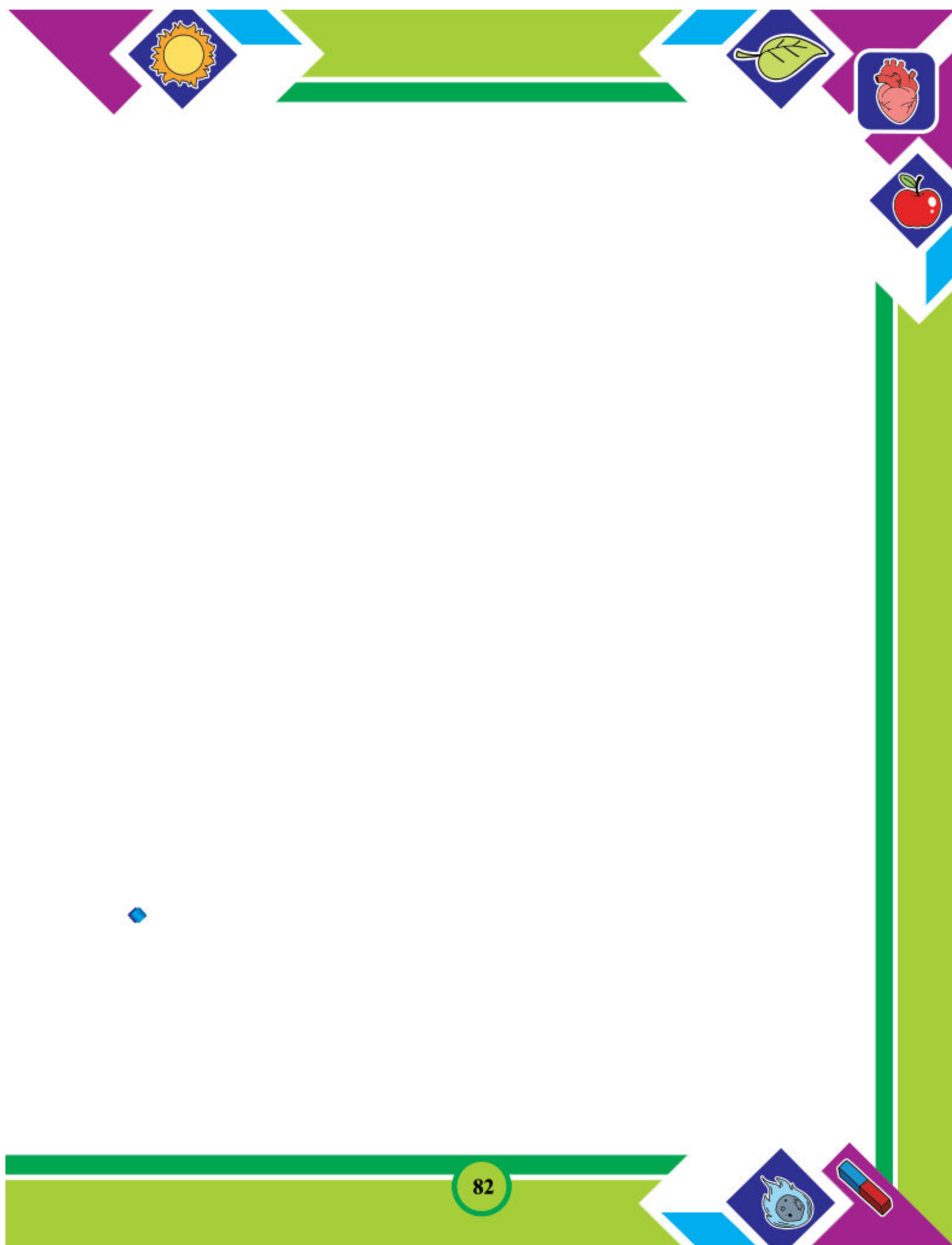
You must have concluded that:

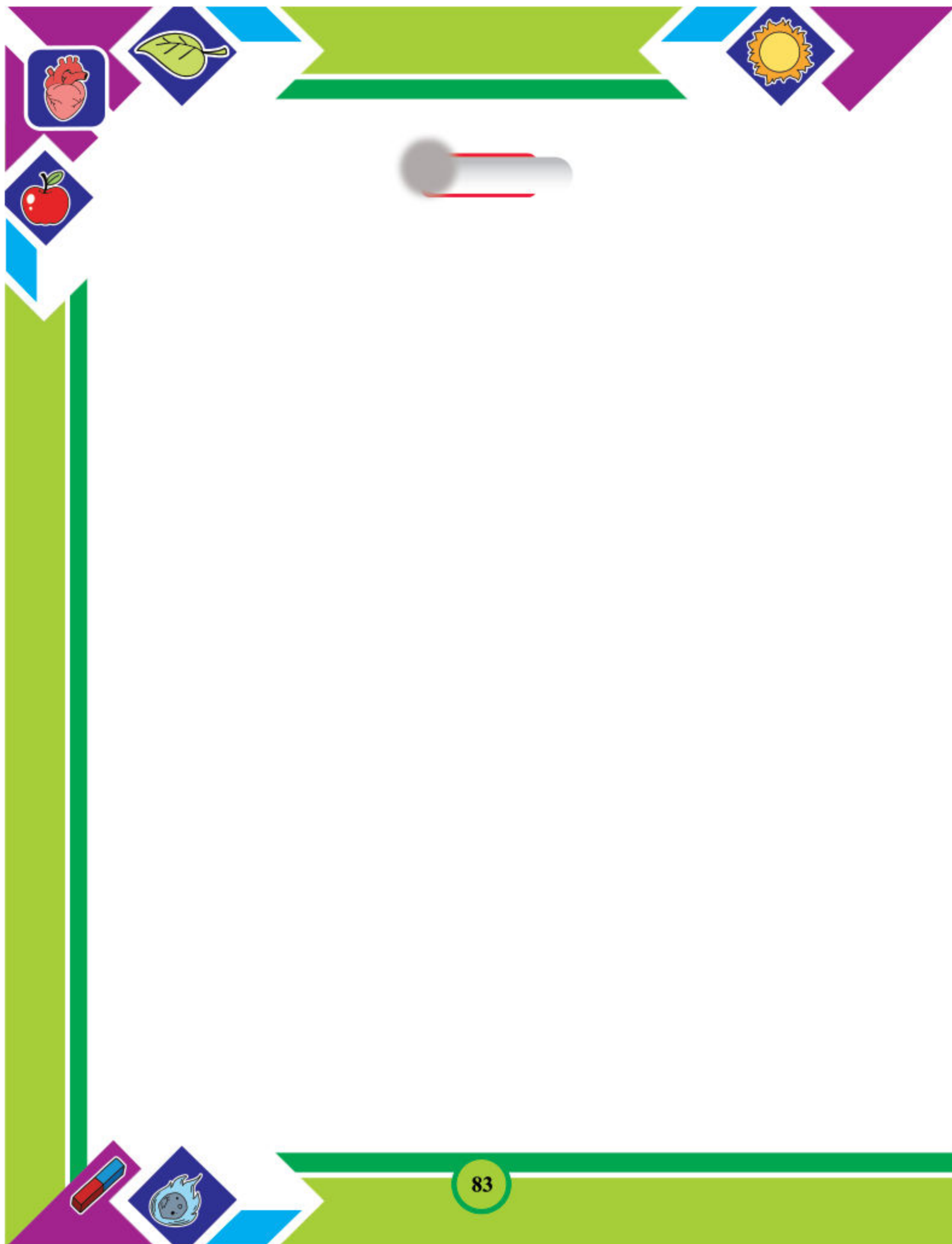
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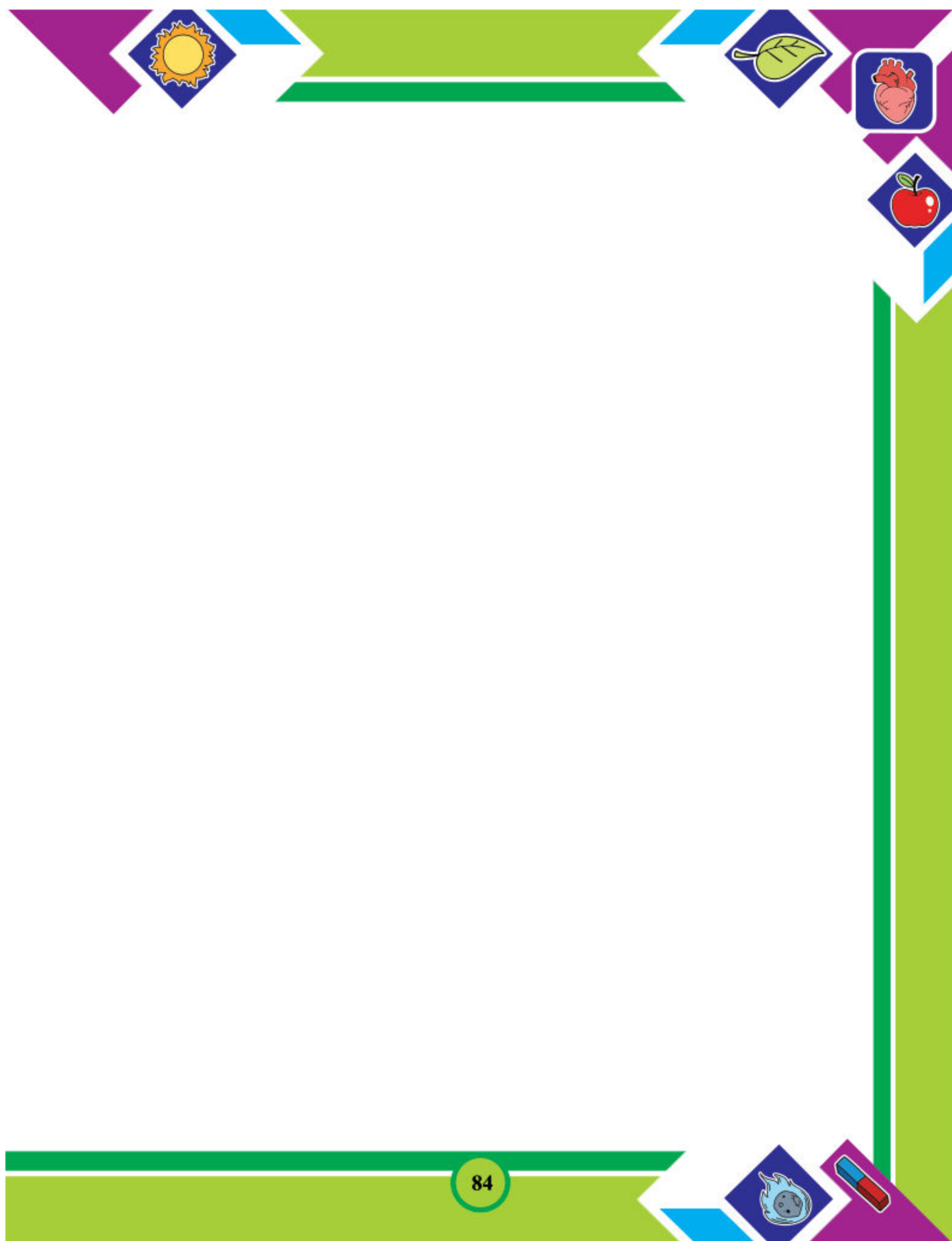












& K D S



) R E V R (Q H U
D Q G
L W V D Q V I H U

10 D M R Q F H S W V

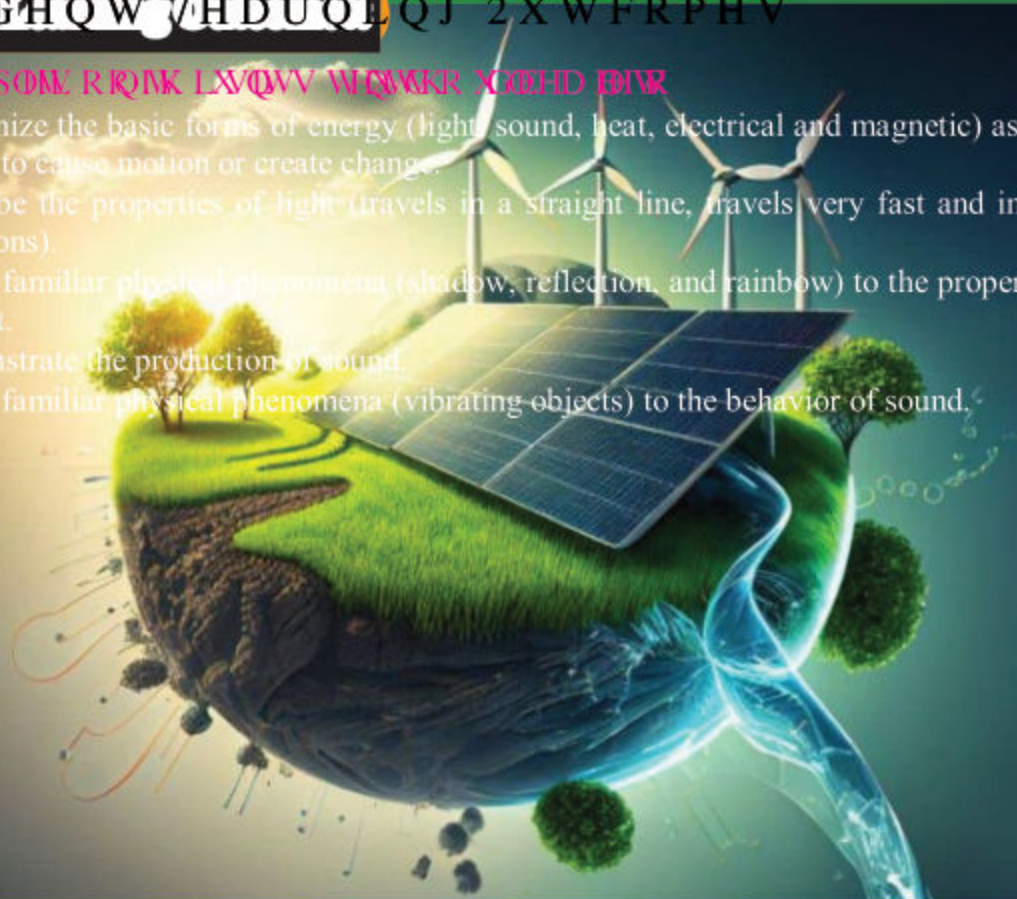
, Q V L X W H Z L O O I D Q D E R X W

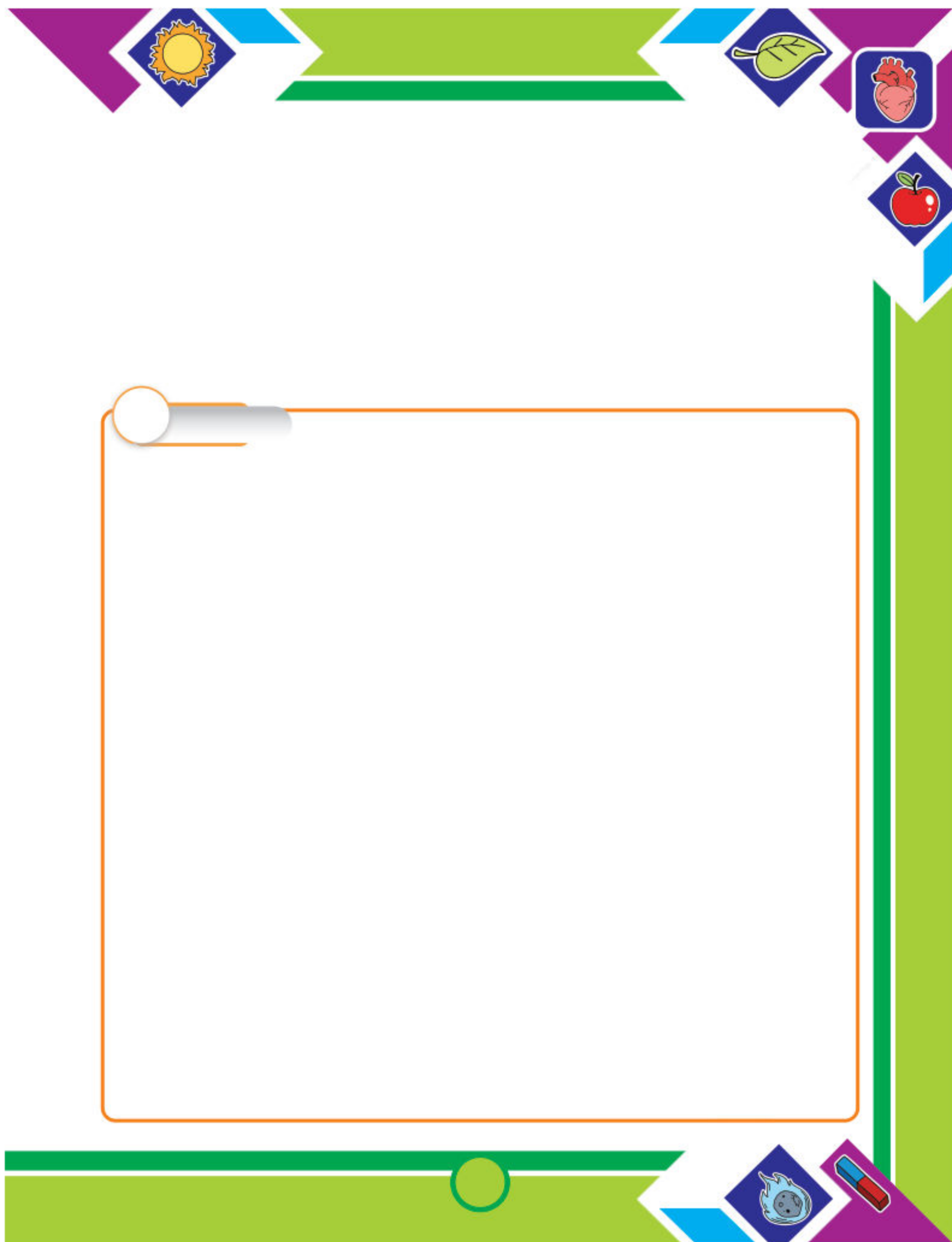
- 6.1 Forms of Energy and its Transfer
- 6.2 Light energy
- 6.3 Sound energy
- 6.4 Heat energy
- 6.5 Electrical energy

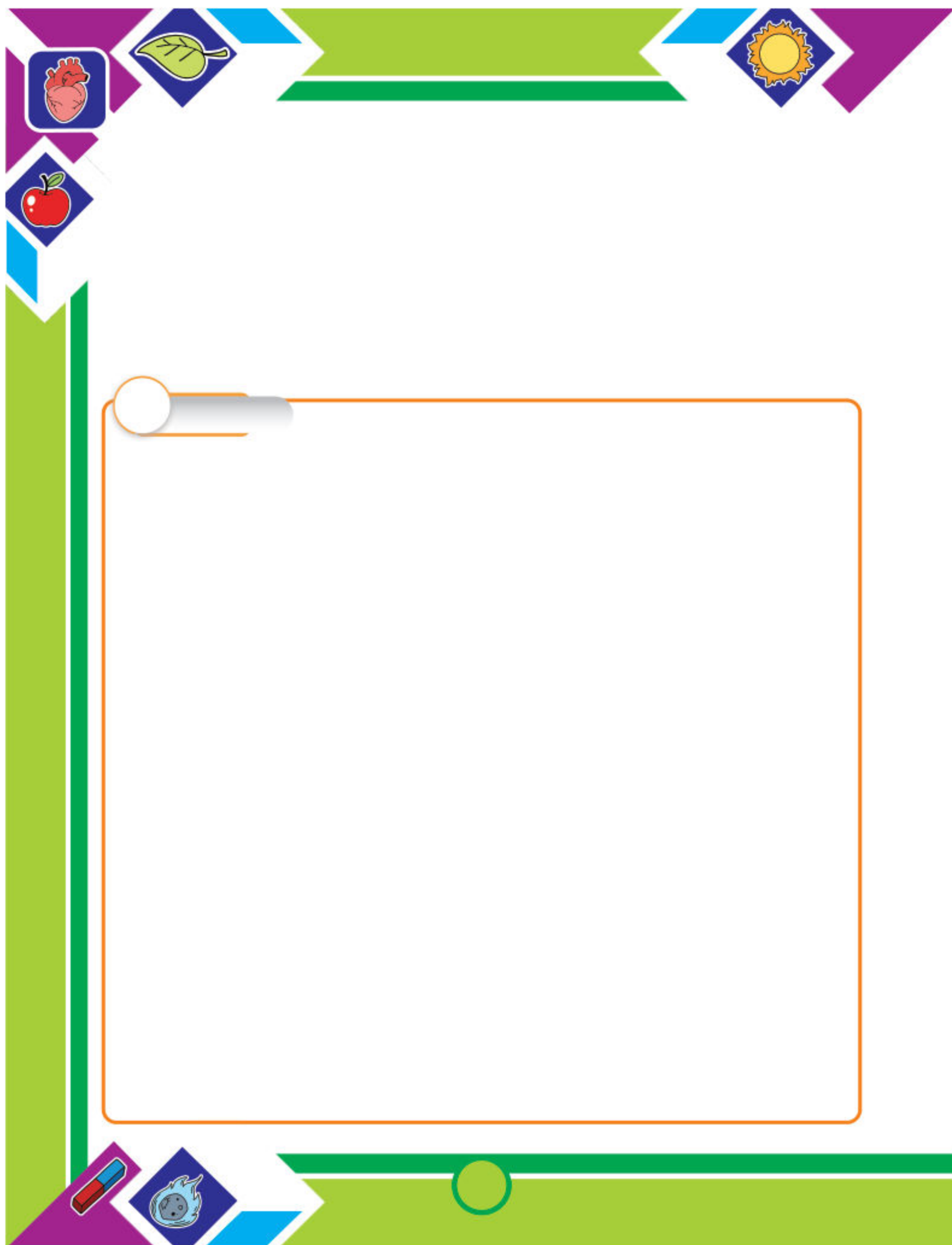
6 W X G H Q W H D U Q L Q J 2 X W F R P H V

S T W L R P S O M R Q N K L X Q V V W Q N K R X G E H D B I W

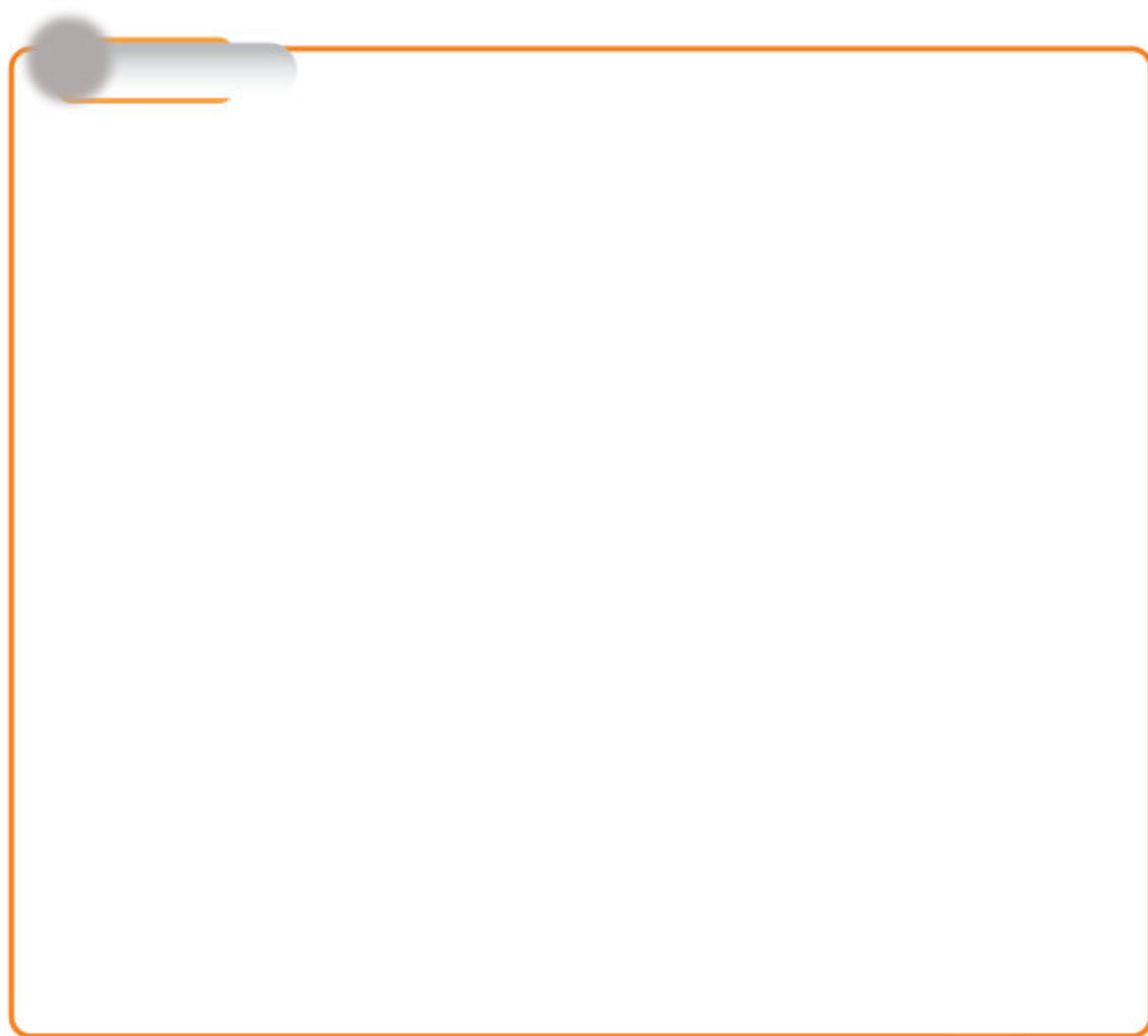
- ◆ Recognize the basic forms of energy (light, sound, heat, electrical and magnetic) as the ability to cause motion or create change.
- Describe the properties of light (travels in a straight line, travels very fast and in all directions).
- Relate familiar physical phenomena (shadow, reflection, and rainbow) to the properties of light.
- Demonstrate the production of sound.
- Relate familiar physical phenomena (vibrating objects) to the behavior of sound.

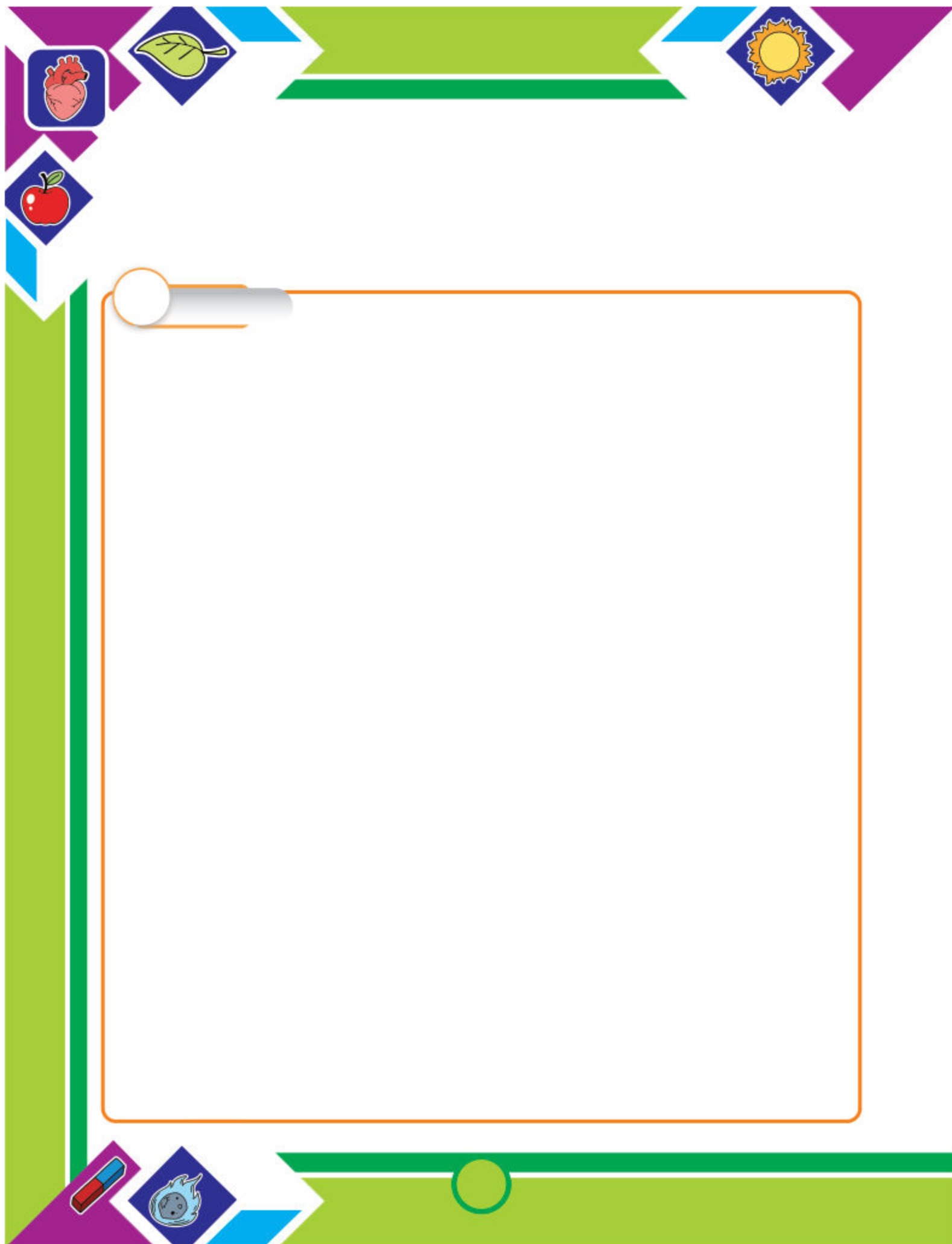






6.2 LIGHT ENERGY





6.2.2 Reflection of light:

Fig 6.3 Reflection of light

6.2.2 Rainbow:



6.3.1 Production of sound:

7 K M R X I Q U R G Z K H C R P H W K L Q J



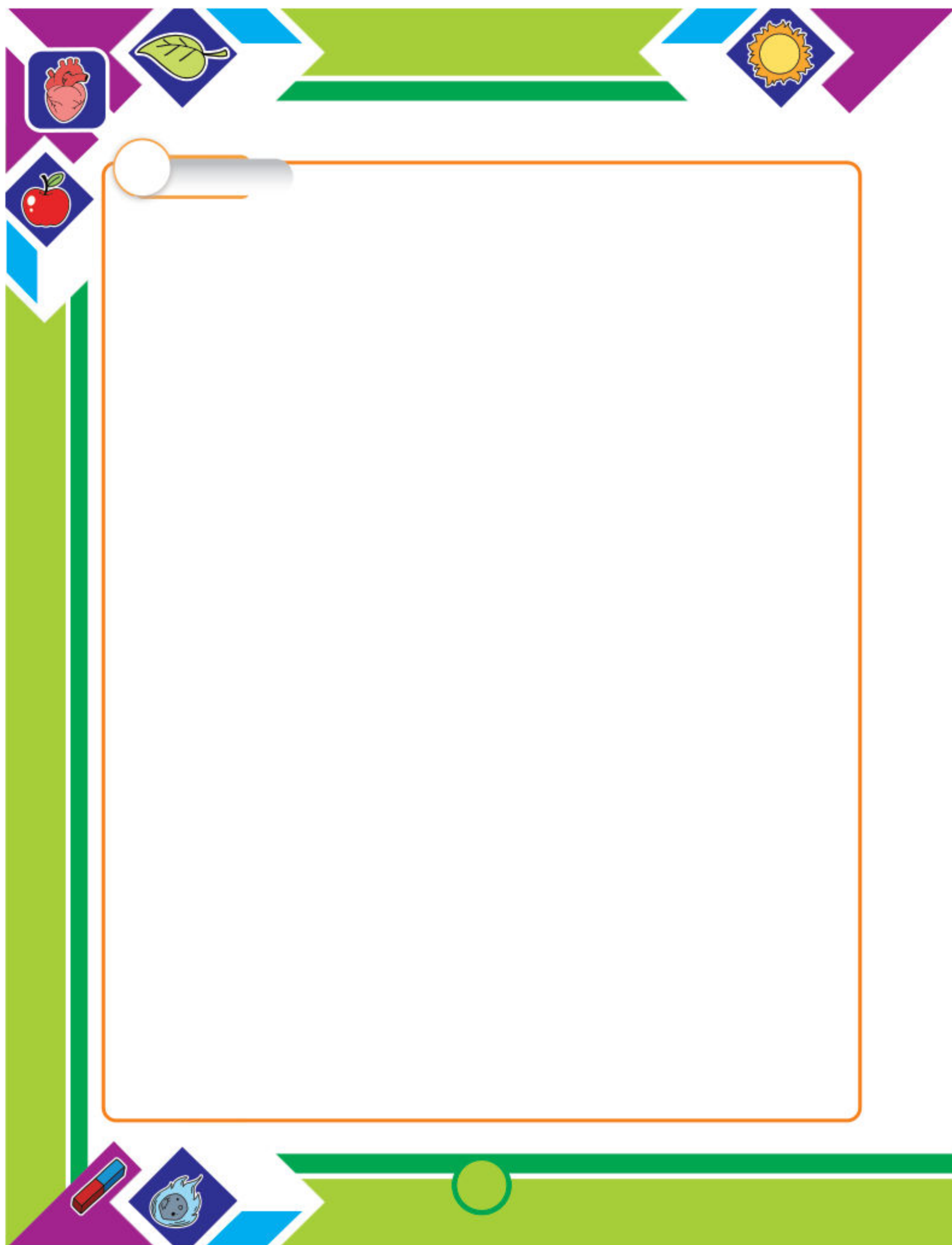
Test yourself:

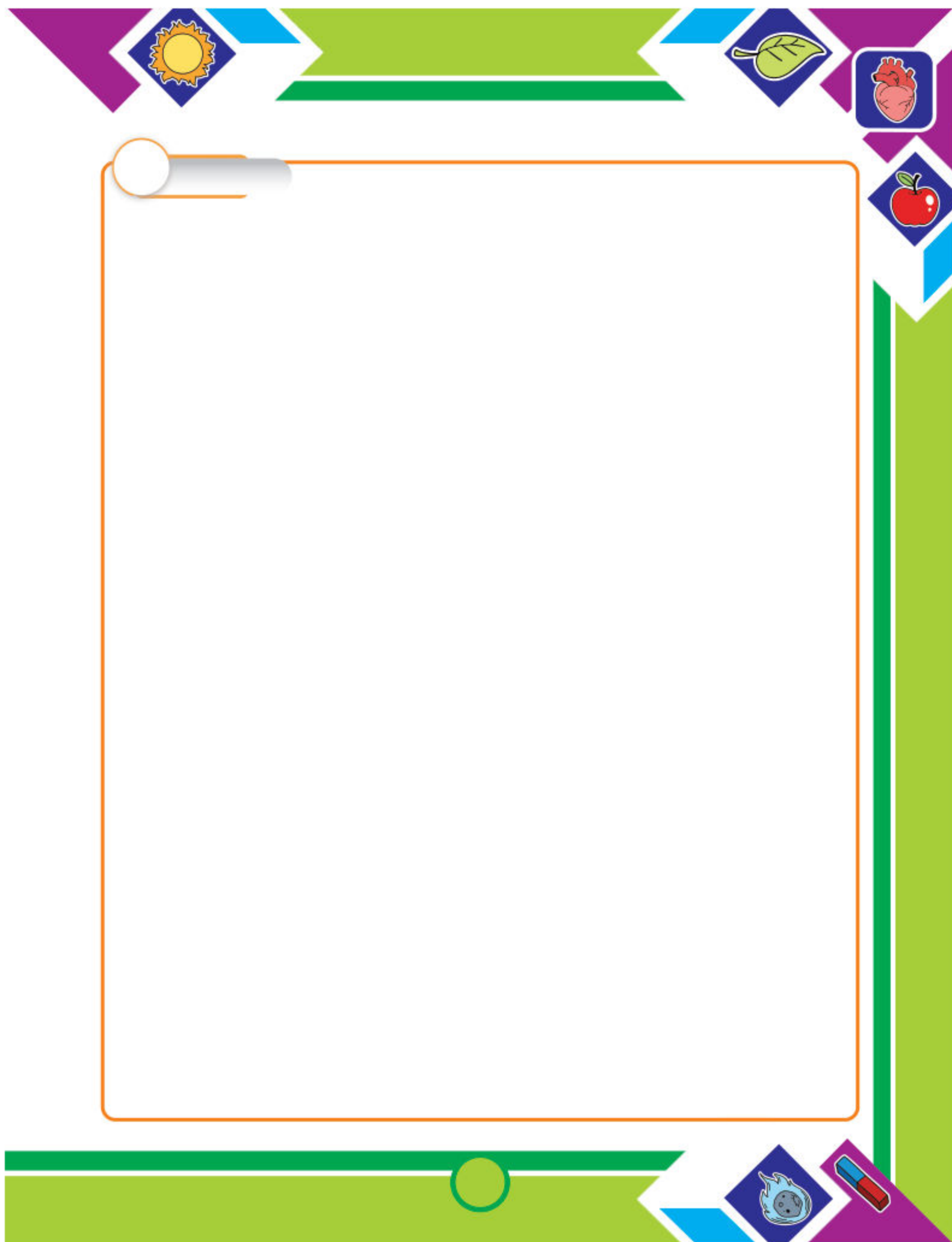
Explain what is going on in the following picture.

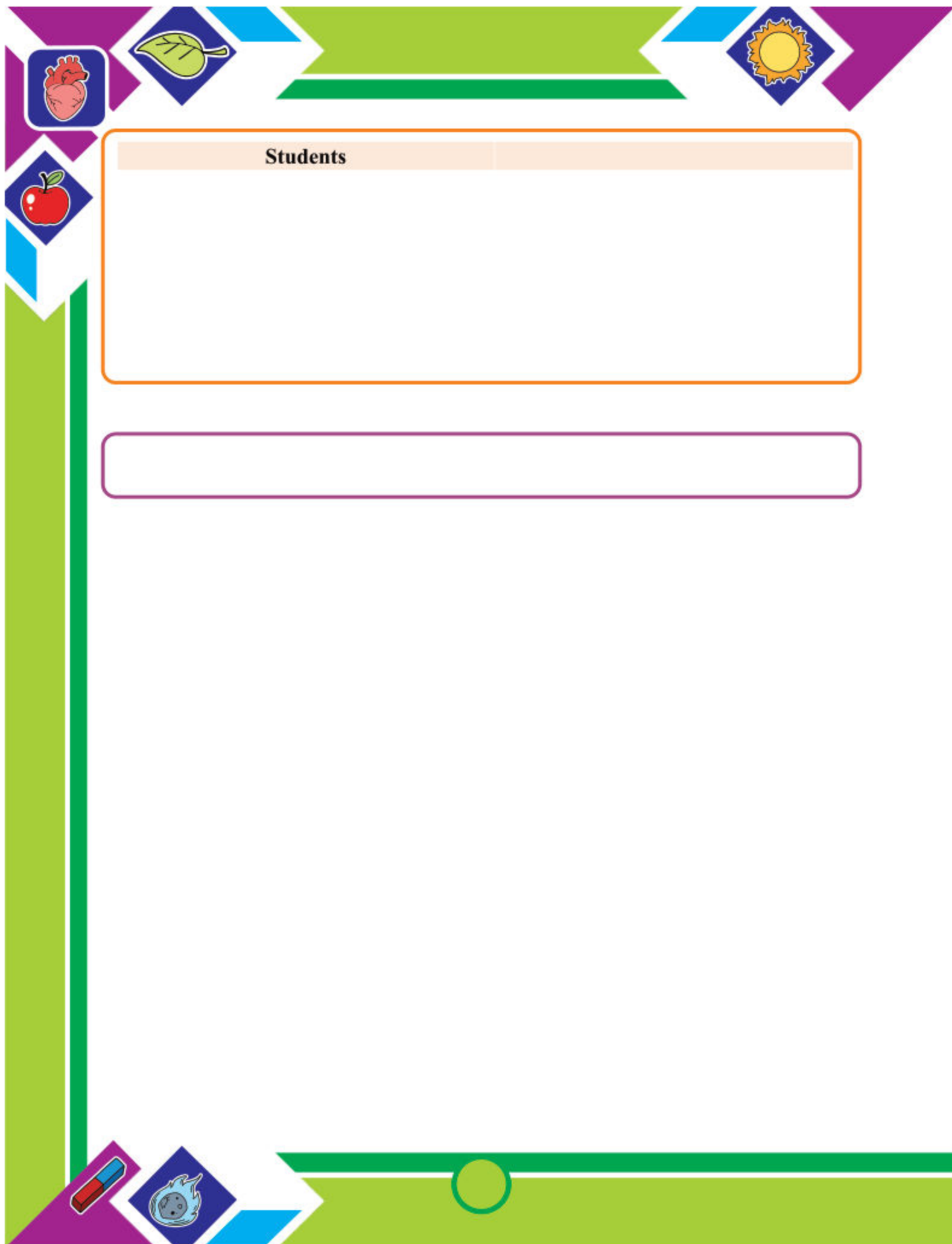


6.4 HEAT ENERGY:

Whenever you have a fever your mother or doctor takes the temperature of



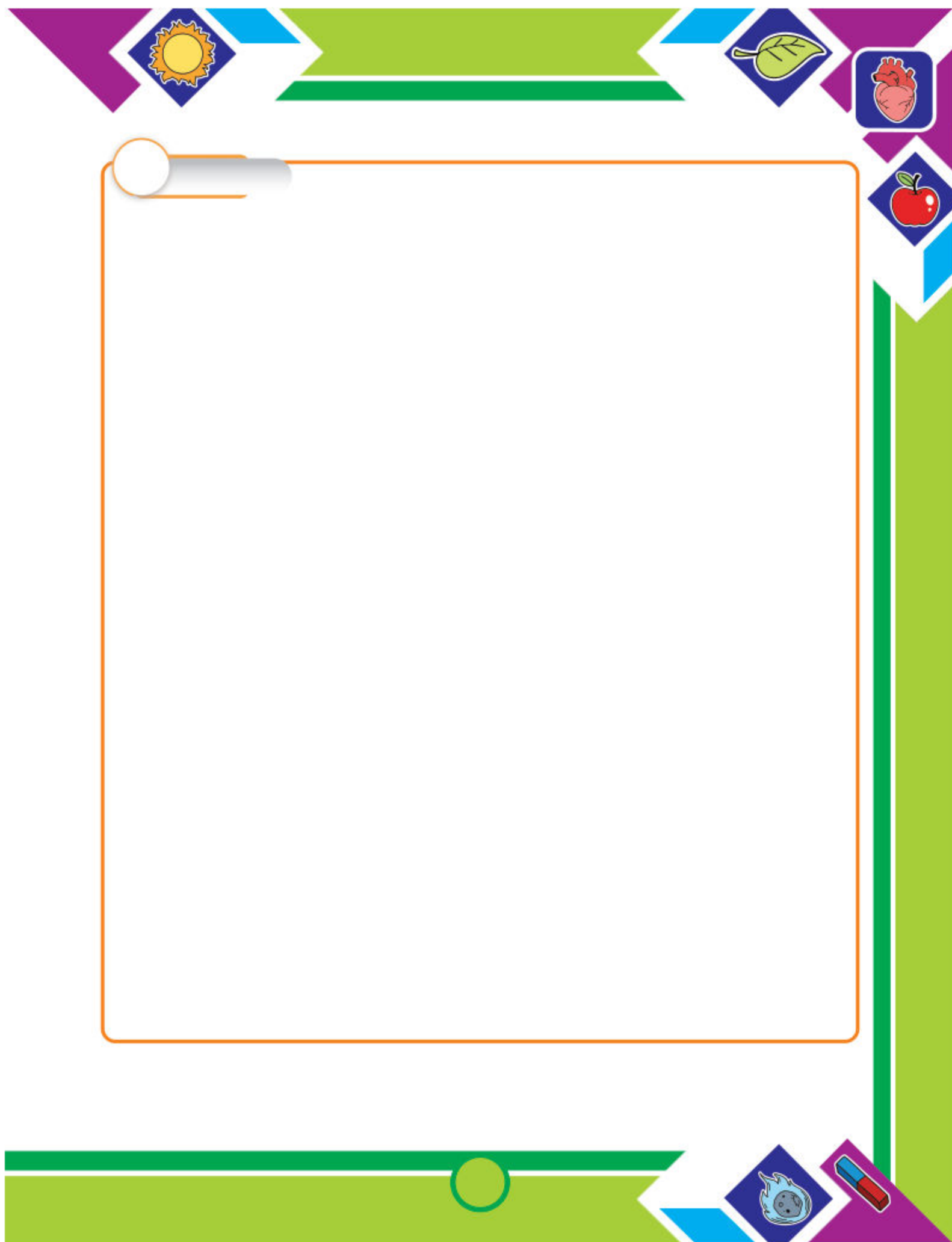


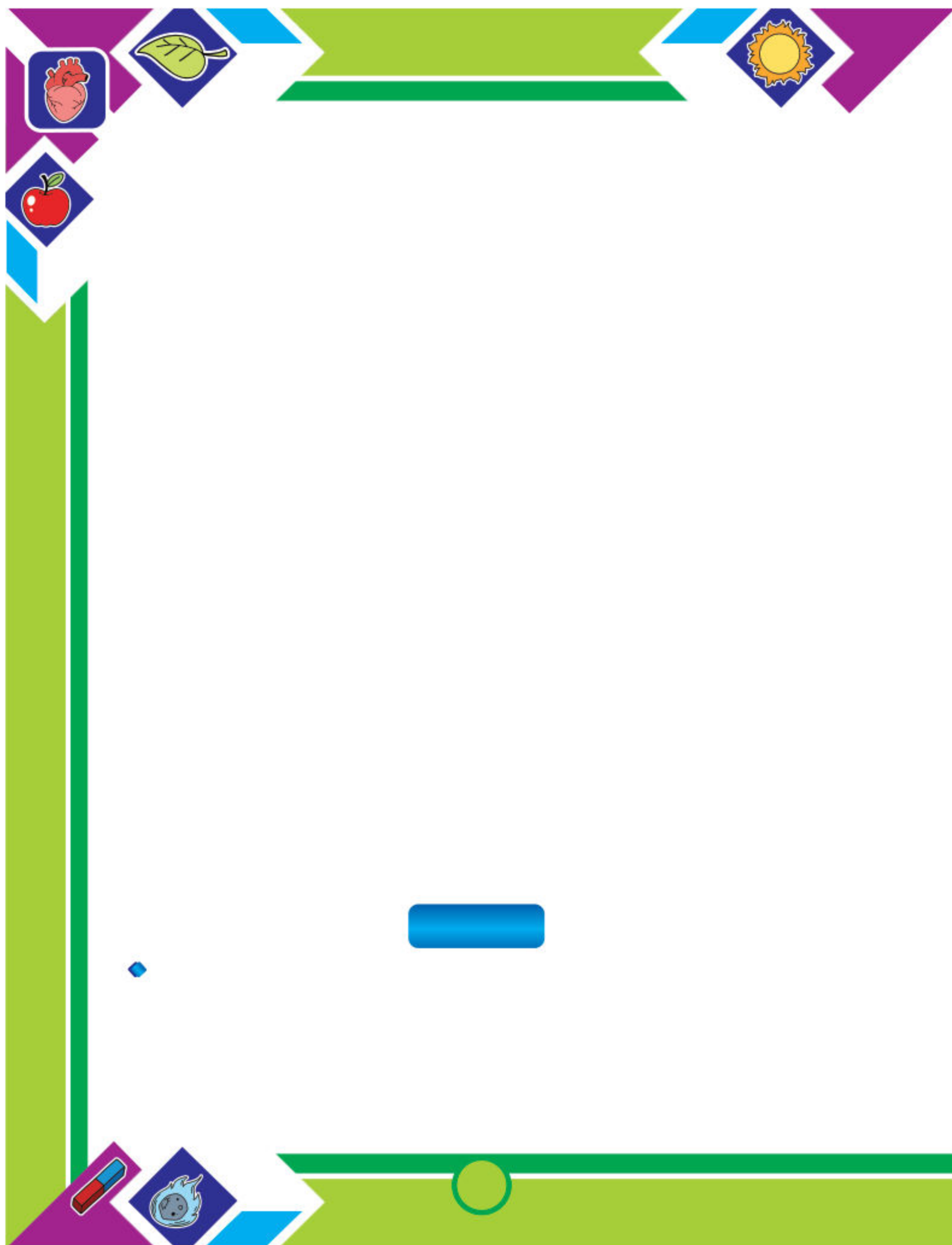


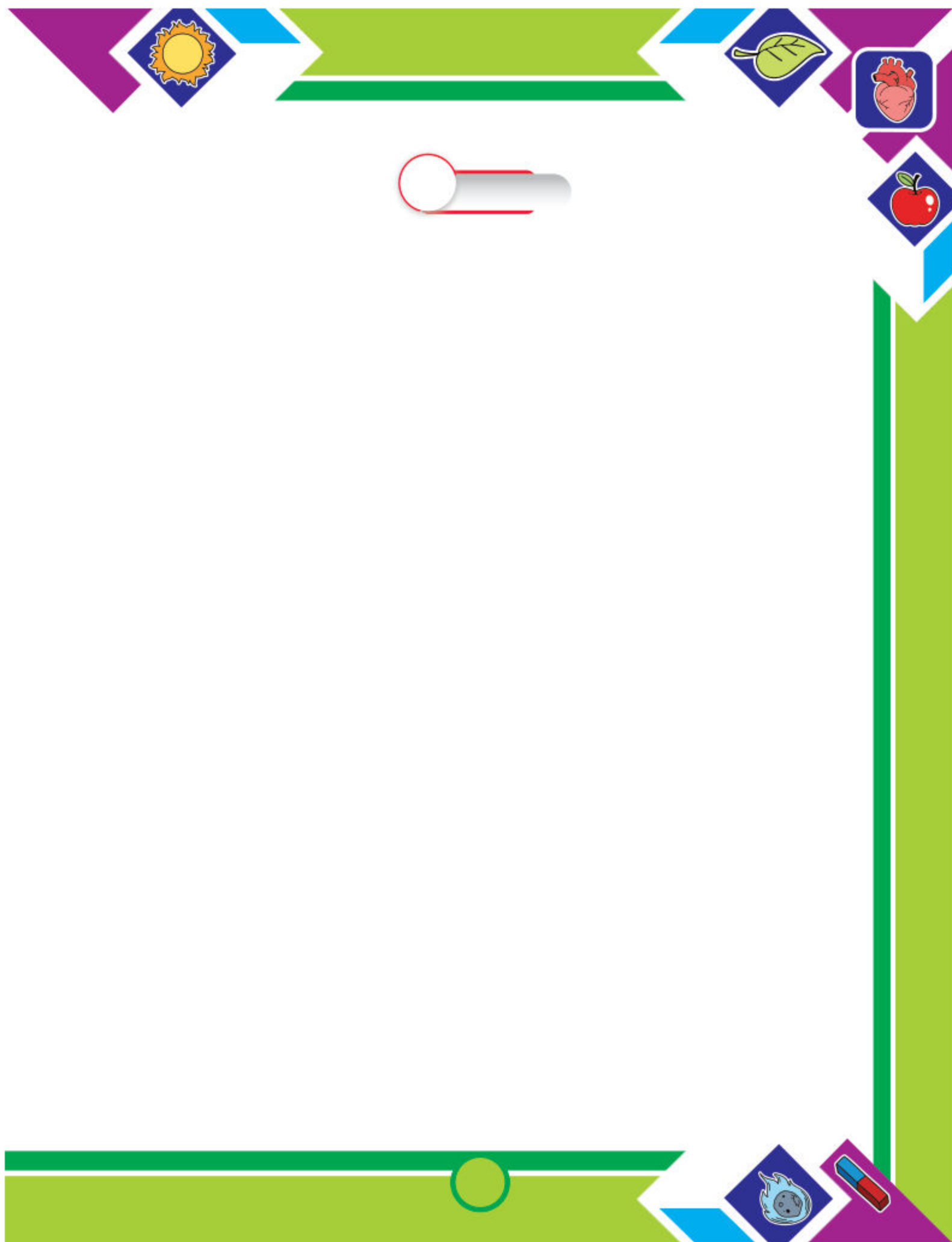
Students

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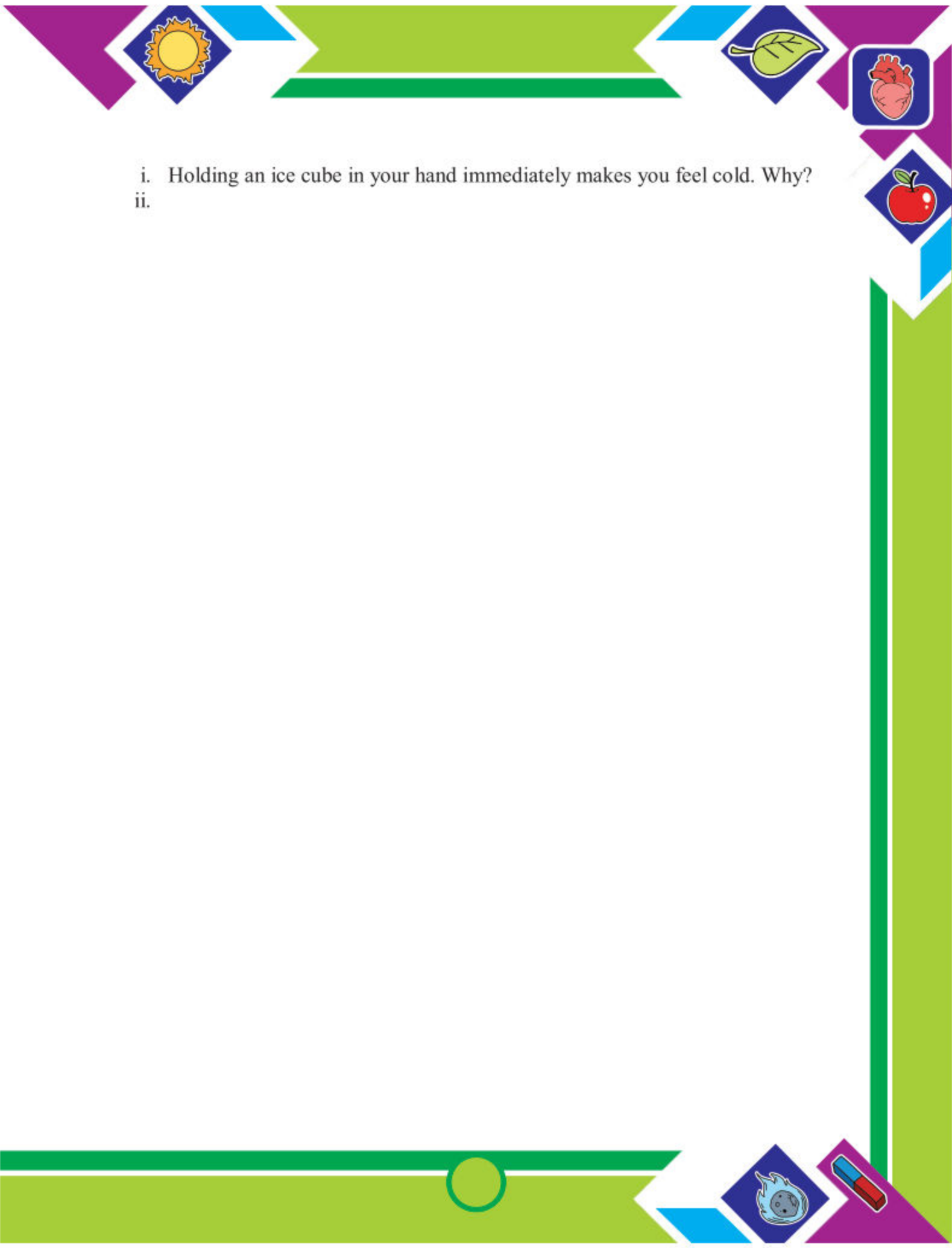






vi) The following is not a conductor of electricity:
A.



- 
- i. Holding an ice cube in your hand immediately makes you feel cold. Why?
- ii.

& KDS



)REH DQG
6LPSOH ODFKLQ

10DMRQFHSWV

,QVKLXQW HZLOOIQDE RXW

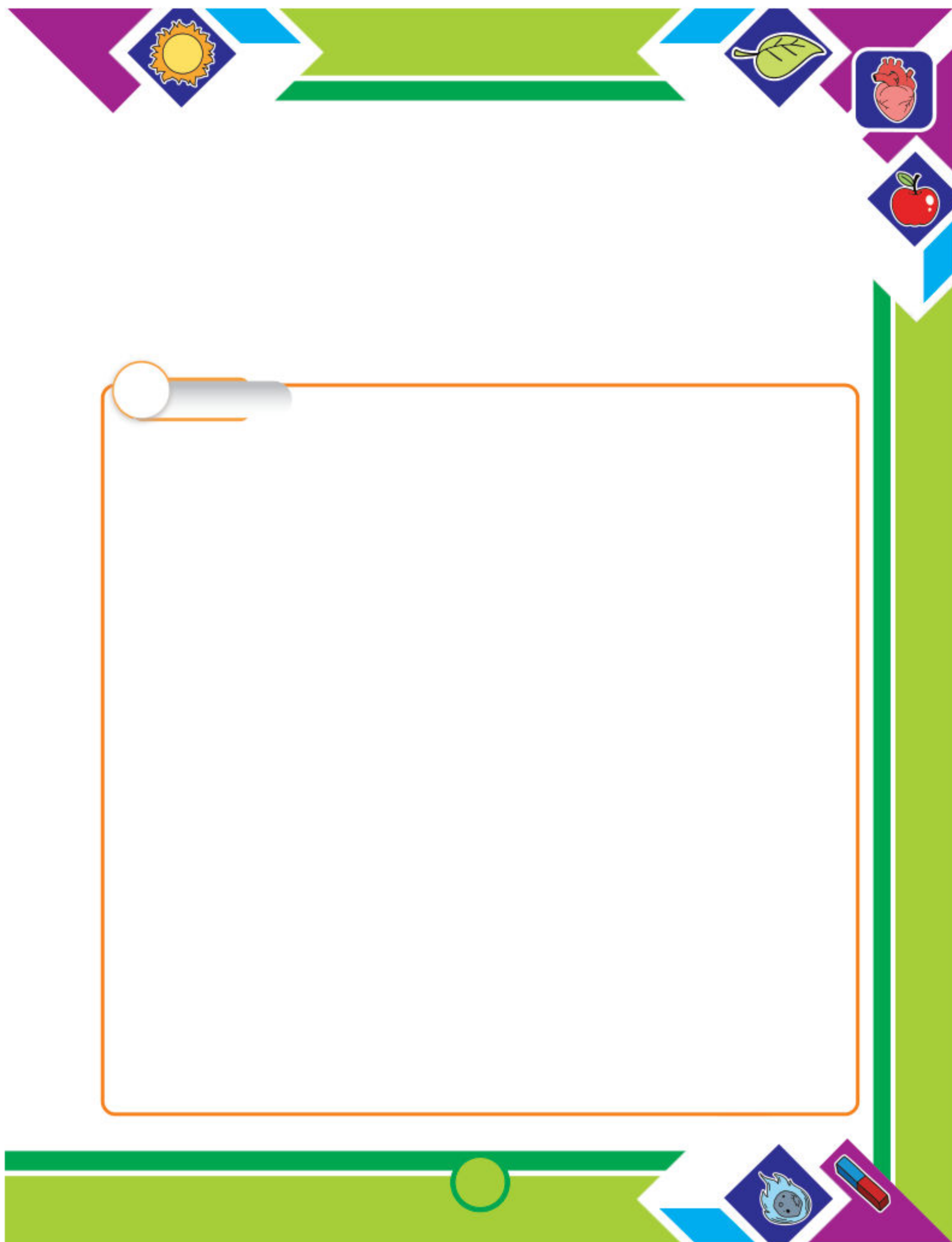
- 7.1 Force and their types
- 7.2 Multiple forces acting on an object
- 7.3 Simple Machines

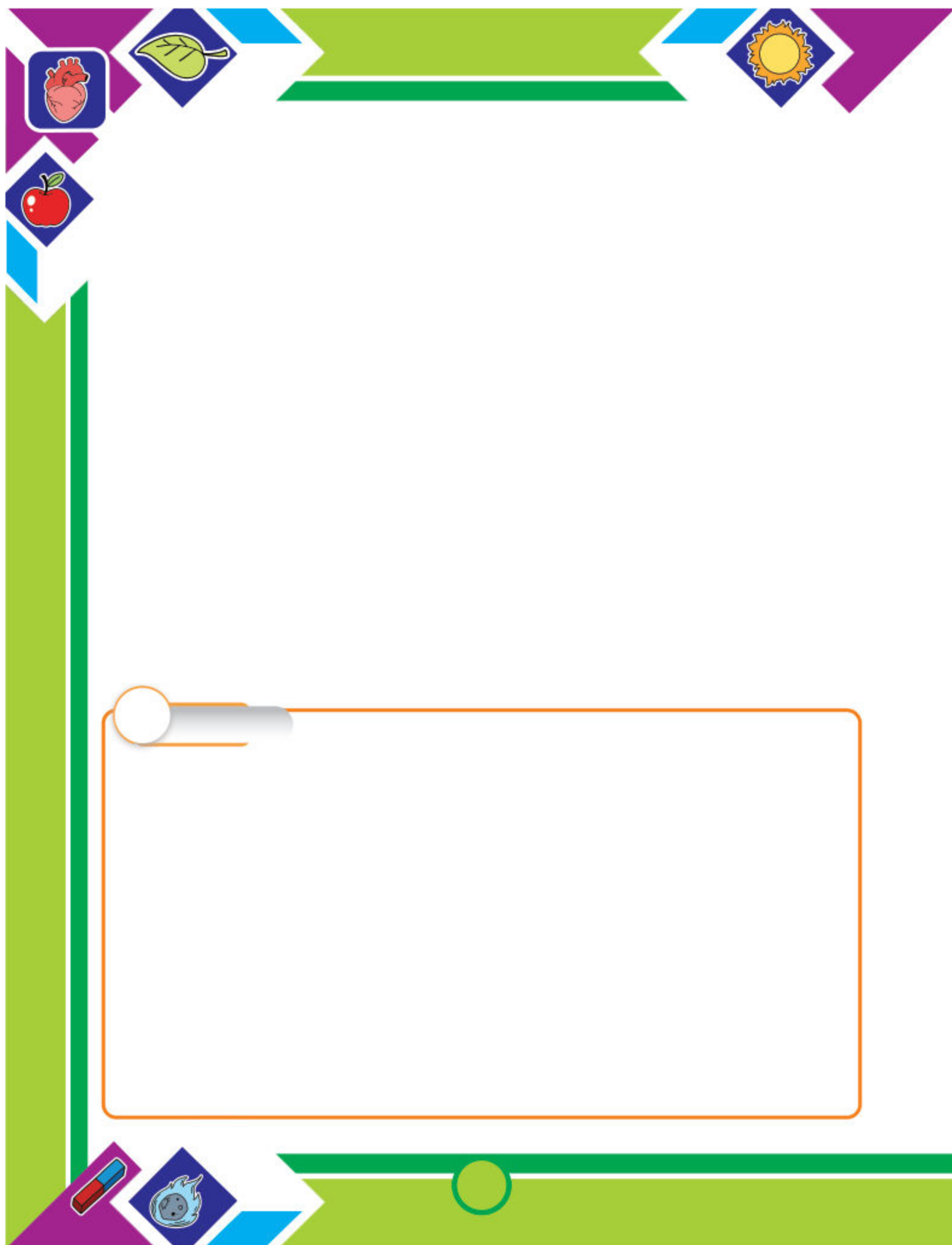
6WXGHOWH DUOLQJ 2XWFRPHV

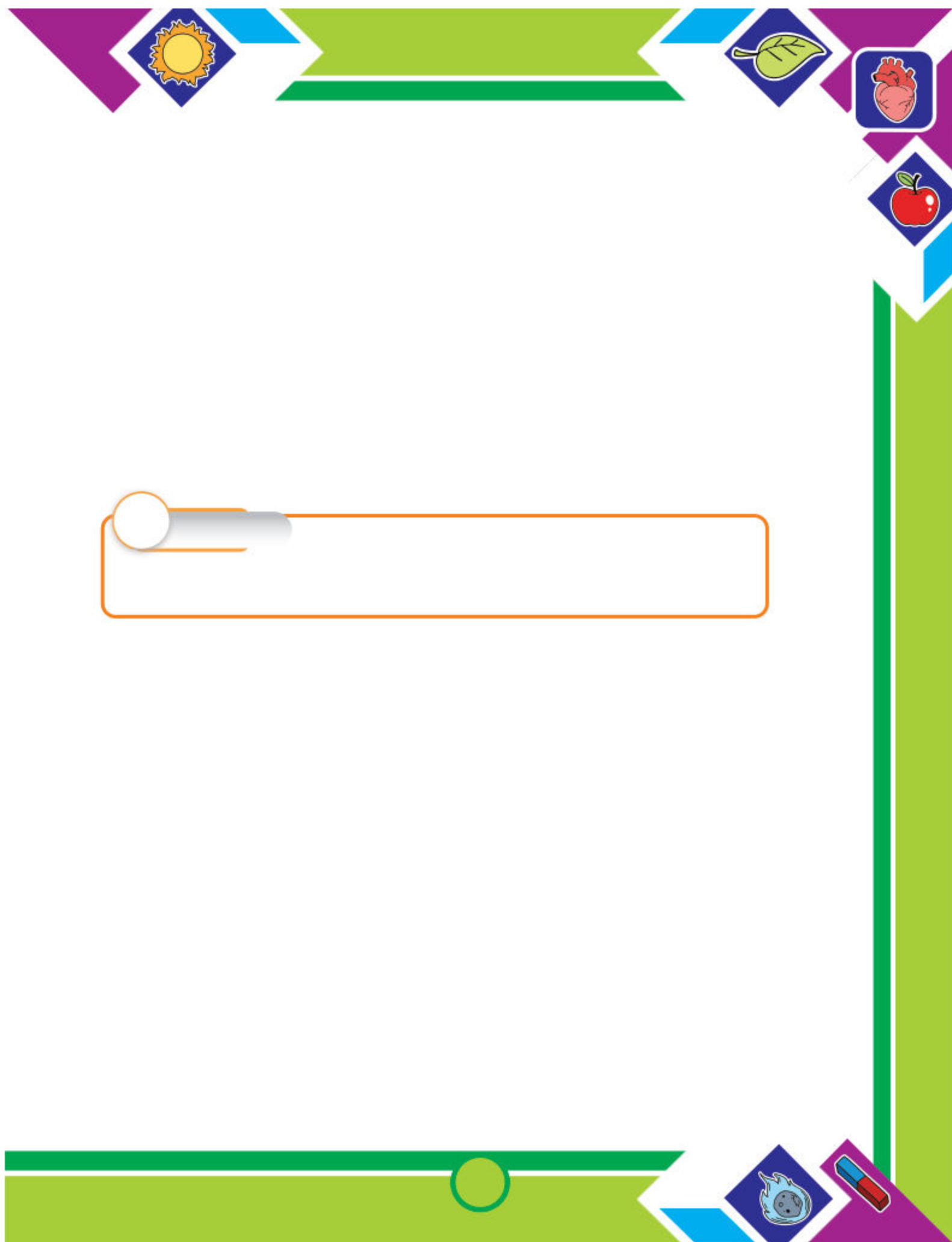
SIWUHRPSOMRQVKLXQVVWQVKR XGHDIVR

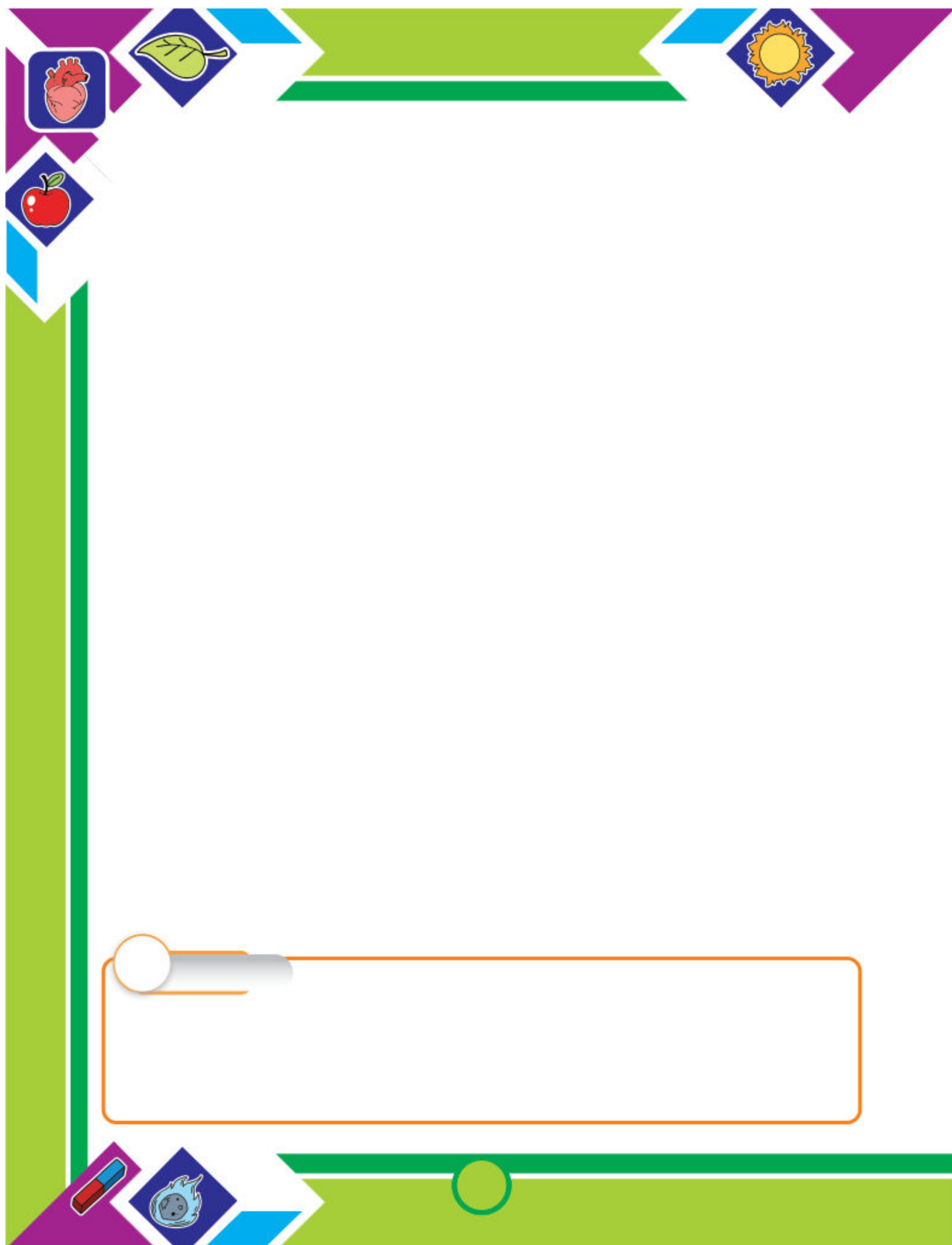
- ◆ Describe the different types of force (friction, applied, gravitational, magnetic). Investigate that friction can either be detrimental or useful under different circumstances. (Ways to reduce the friction). List uses of different types of forces in our daily life. Explore how force can move or stop objects, change directions, shape and speed. Describe that an object may have multiple forces acting on it even when at rest. (free body diagram). Compare the effect of force of different strengths in the same or opposite directions acting on an object. Design hammer, wheels, roller and gears using clay or playdough/environment-friendly material (STEAM).











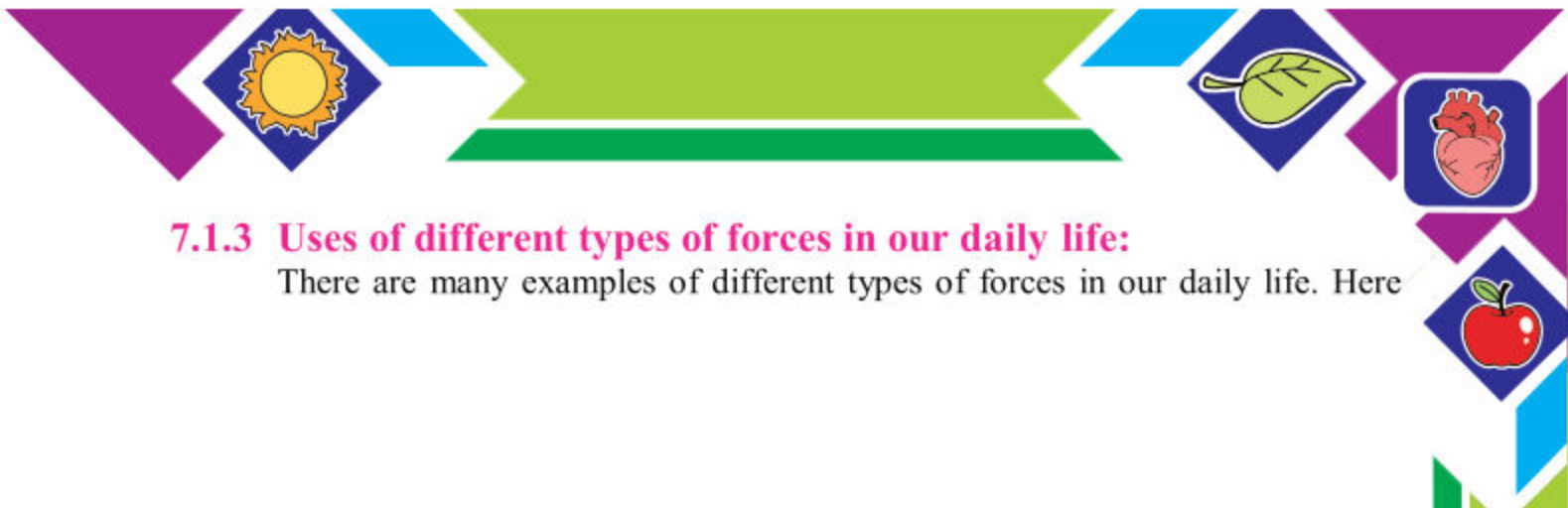


7.1.2 Ways to reduce the friction:



1. Sprinkling of powder on the

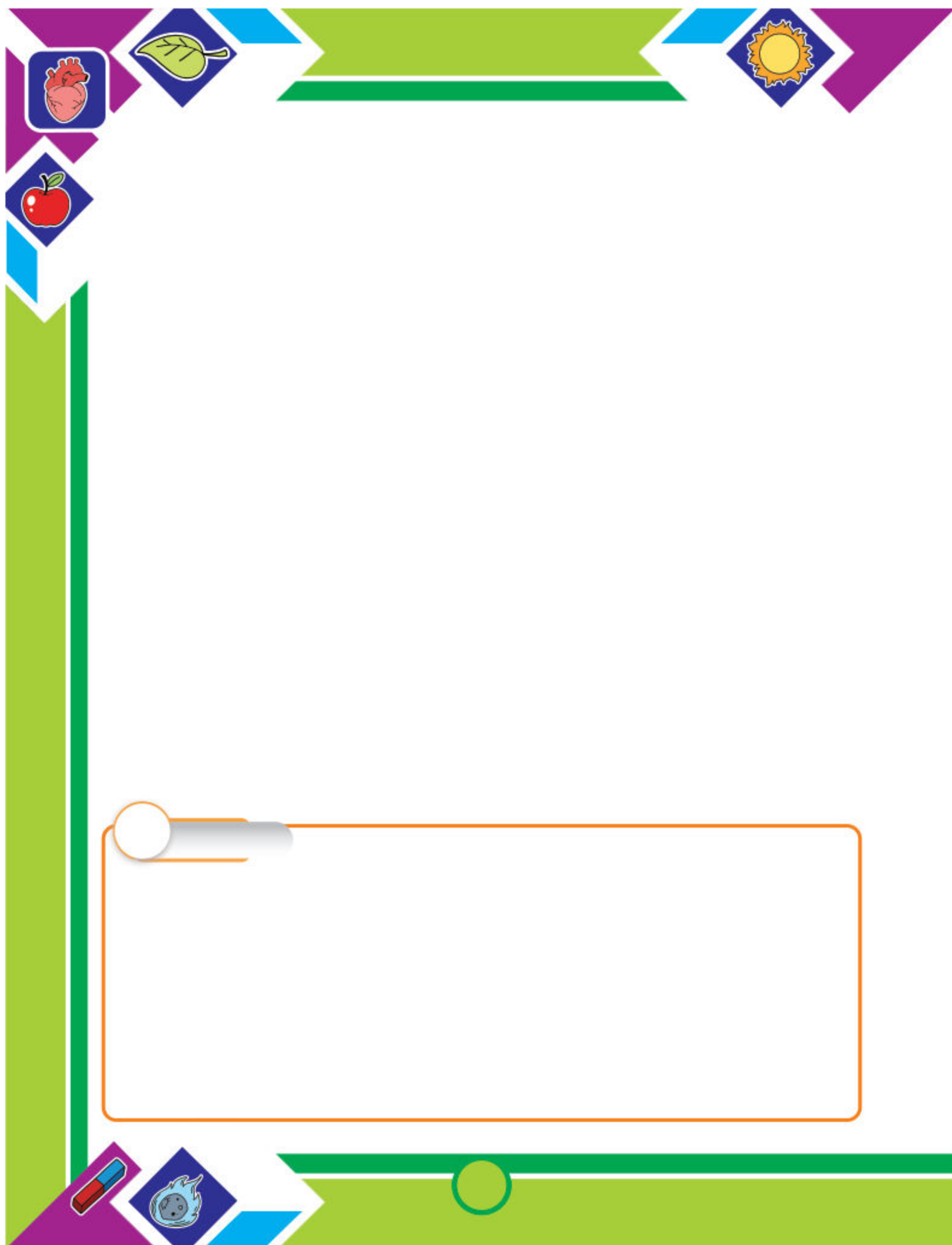


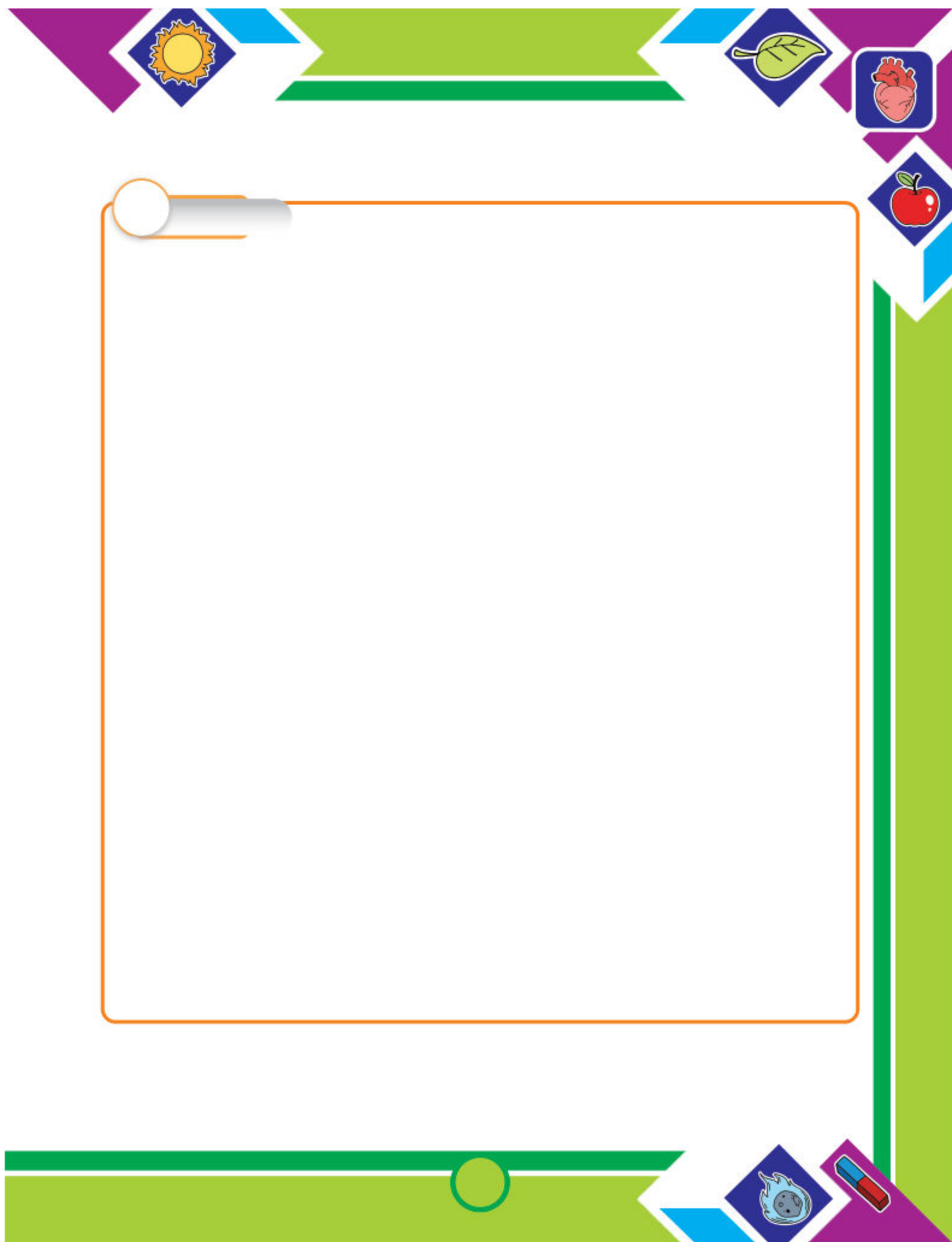


7.1.3 Uses of different types of forces in our daily life:

There are many examples of different types of forces in our daily life. Here









A machine can perform the following tasks:



- (1) **Lever:**
A lever is a simple machine that helps us

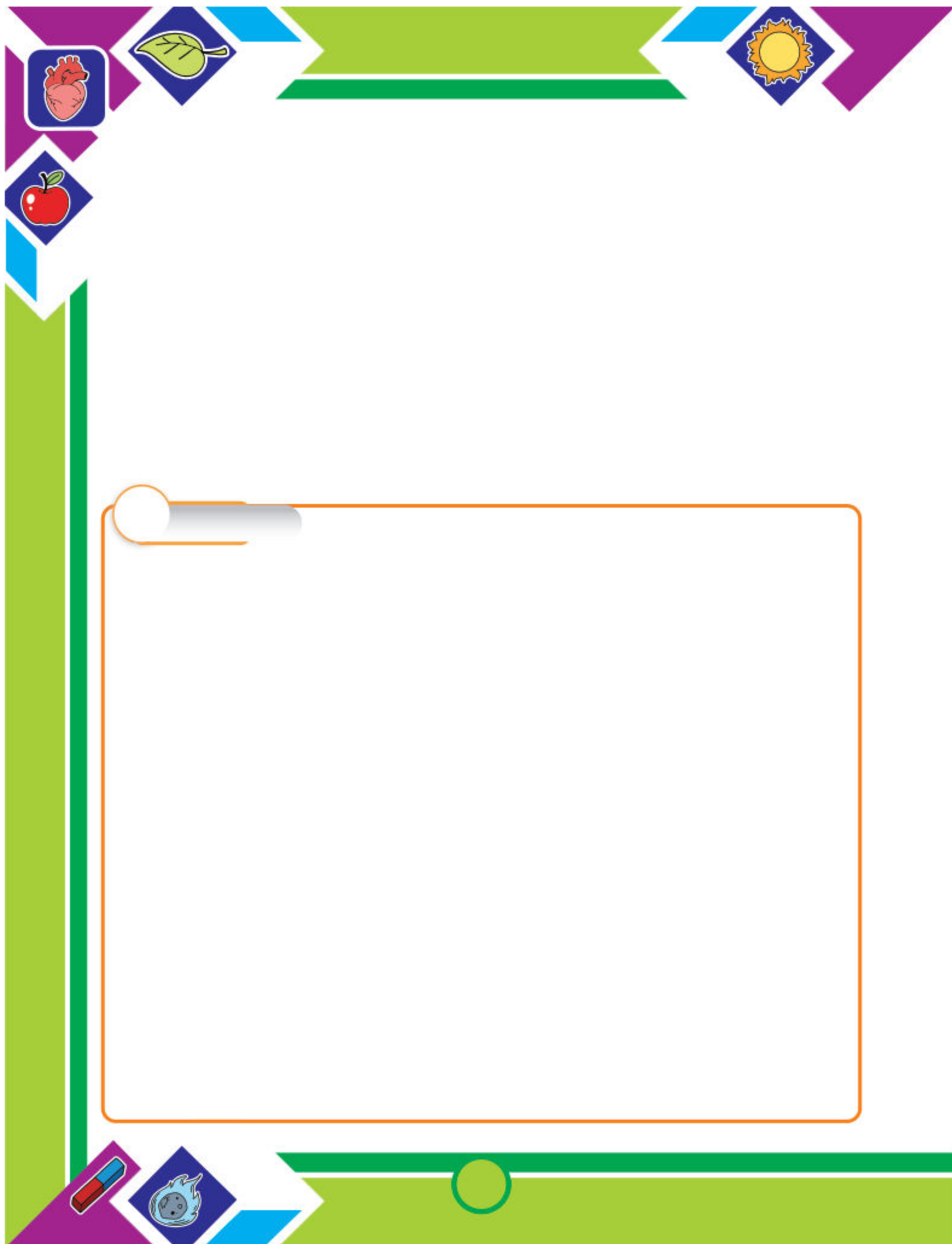


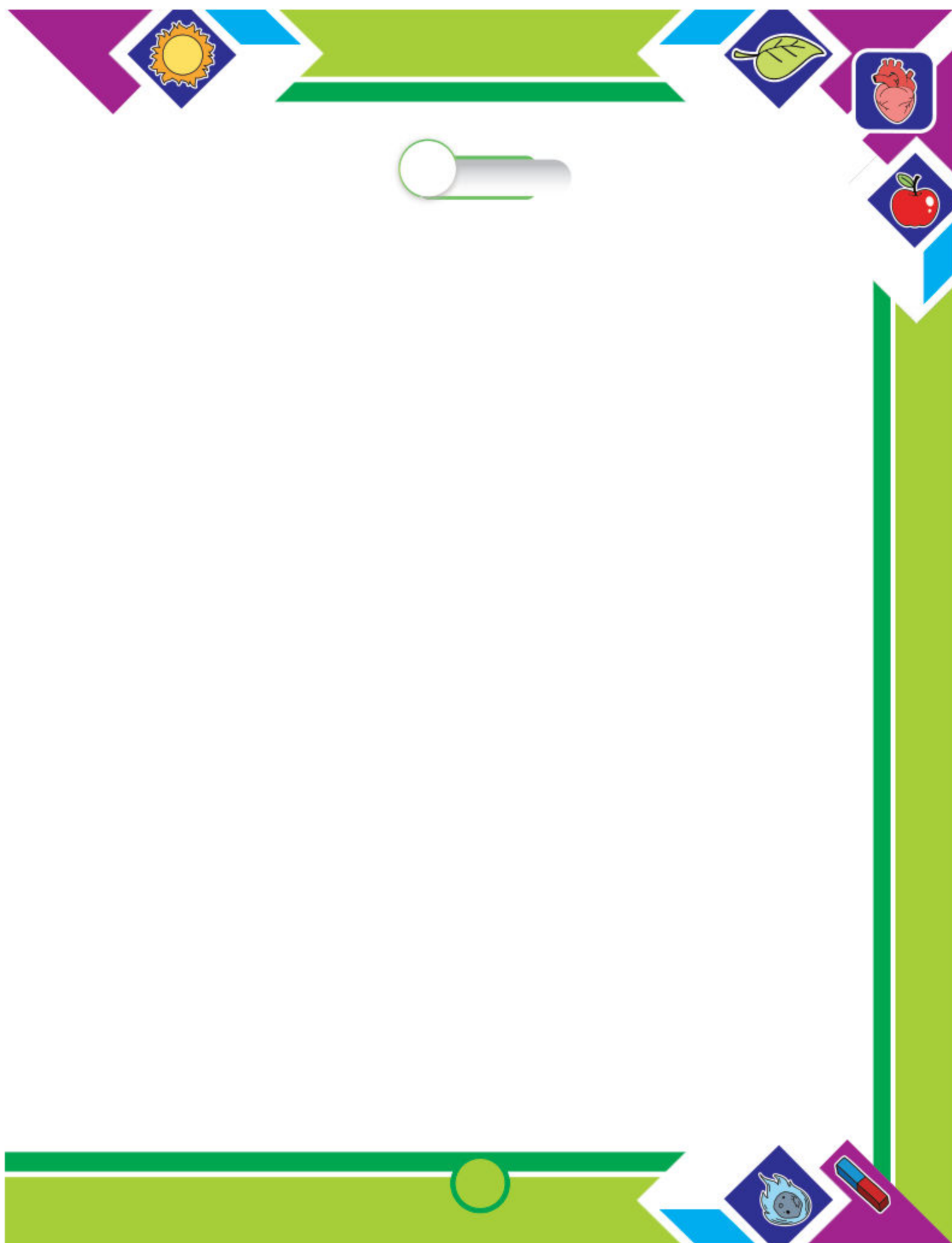
(2) Pulley:

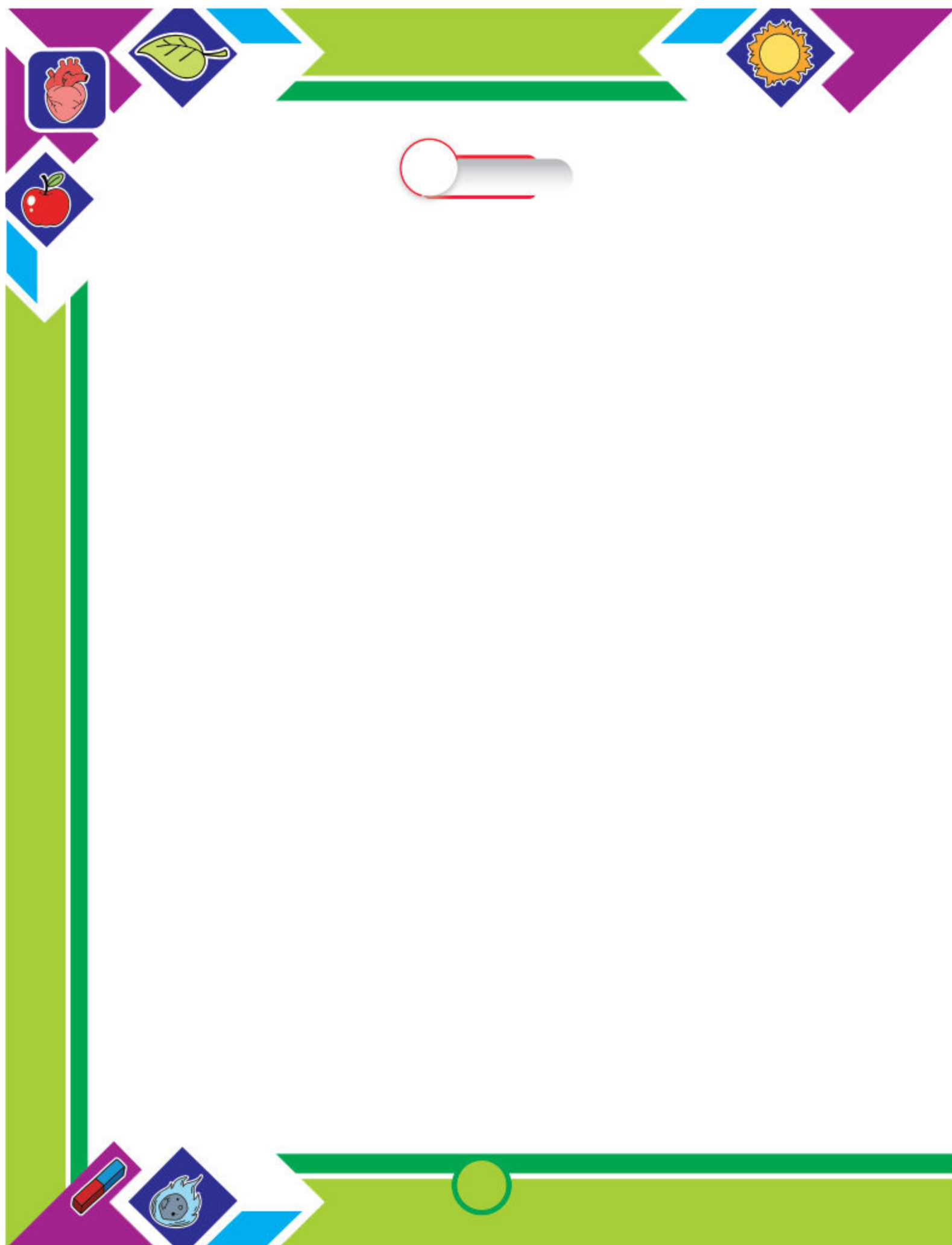
Water drawn from well

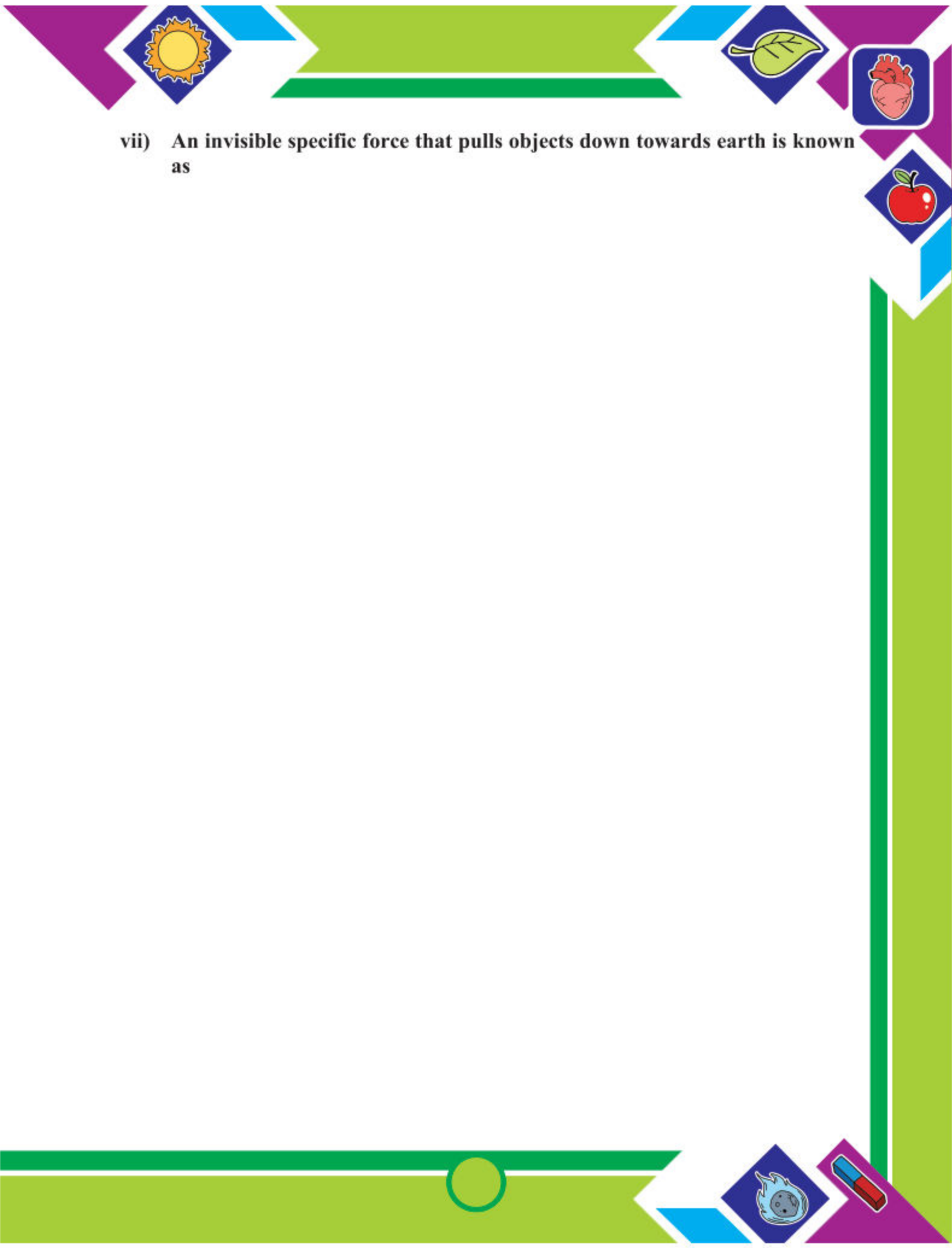
Crane for heavy load

(3)

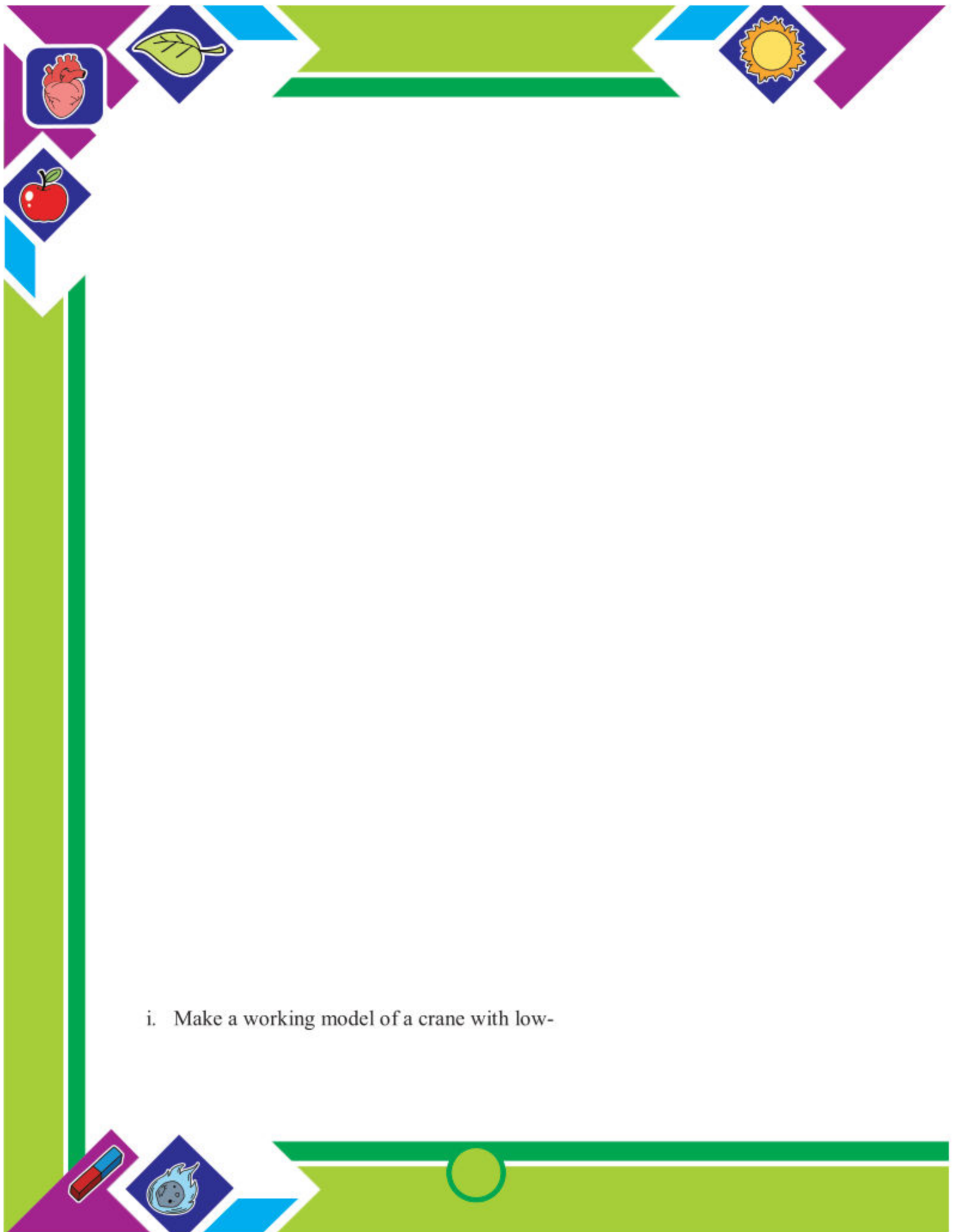








vii) An invisible specific force that pulls objects down towards earth is known as



- i. Make a working model of a crane with low-

& KDS



7HFKQRORJ\ LQ
(YHUGD\ /LIH

10DMRQFHSWV

,QVK LXQW HZ LOOIQDERXW

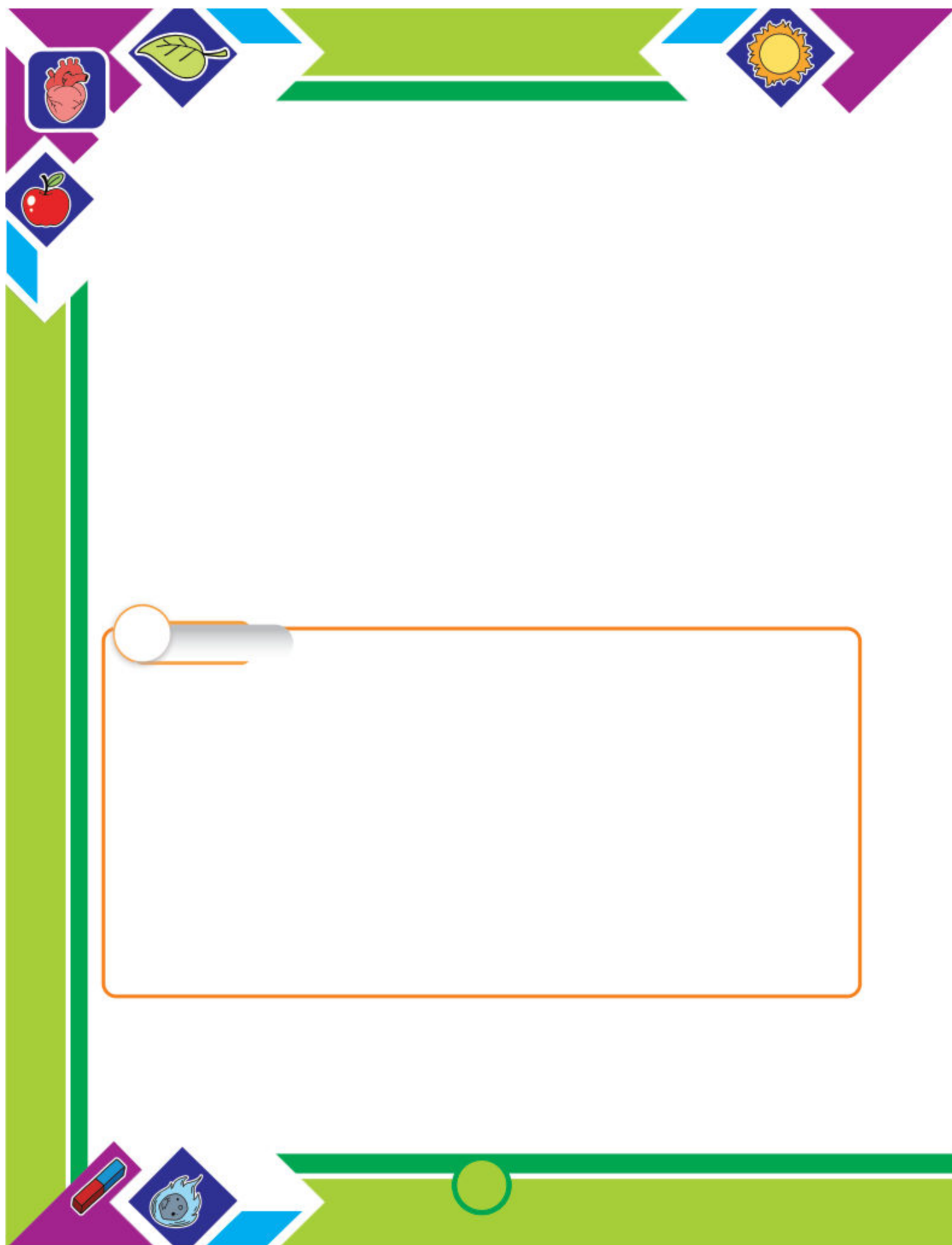
8.1 8VH RI VFLHQWL F LQVWUXPHQWV

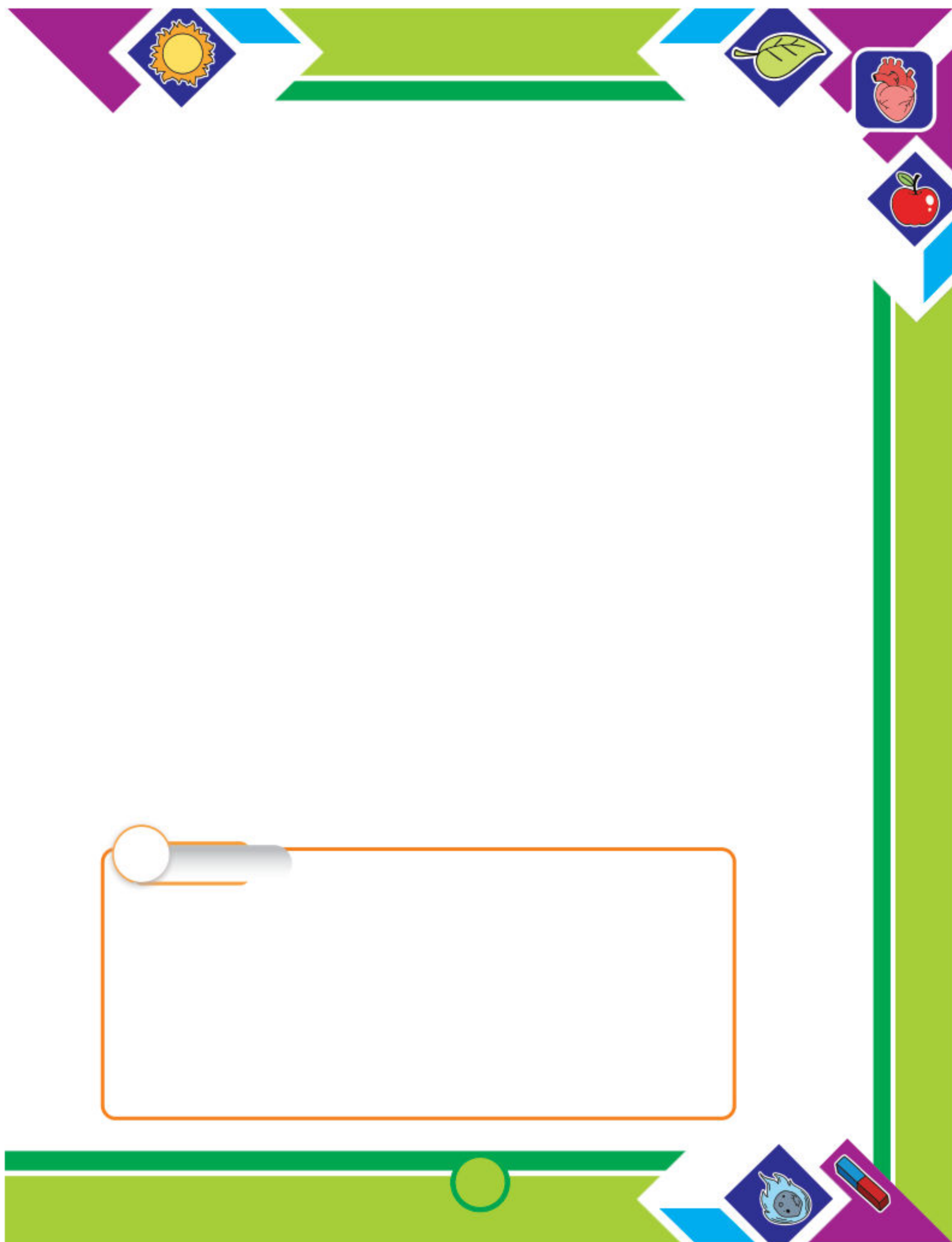
6WXGHQW/HDUQLQJ 2XWFRPHV

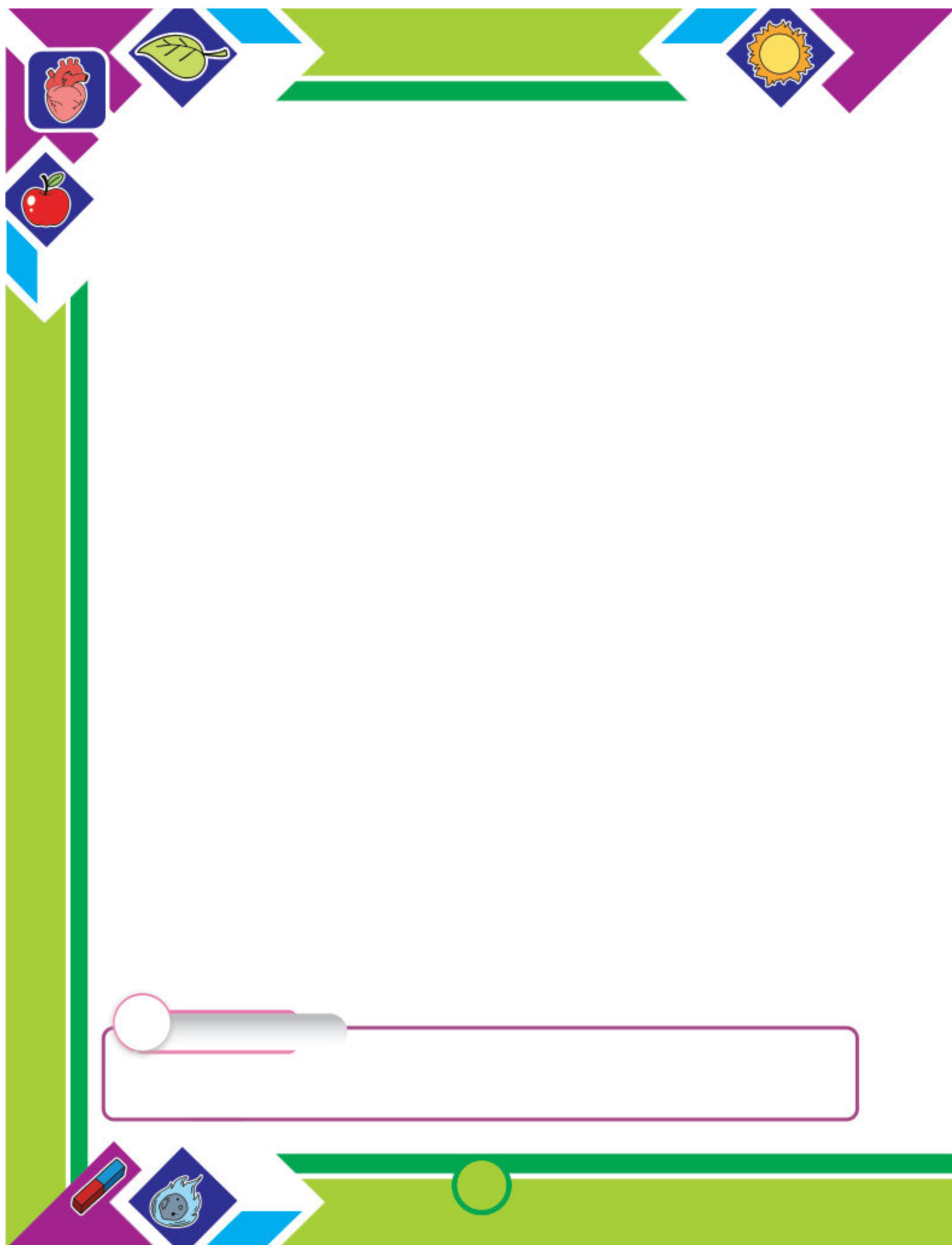
\$IWUHRPSOMRQVK LXQVV WQVKR XCHD BIVR

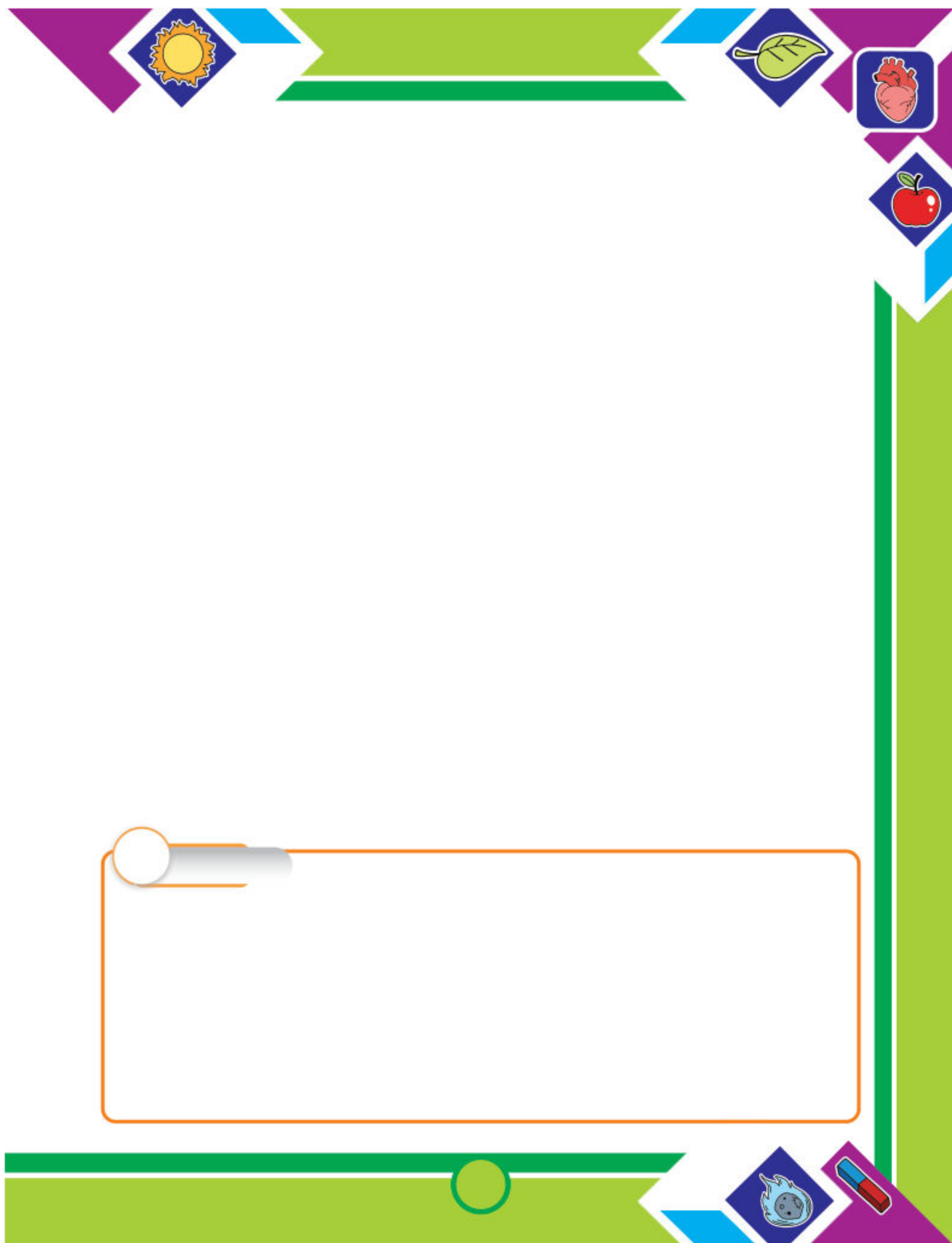
- ◆ Use scientific instruments/apparatus in everyday life (e.g. thermometer, blood pressure apparatus, digital balance, stopwatch, calculator, available digital device).
- ◆ Use a plumb line to install a flagpole vertically.
- ◆ Design models of sphere, cube, prism, cylinder and cone with clay or playdough/ environment friendly materials (STEAM)







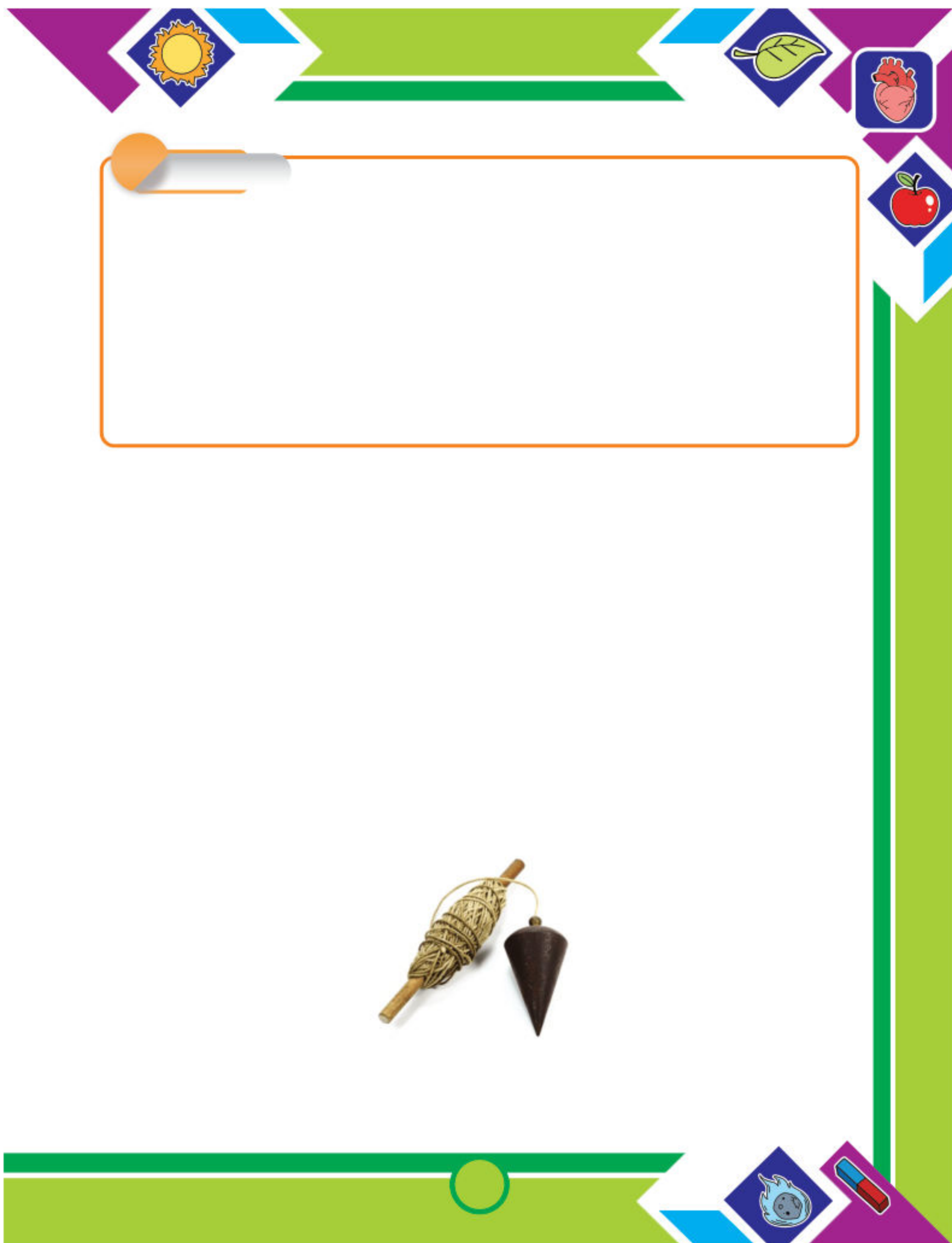


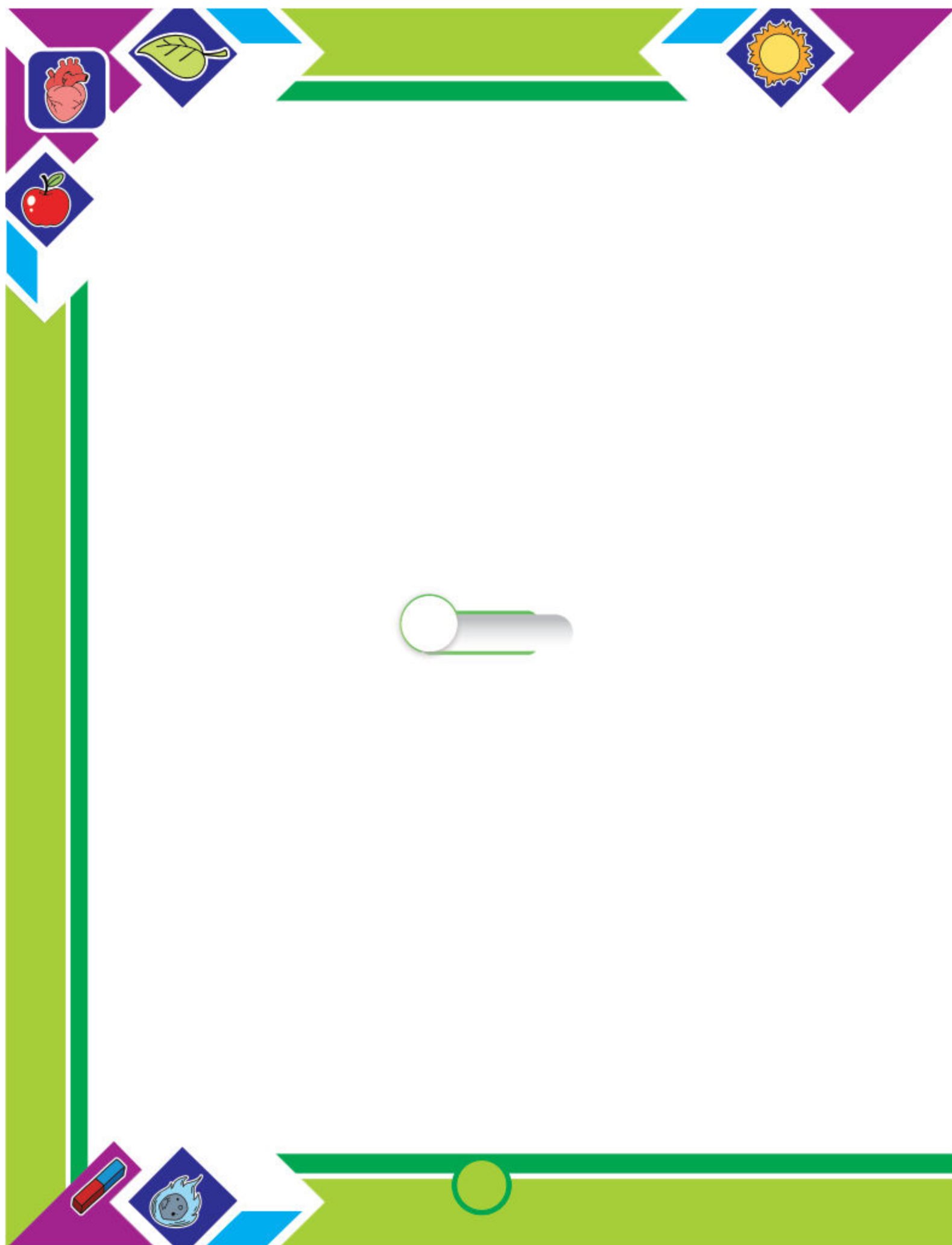


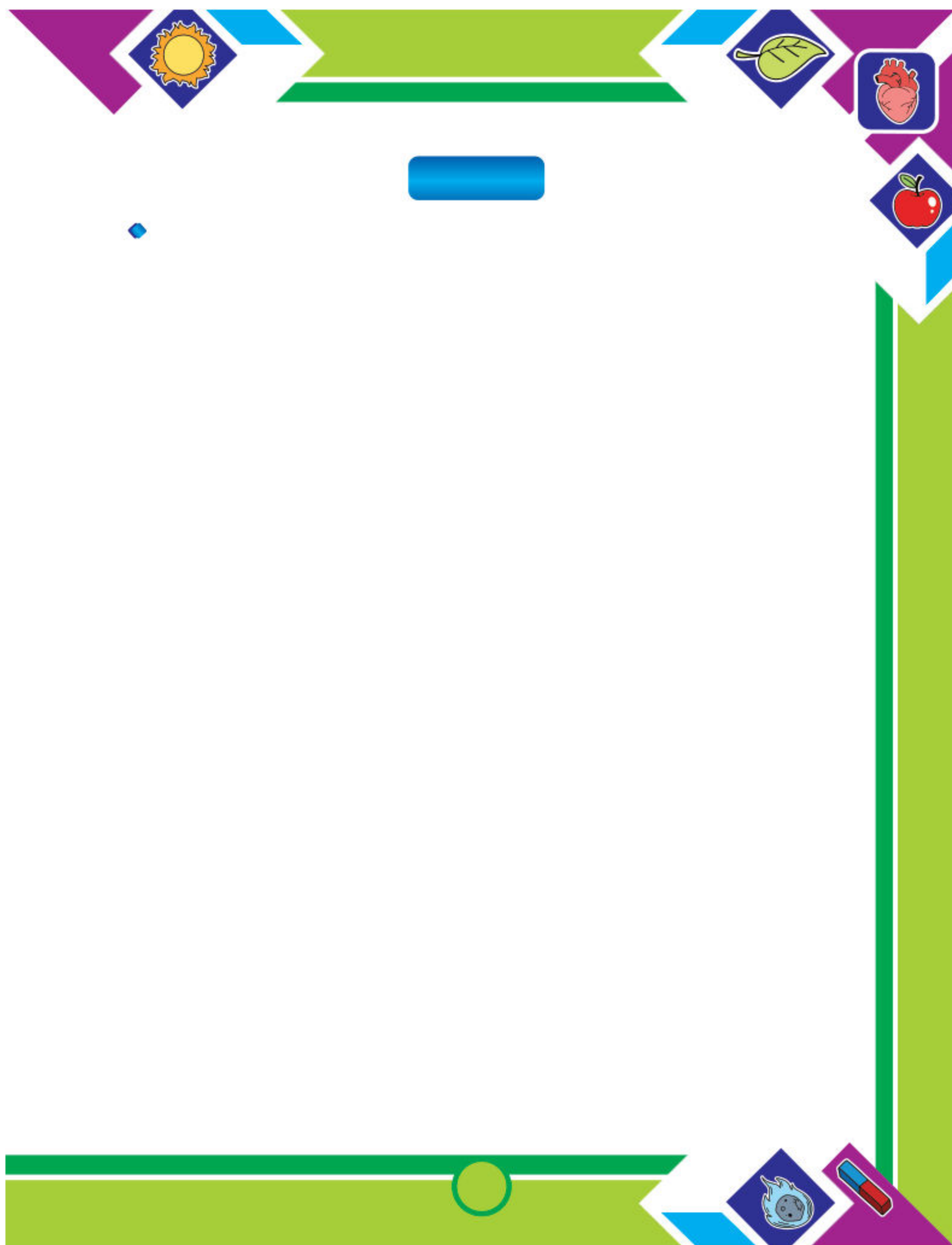


8. If multiple measurements are needed, remove the object from the pan and clean the pan before weighing the next time.











v) **It can measure fast and accurate measurement of mass:**

A. stopwatch

B. t



i.

i. Make a plumb line with the help of no-cost & low-cost materials.

& KDS



(DWK DQG LWV
5HVREHV

10DMRQRQH SWV

, QVK LXQW HZ LOO IQD ERXW

9.1 Composition of earth and atmosphere

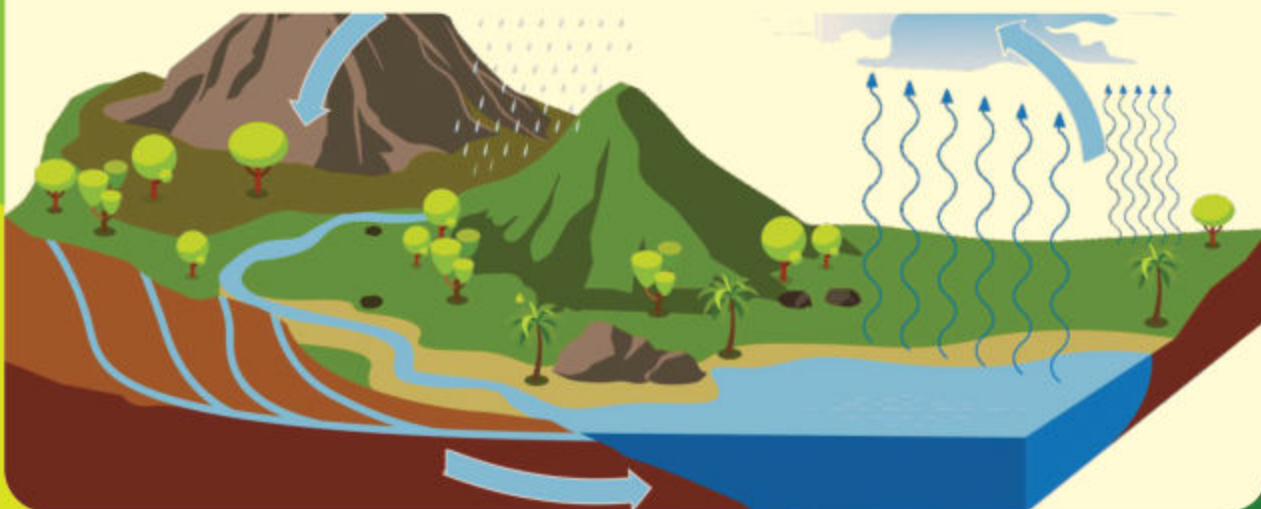
9.2 Water Cycle

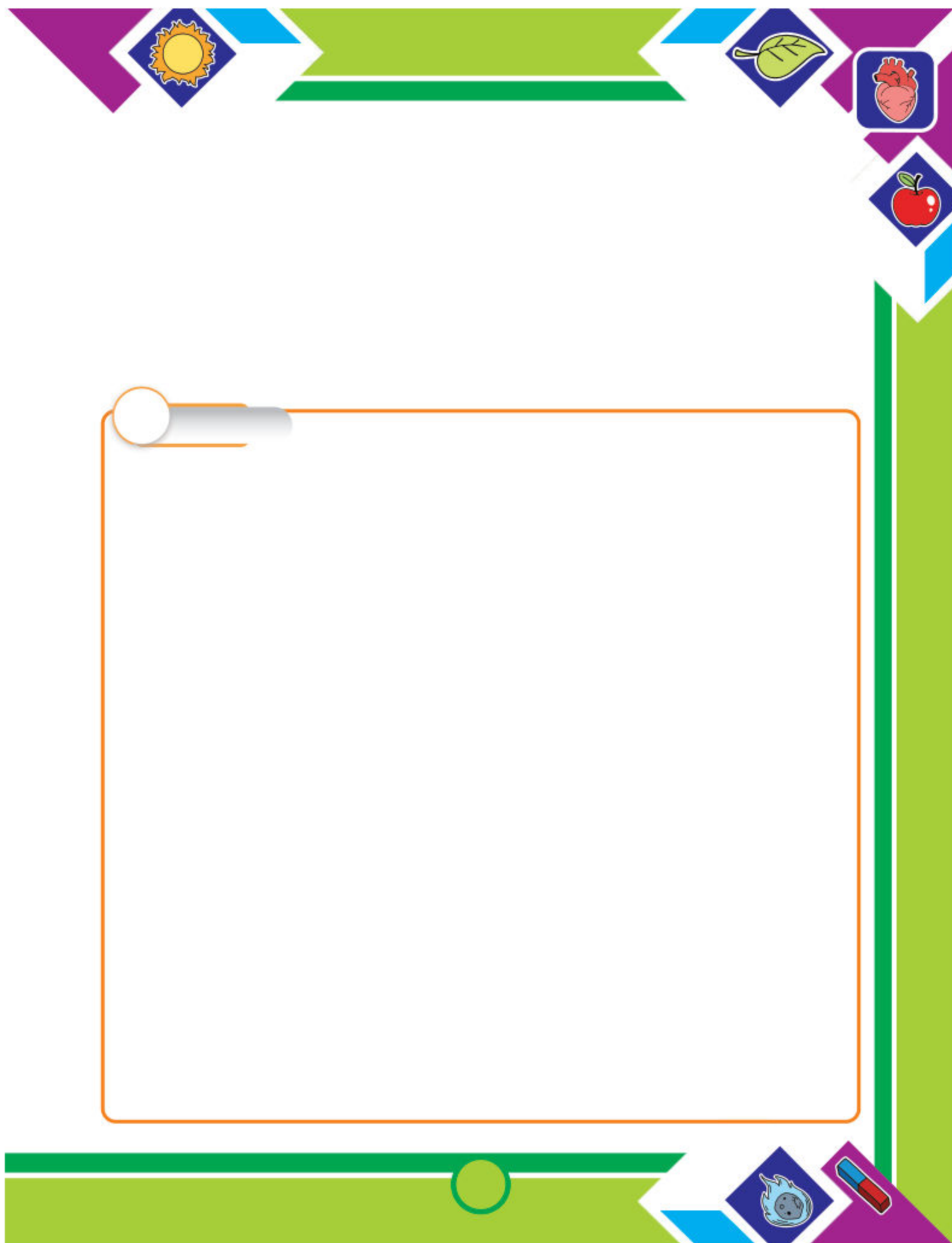
9.3 Types of water

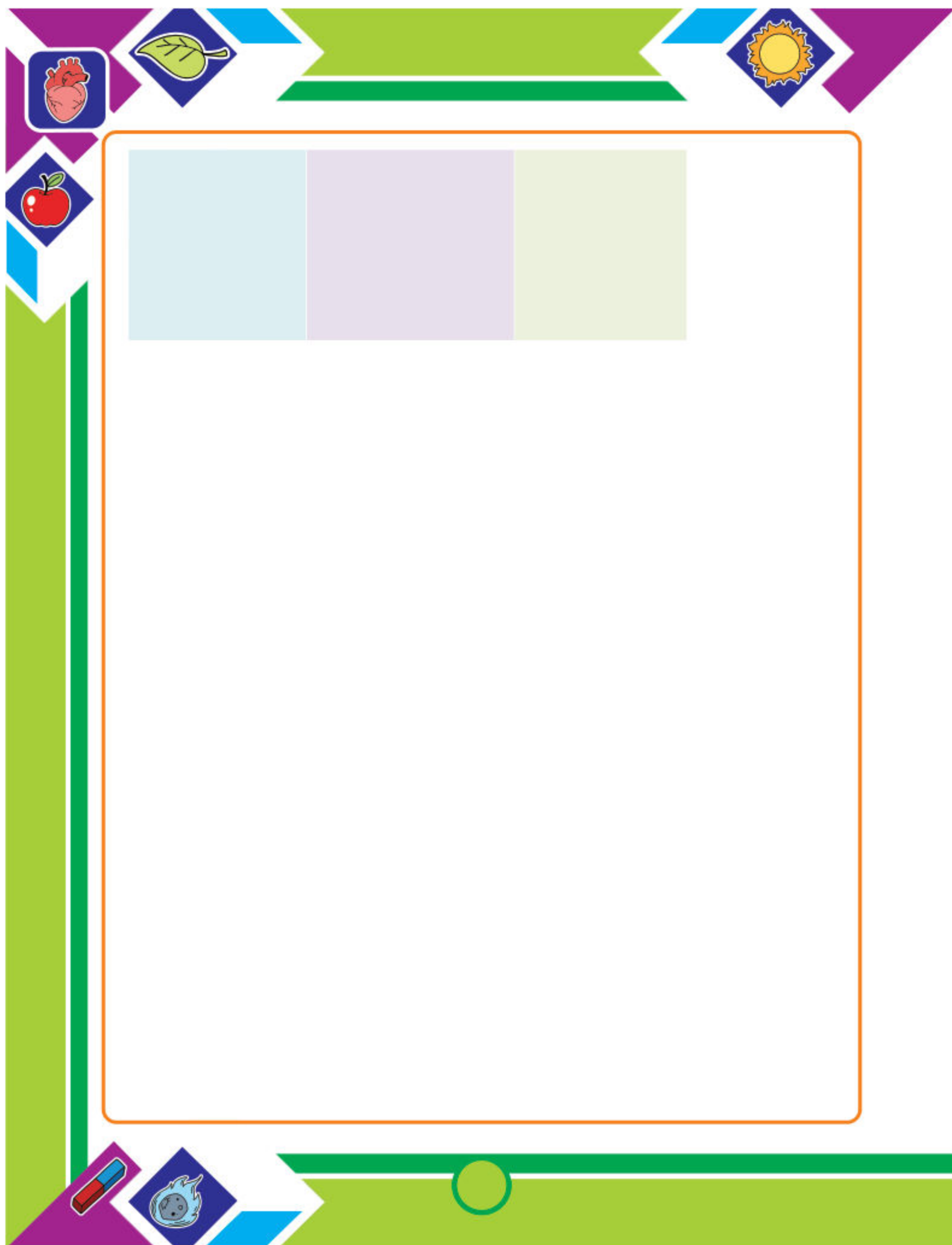
6WXGHQW/HDUQLQJ 2XWFRPHV

SWUHRPSOMRQVK LXQVV WHQVKR XGHQD DQV

- ◆ Define natural resources.
- ◆ Recognize that the Earth's surface is made up of land and water and is surrounded by a layer of air called the atmosphere, which is a mixture of different gases (Nitrogen, carbon dioxide and oxygen).
- ◆ Describe the source of water on earth.
- ◆ Apply knowledge of changes of state of water to common weather events (e.g. cloud formation, dew formation, evaporation of puddles, snow, and rain) and understand the water cycle.



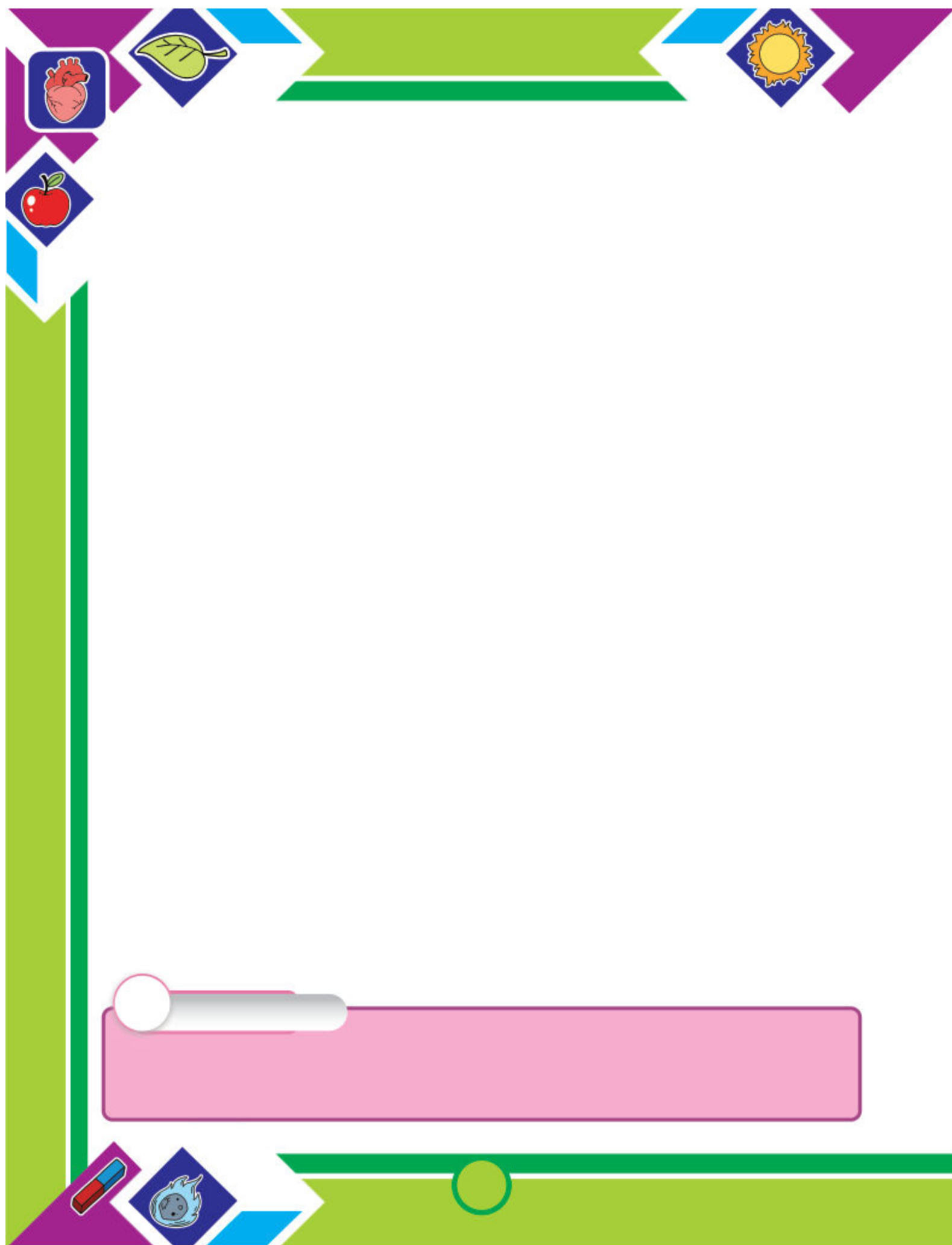


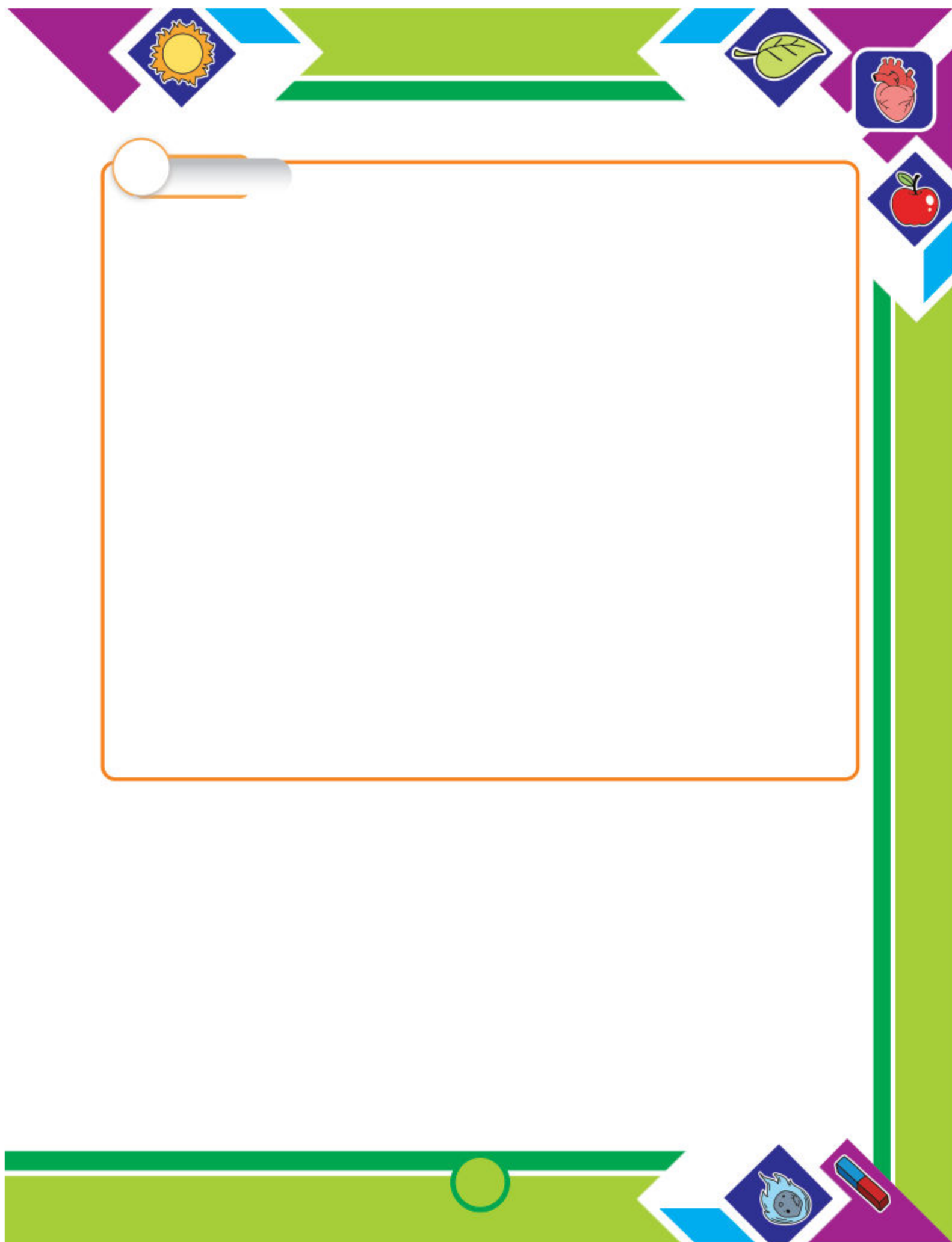


9.1 COMPOSITION OF EARTH AND ATMOSPHERE

9.1.1 Composition of Earth:

Land:







Condensation:

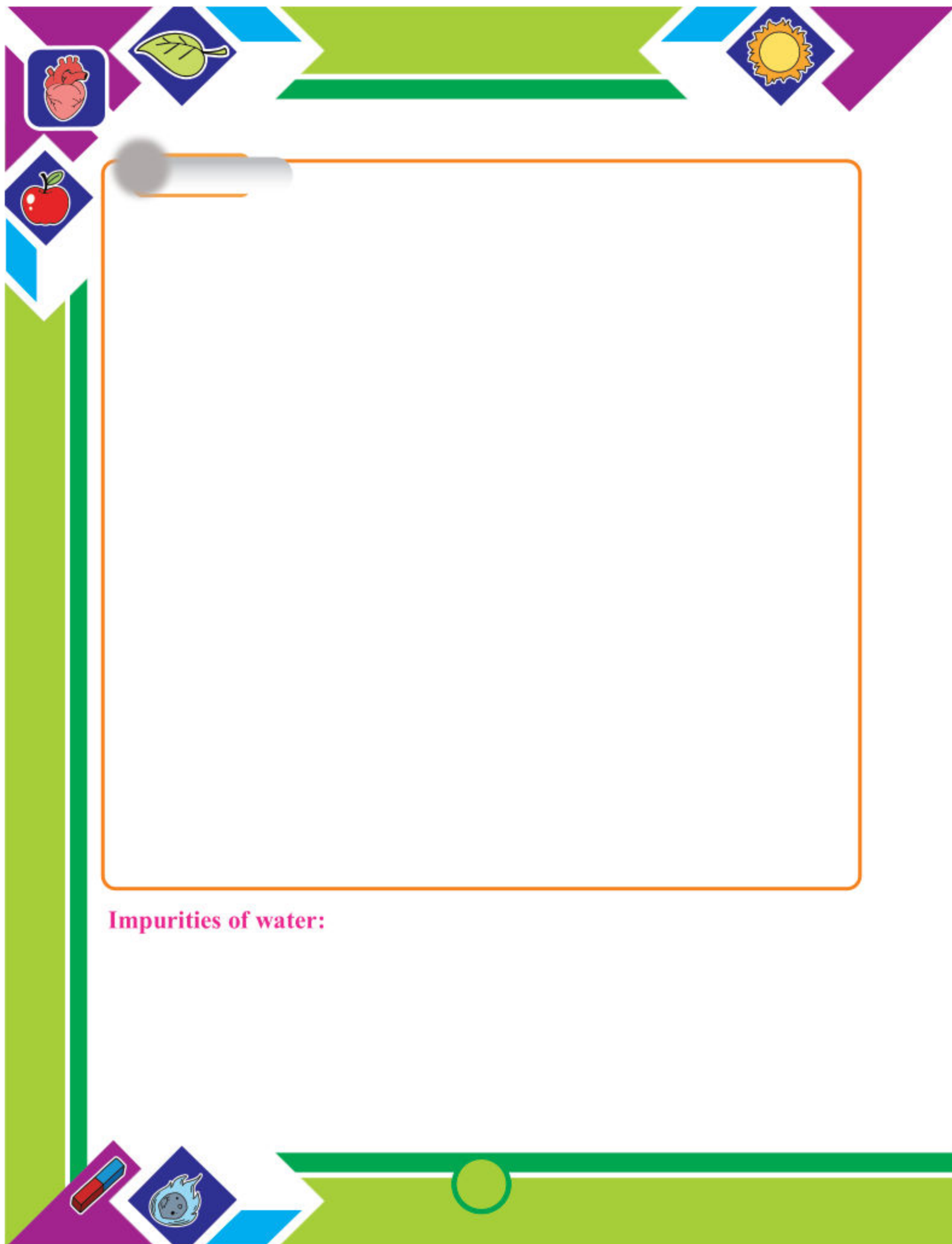
When the water vapours reach up in the
water, these droplets of water combine with the



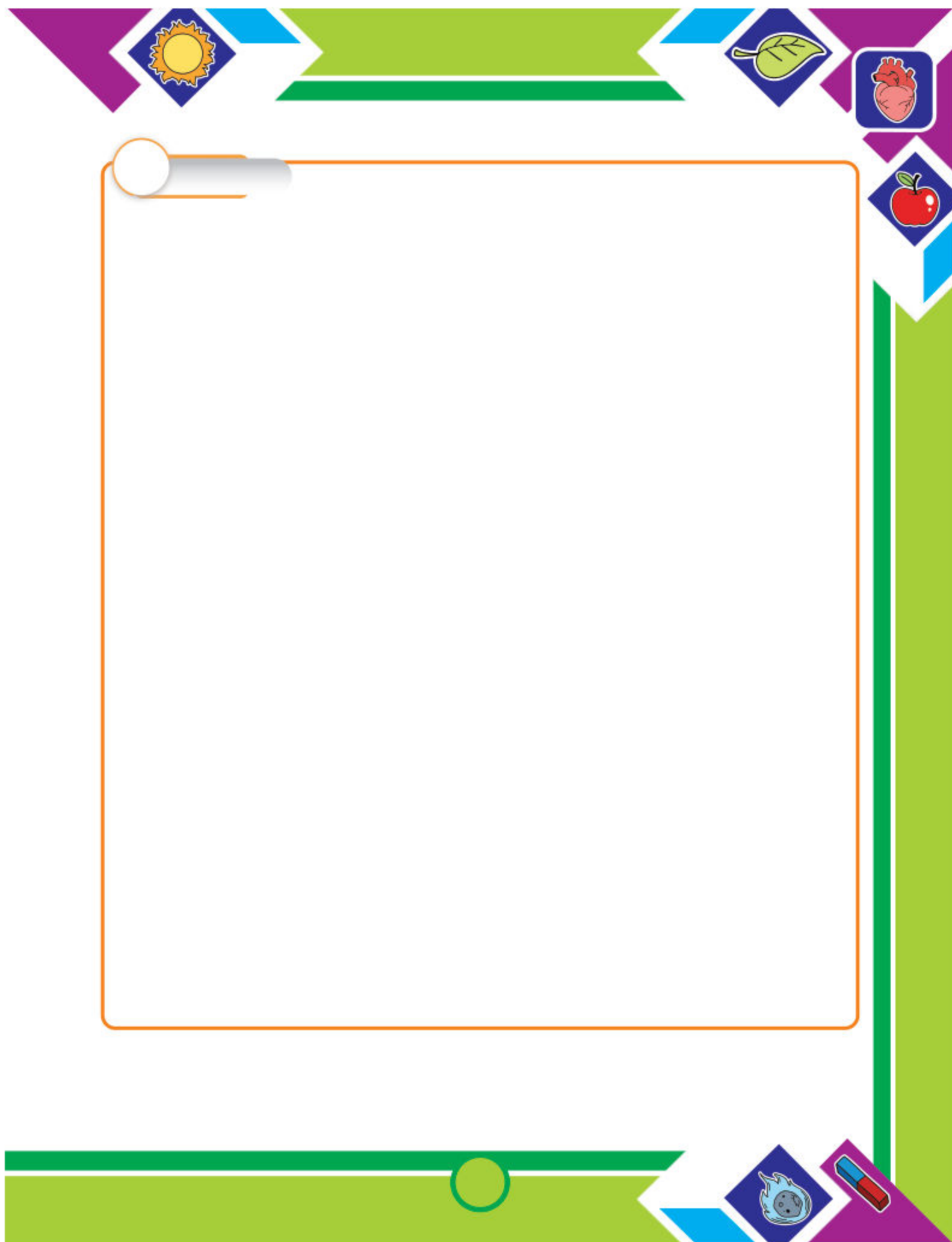
Test yourself:

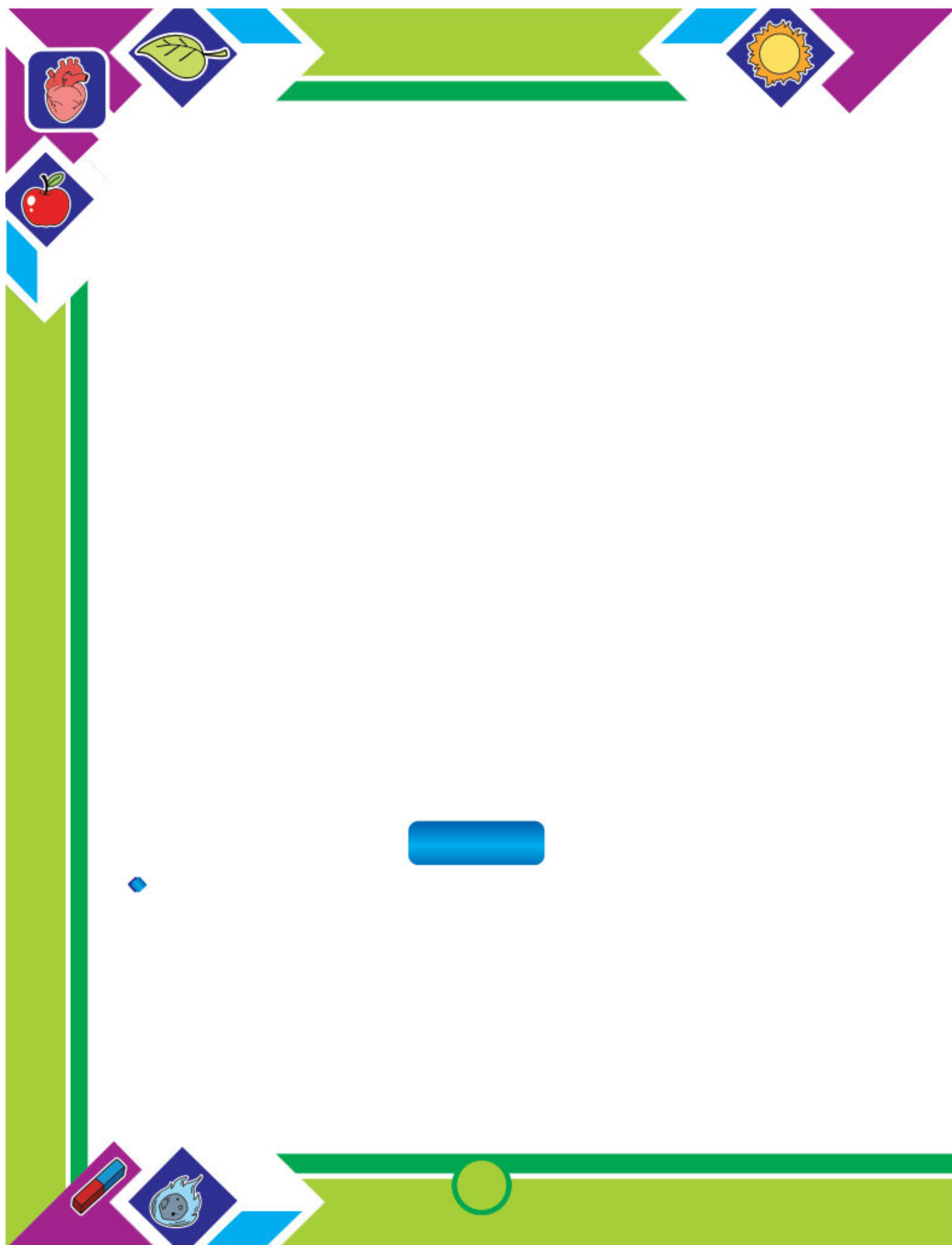
Label the following water cycle properly and check your understanding.

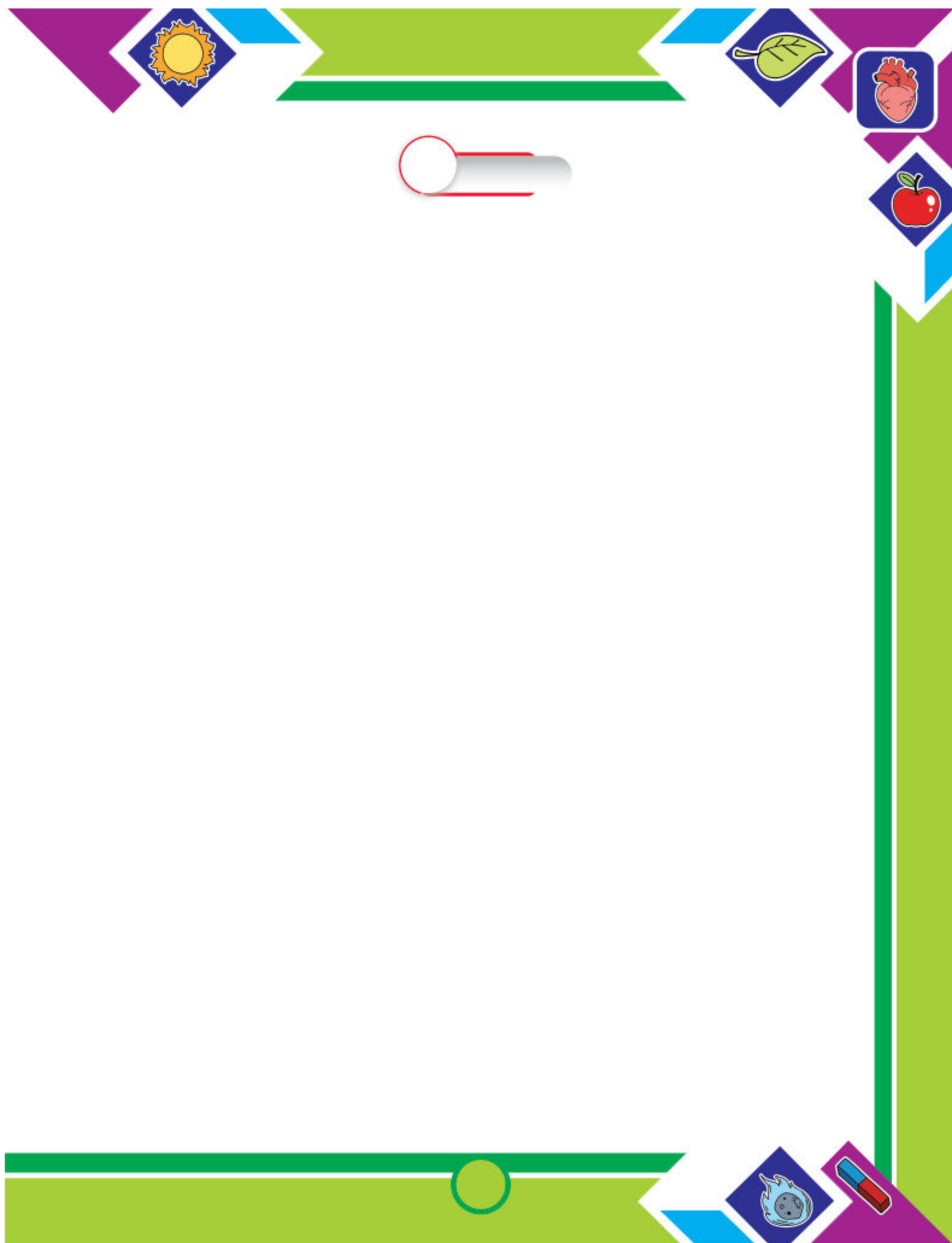
9.3 TYPES OF WATER

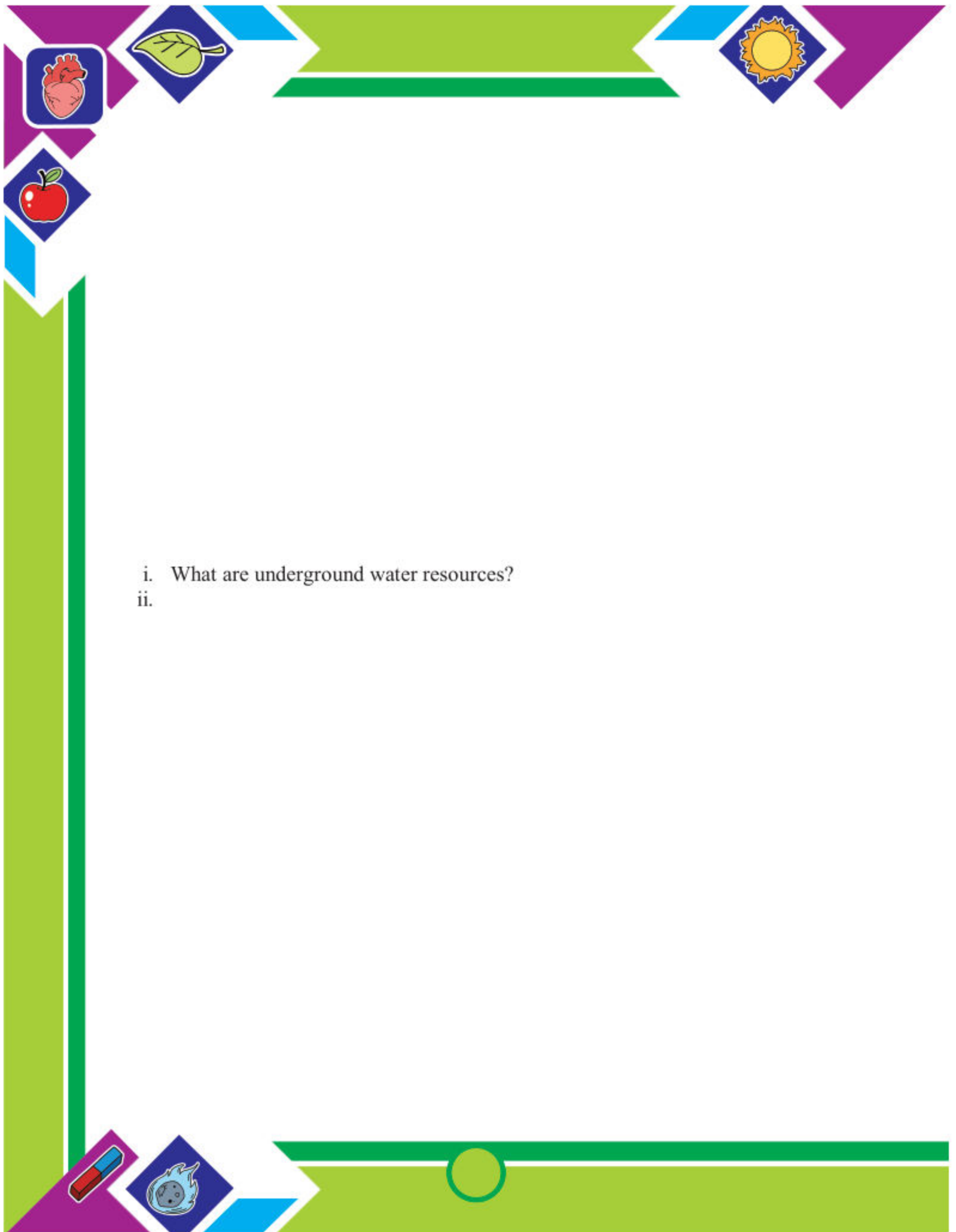


Impurities of water:









- i. What are underground water resources?
- ii.

& KDS



(DWK LQ
6RODVP

WKH

10DMRQFHSWV

,QVK LXQW HZ LOOI QD ER XW

10.1 Position of Sun and Planets

10.2 Planetary System

10.3 ORRQ DQG LWV 3KDVHV

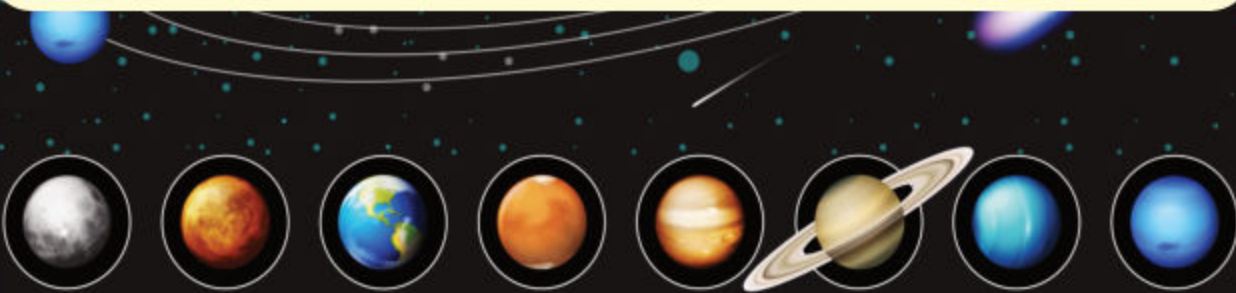
10.4 Earth Rotation

10.5 Lunar and Solar Eclipse

6WXGHQW/HDUQLQJ 2XWFRPHV

SIWUFR PSOM RQNK LXQVV WQWKR XGHD DNV

- ◆ Describe the Solar System with the Sun at the center and the planets revolving around the sun.
 - ◆ Understand the planetary system can contain stars, planets, asteroids, and comets.
 - ◆ Recognize that the earth has a moon that revolves around it, and from earth the
 - ◆ Investigate and describe how day and night are related to Earth's daily rotation about its axis, and provide evidence of this rotation from the changing appearance of shadows during the day.
- Illustrate and explain how Solar and Lunar Eclipses occur.





Introduction:



10.1 POSITION OF SUN AND PLANETS

around the sun. Our solar system consists of eight planets, several moons, millions of



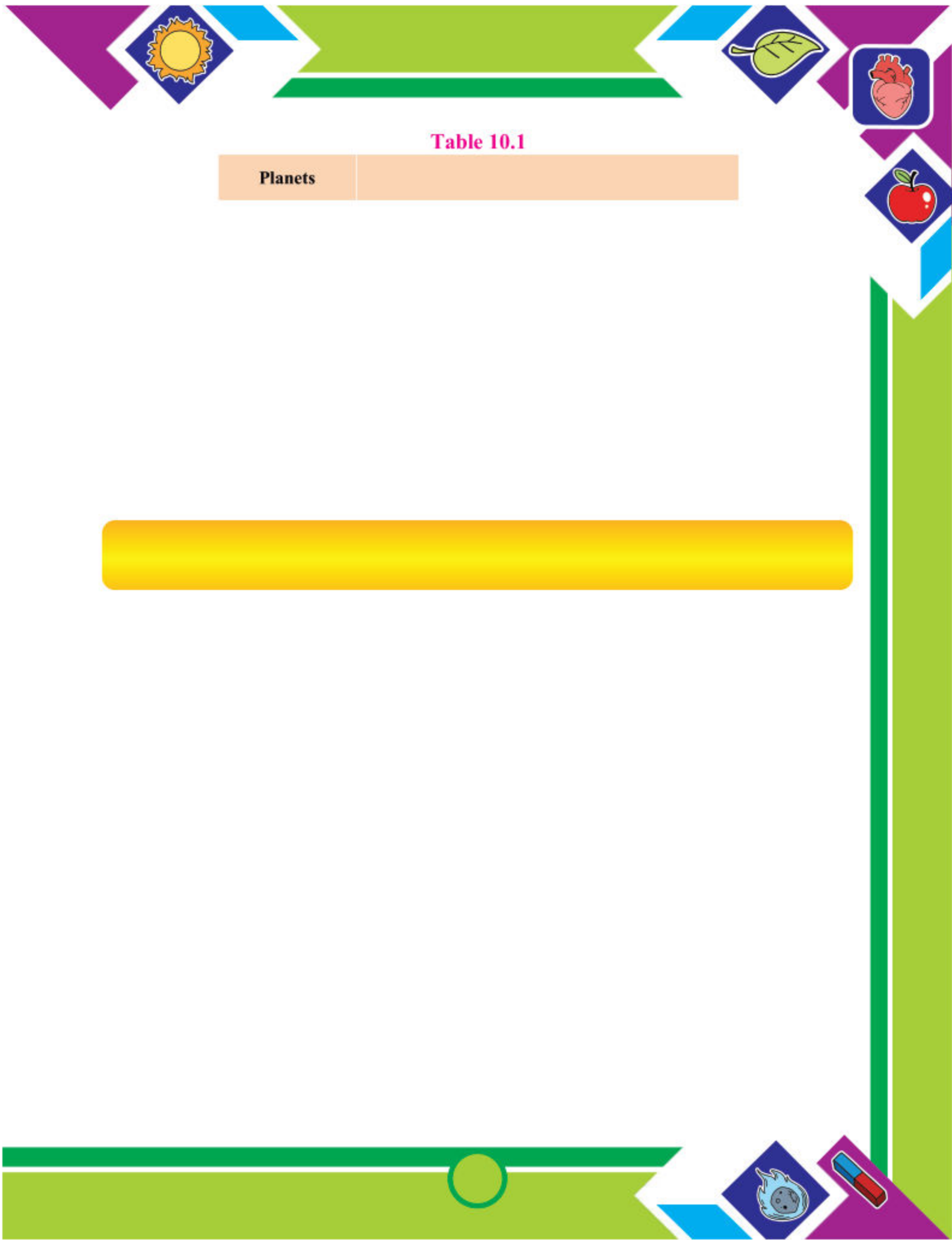
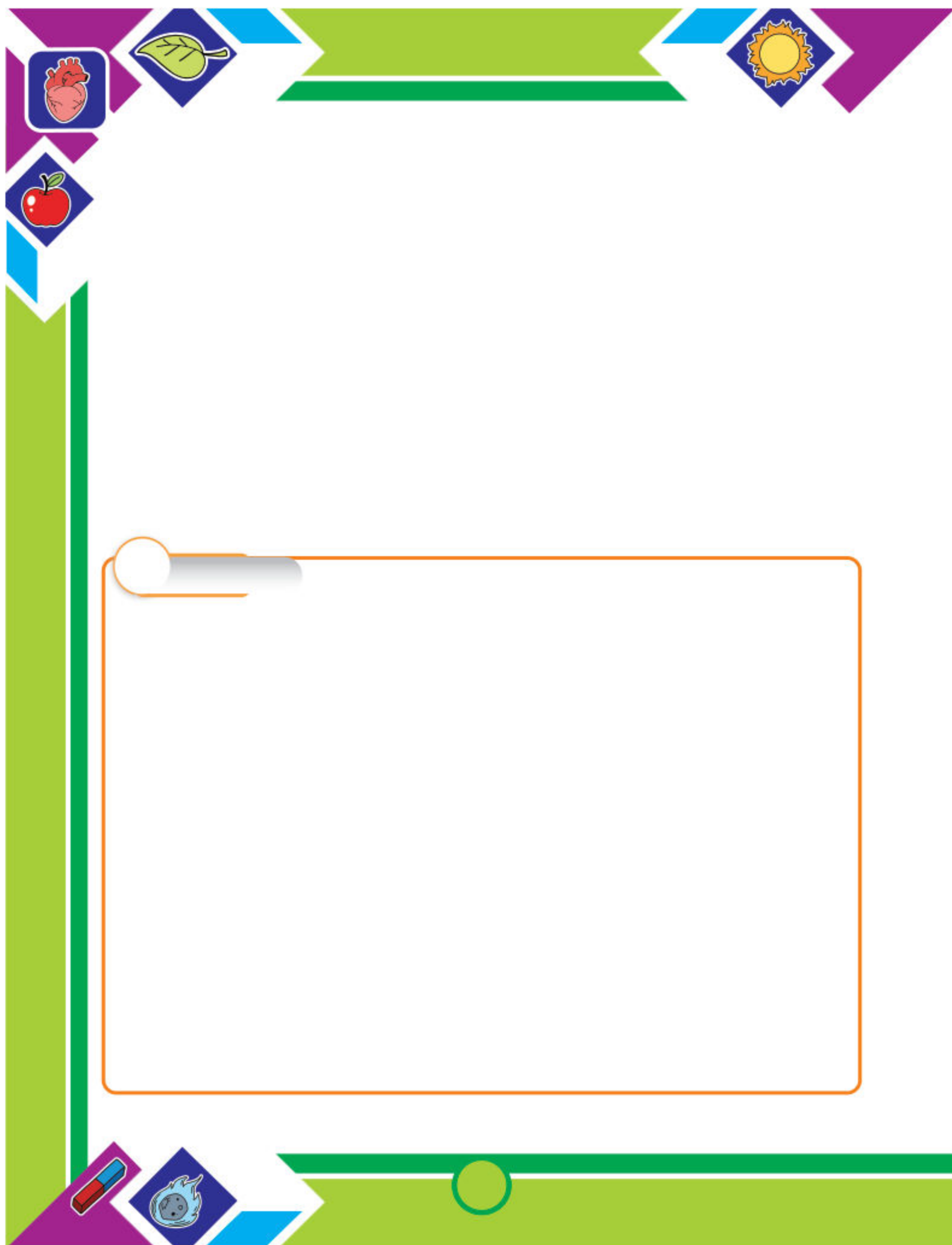


Table 10.1

Planets	
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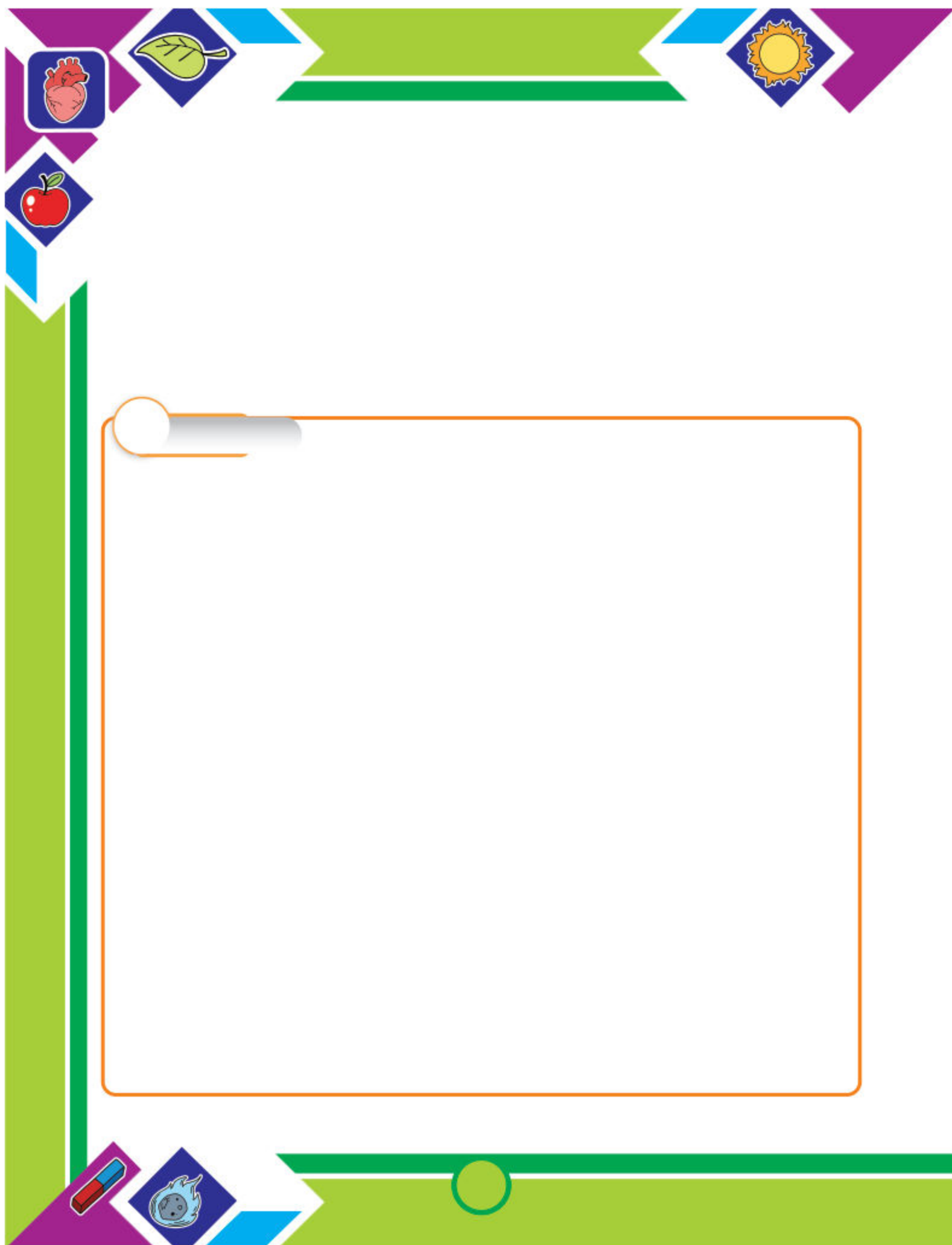


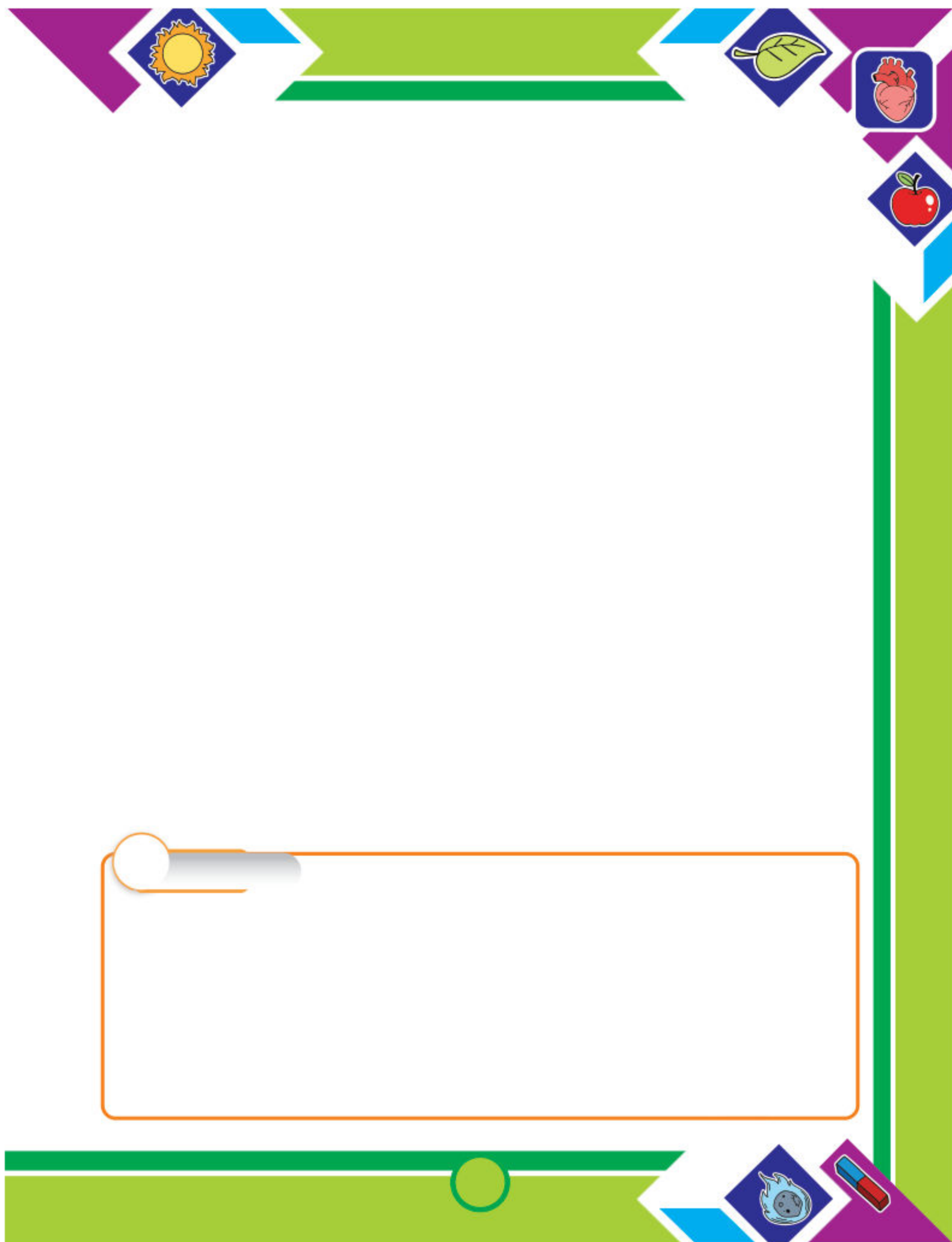
10.3 MOON AND PHASES:

The Moon:



Fig 10.7 Moon

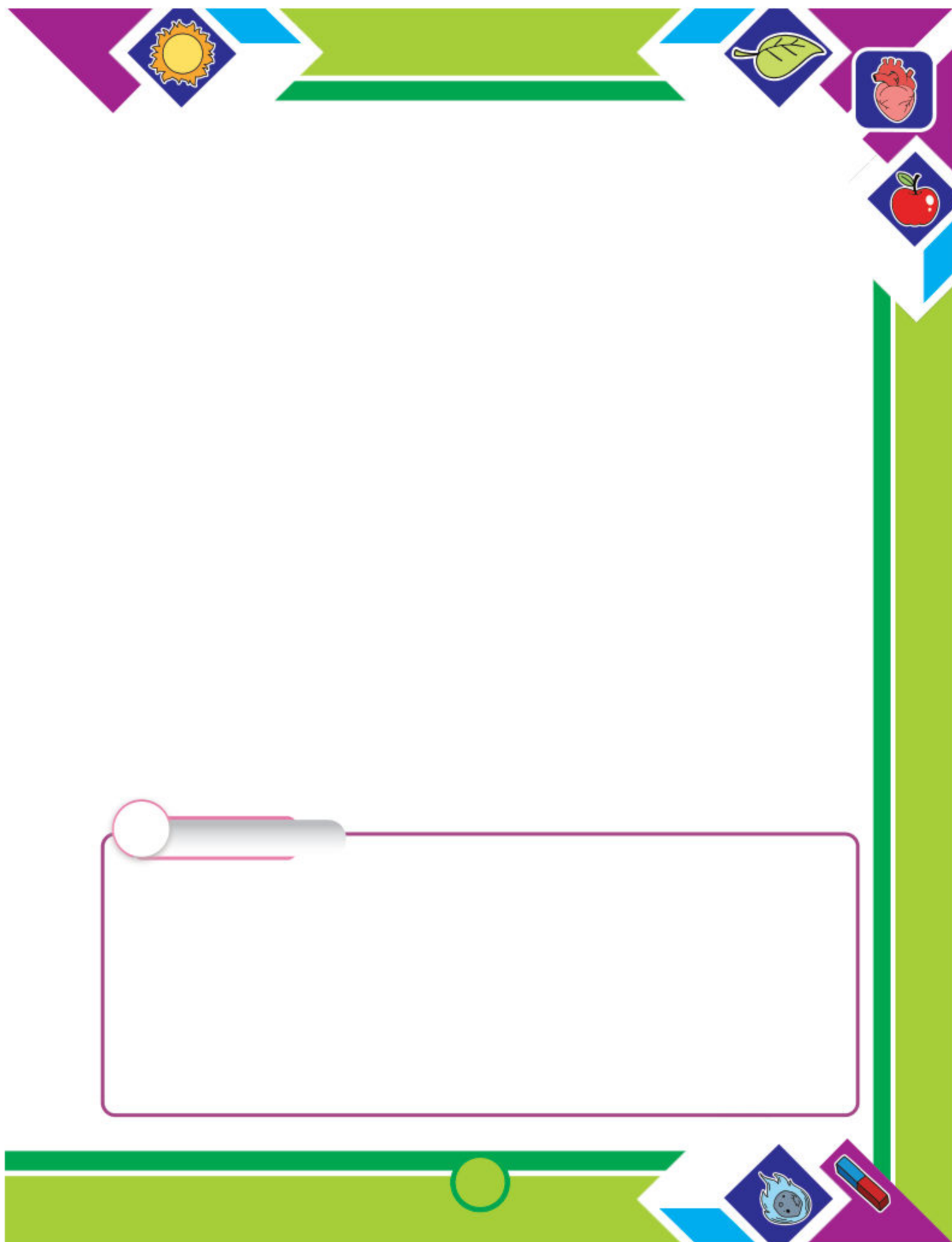






-
- Light the torch and focus the beam on the ball from







10.5 Lunar and Solar Eclipses:

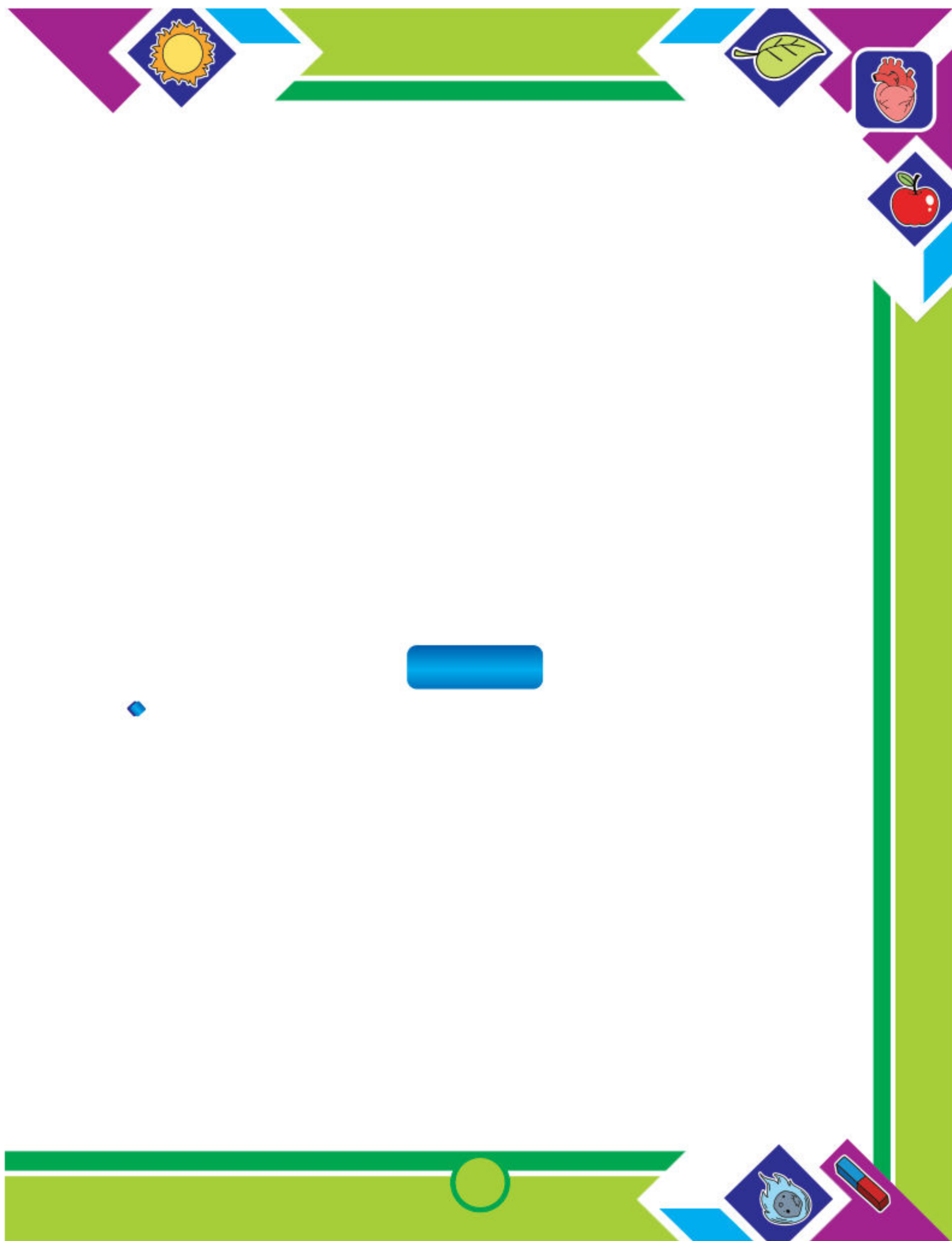
Lunar Eclipse:



Lunar Eclipse

Solar Eclipse





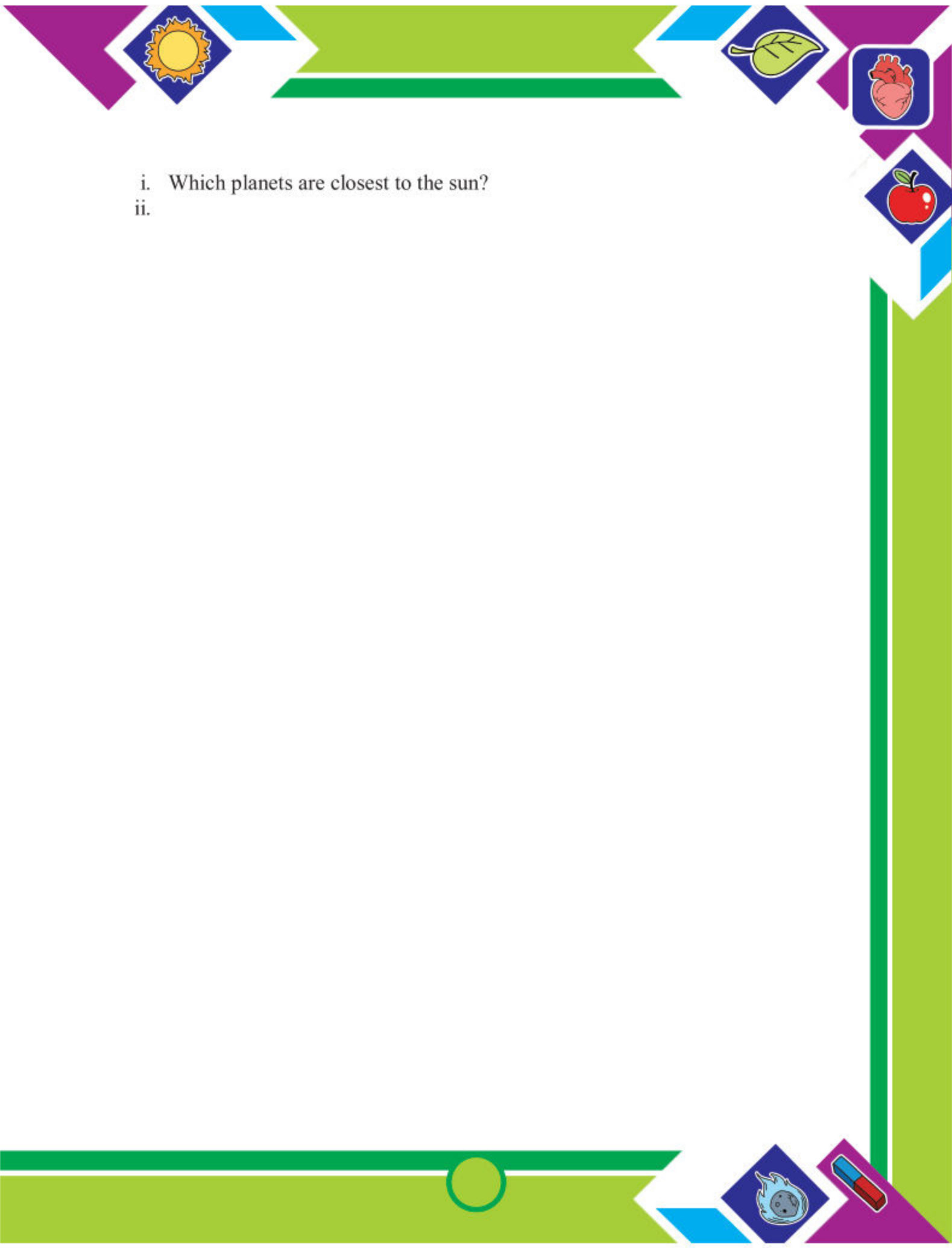


ii)



- A. new-moon
- B. h



- 
- i. Which planets are closest to the sun?
ii.



Air: It is consists of a mixture of gases.
These gases consist of tiny particles

